

Biofísica

Proteinas y ácidos nucleicos

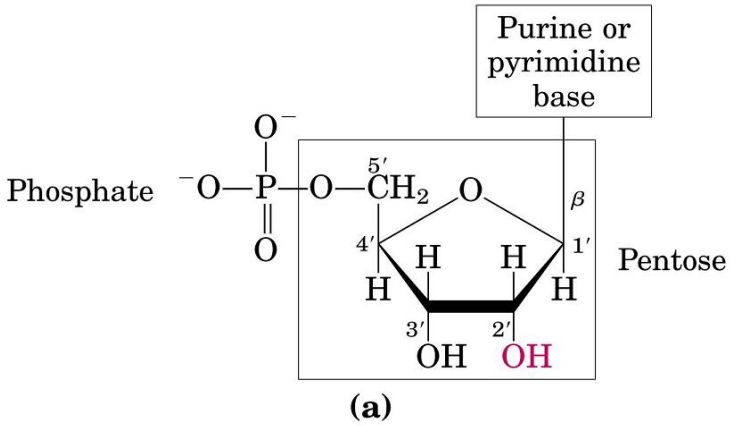
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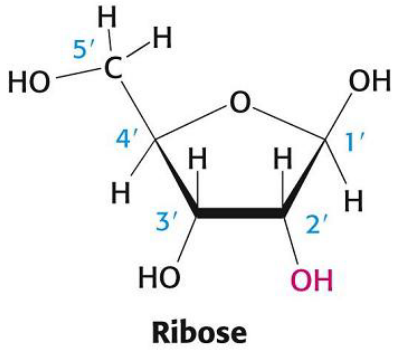
ESTRUCTURA DE LOS ÁCIDOS NUCLEICOS

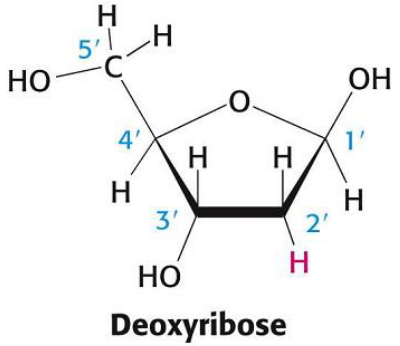
1. NUCLEOTIDOS

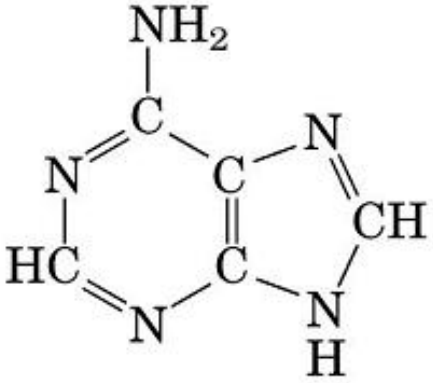
1.1. ESTRUCTURA GENERAL

Los nucleótidos constan de una pentosa (ribosa en RNA y desoxirribosa en DNA), una base (púrica (A, G) o pirimídica (T, C en DNA y U,C en RNA)) y de 1 a 3 fosfatos. La unión de pentosa y base (sin fosfato) se denomina nucleósido. Los nucleósidos y nucleótidos toman nombres derivados de los de la base que contienen.

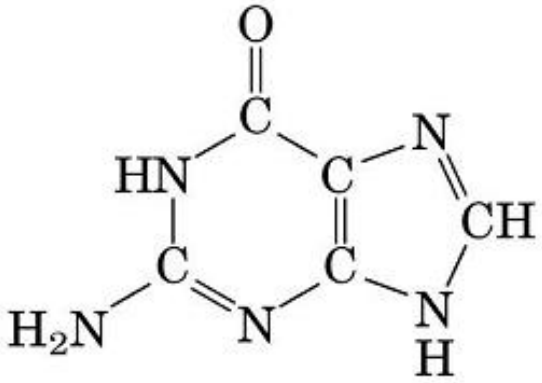






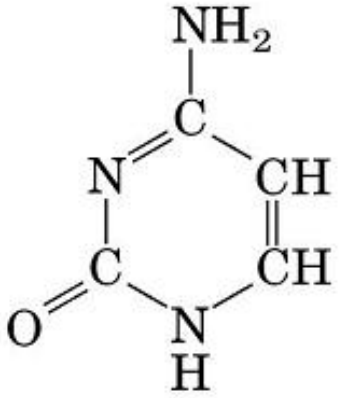


Adenine

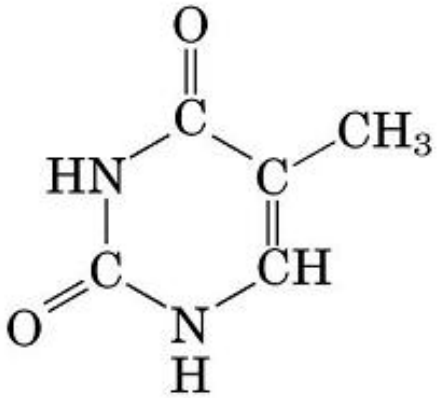


Guanine

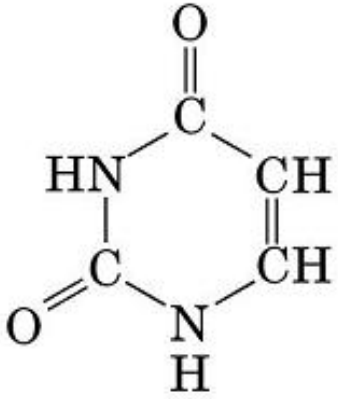
Purines



Cytosine



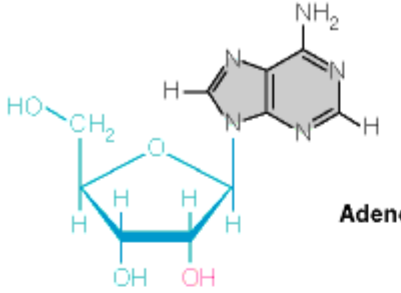
Thymine
(DNA)



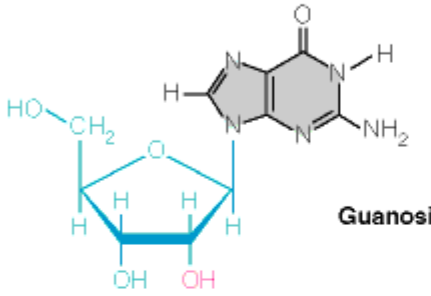
Uracil
(RNA)

Pyrimidines

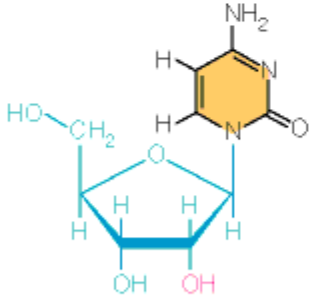
NUCLEOSIDES



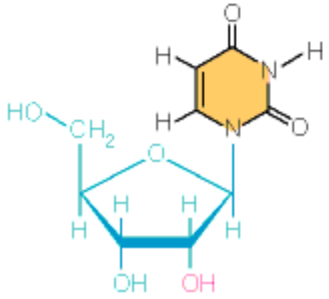
Adenosine



Guanosine

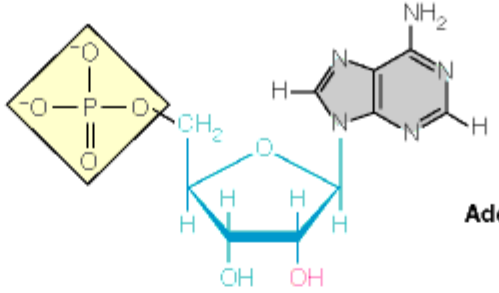


Cytidine

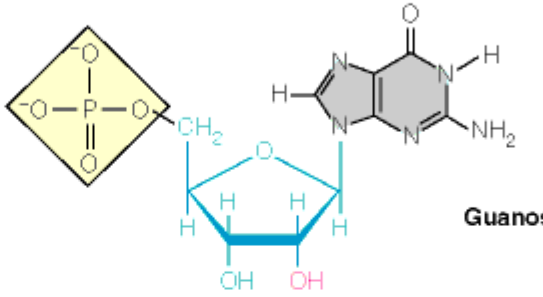


Uridine

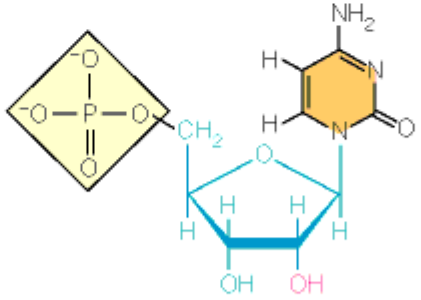
NUCLEOTIDES



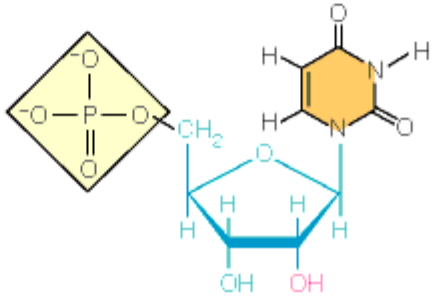
Adenosine 5′-monophosphate (AMP)



Guanosine 5′-monophosphate (GMP)

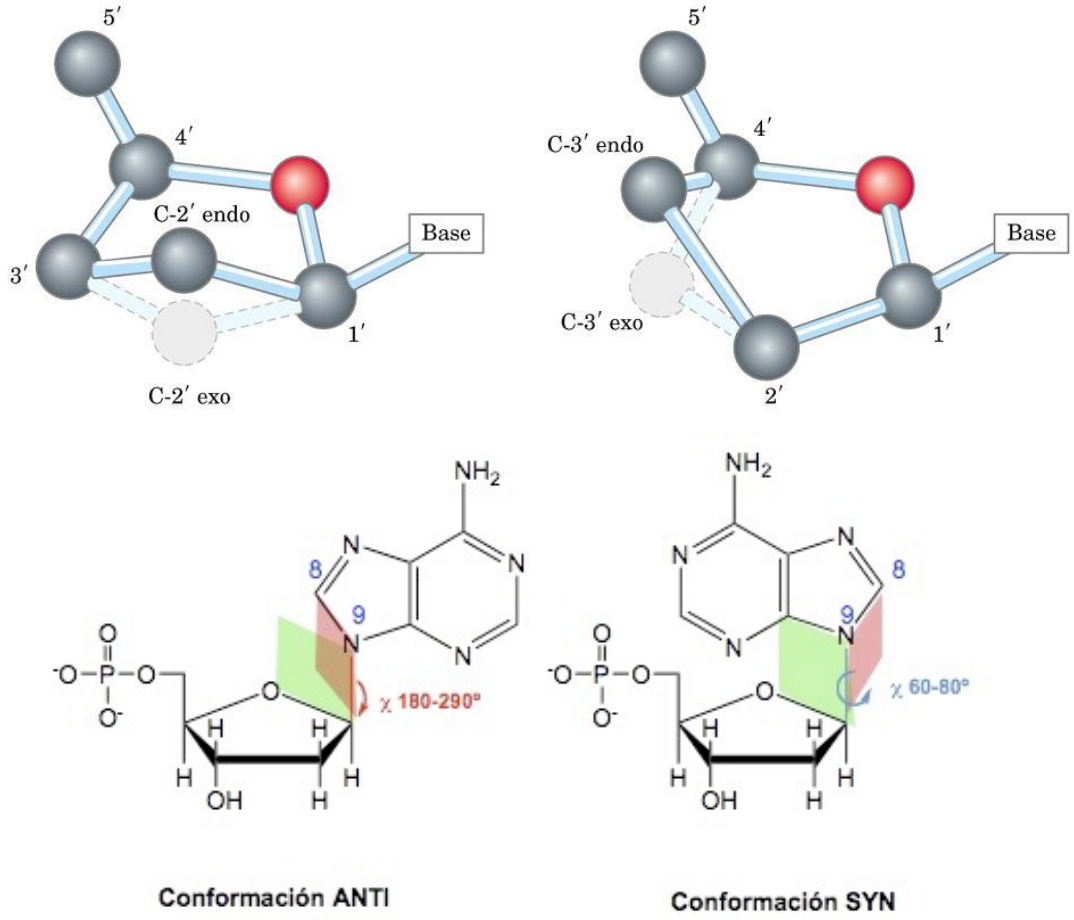


Cytidine 5′-monophosphate (CMP)



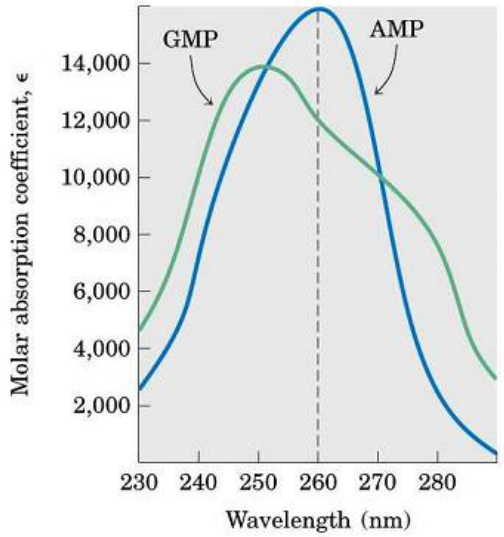
Uridine 5′-monophosphate (UMP)

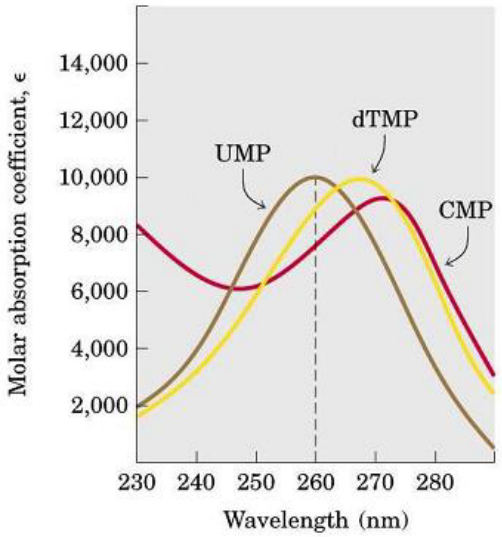
La pentosa forma un ciclo de 5 átomos (anillo de furanosa) que presenta dos conformaciones preferentes según el átomo (2' o 3') que queda fuera del plano y en el mismo lado que el carbono 5 (conformaciones 2' endo o 3'endo). En nucleósidos las dos conformaciones se interconvierten.



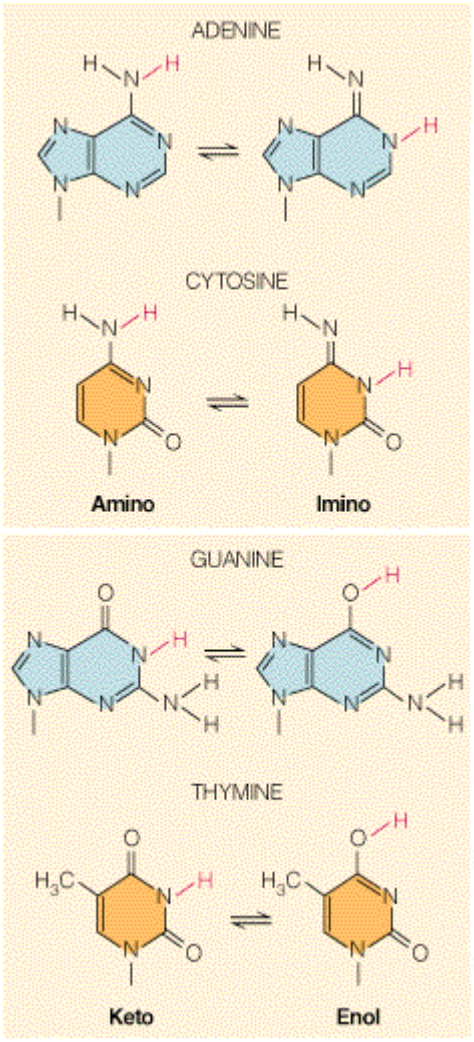
1.2. PROPIEDADES DE LAS BASES

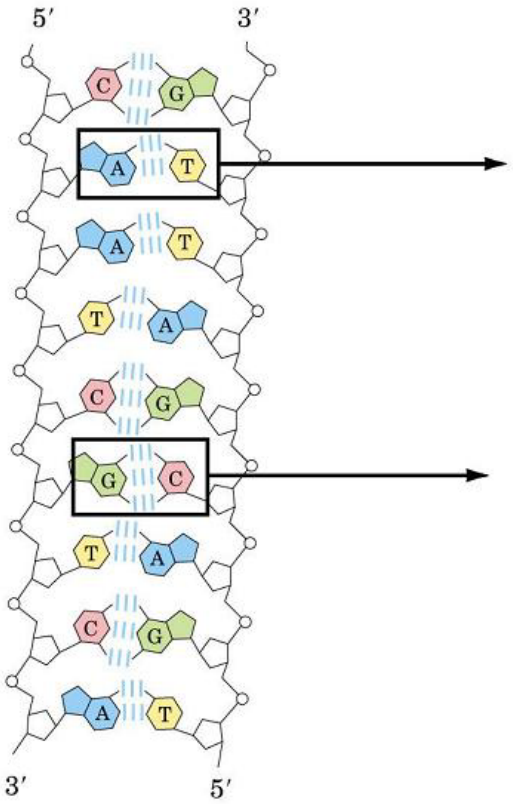
Las bases absorben luz ultravioleta lo que sirve para cuantificar ácidos nucleicos espectrofotométricamente.

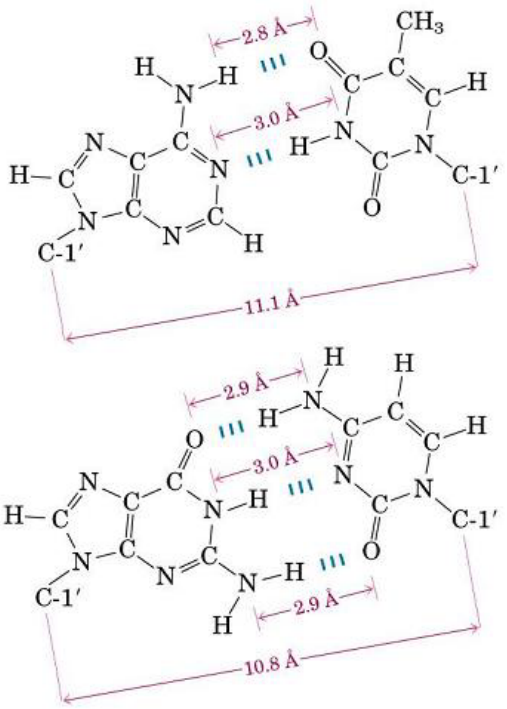




Molar absorption coefficient at 260 nm, ϵ_{260} ($M^{-1}cm^{-1}$)	
AMP	15,400
GMP	11,700
CMP	7,500
UMP	9,900
dTMP	9,200

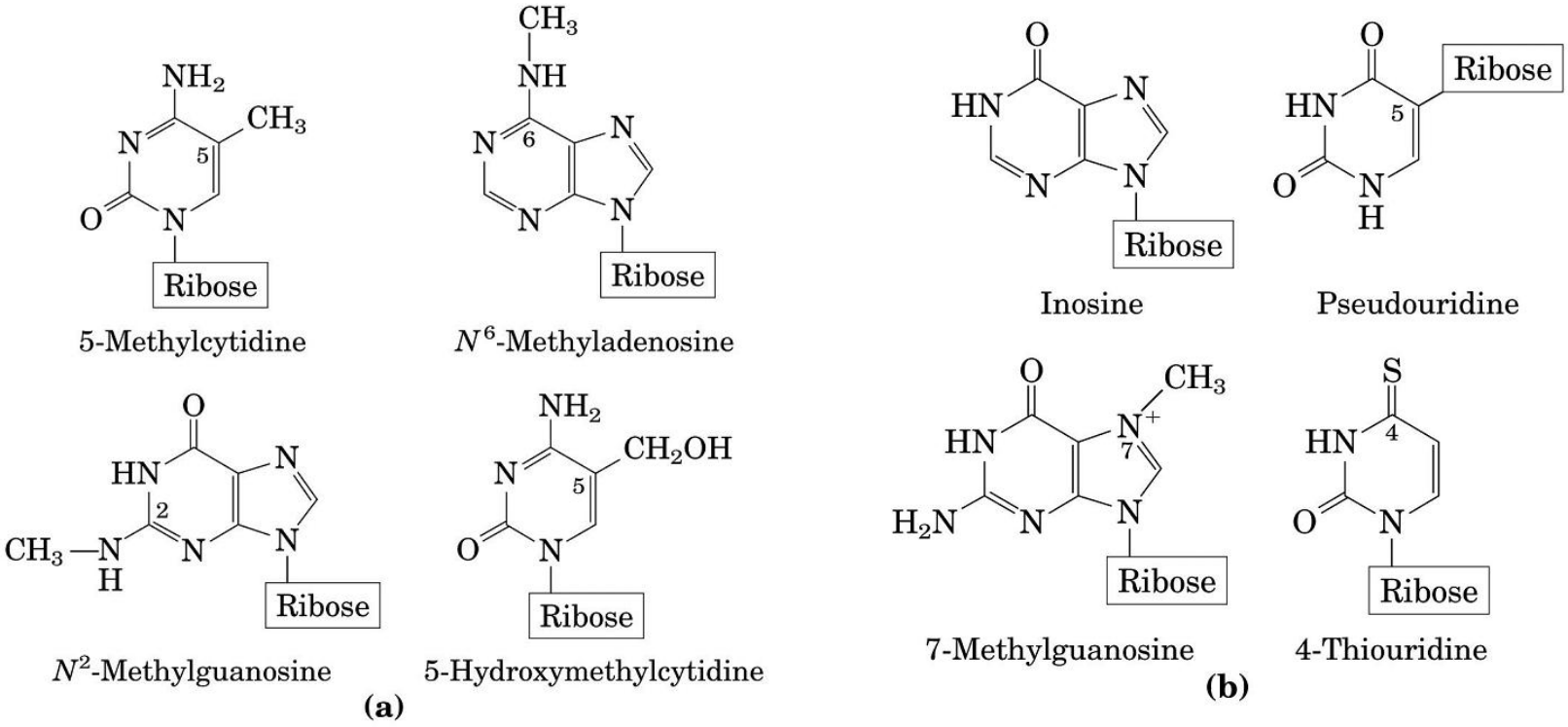

ADENINE
CYTOSINE
Amino
Imino
GUANINE
THYMINE
Keto
Enol


5′
3′
3′
5′
Adenine
Guanine


2.8 Å
3.0 Å
11.1 Å
Thymine
2.9 Å
3.0 Å
2.9 Å
10.8 Å
Cytosine

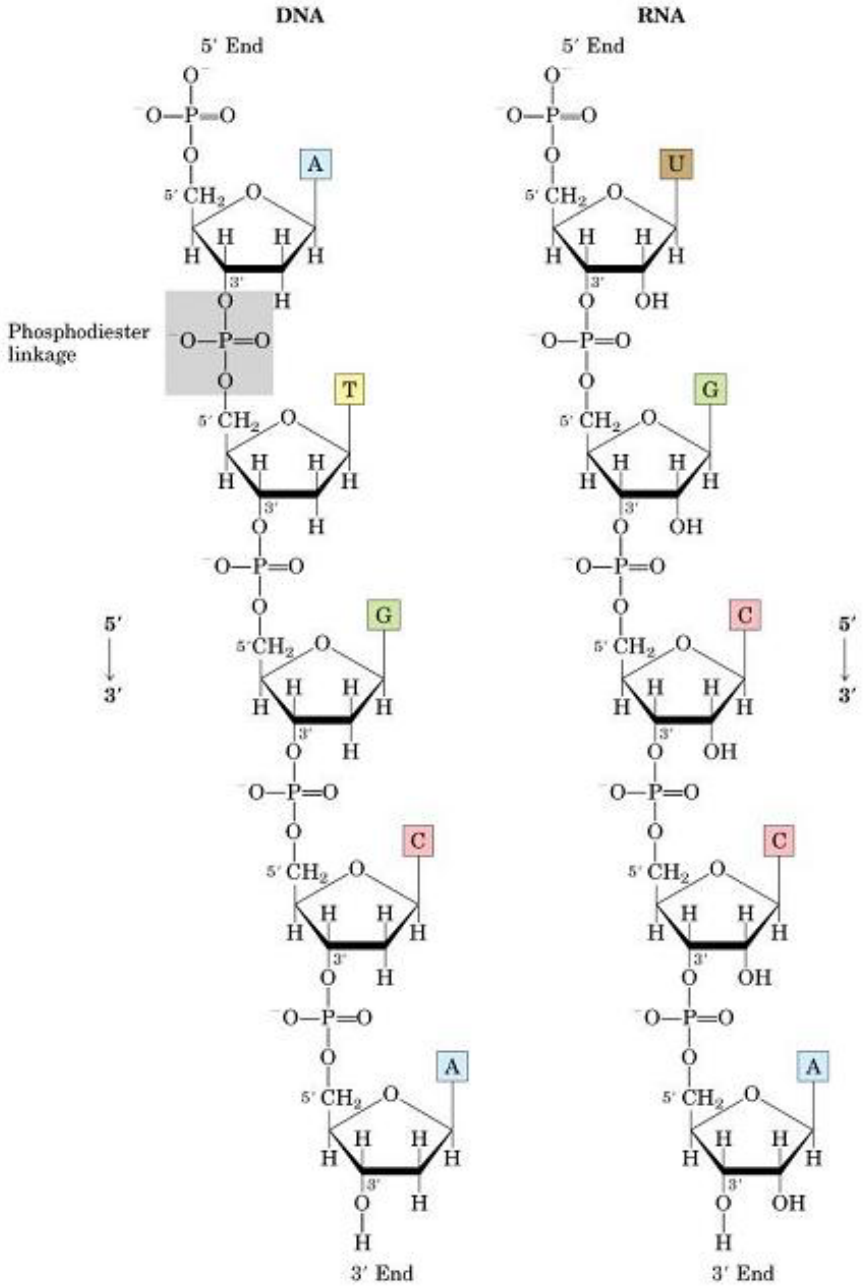
1.3. BASES SECUNDARIAS

Además de las bases principales algunas moléculas de ácidos nucleicos presentan otras bases (secundarias o raras) como bases metiladas (en DNA) u otras bases (en t-RNA).



1.4. POLINUCLEÓTIDOS: ESTRUCTURA, SÍNTESIS Y SECUENCIACIÓN

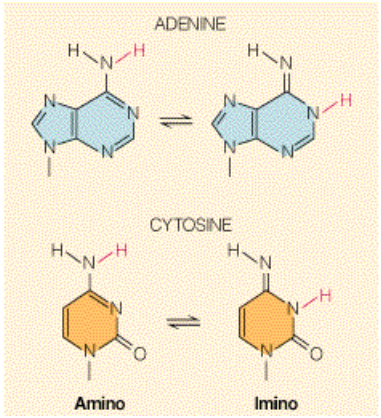
Los polinucleotidos (RNA o DNA) son polímeros lineales en los que aparecen nucleósidos enlazados por puentes fosfato entre oxígenos en 5 y en 3. El extremo 3 se localiza en el nucleótido que utiliza sólo su oxígeno en 5 para enlazarse a otro nucleótido. El extremo 5 en el que sólo utiliza el oxígeno en 3 para enlazarse a la cadena. Un polinucleótido puede concebirse como constituido por un esqueleto de grupos pentosa y fosfato del que cuelgan las bases.

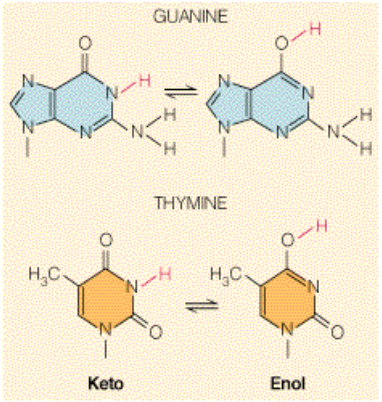


2. DNA

2.1. LA DOBLE HÉLICE: B-DNA

Es el tipo de doble hélice que predomina en el DNA. Su estructura fue modelada por Watson y Crick utilizando información de rayos x, de abundancia relativa de bases y de las estructuras tautomericas de las mismas.





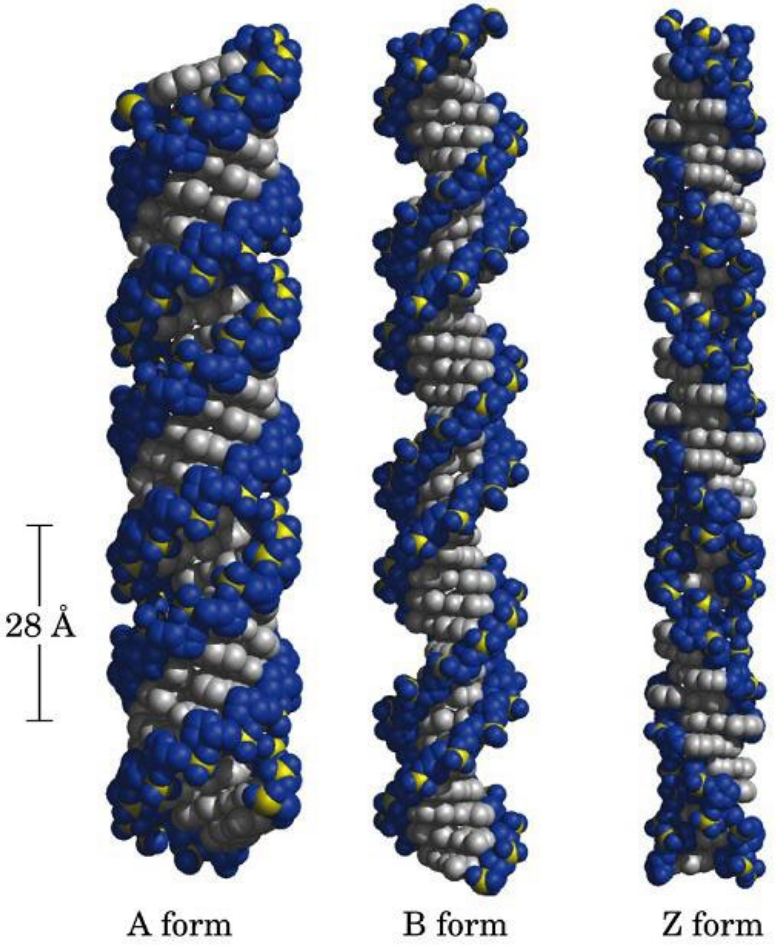
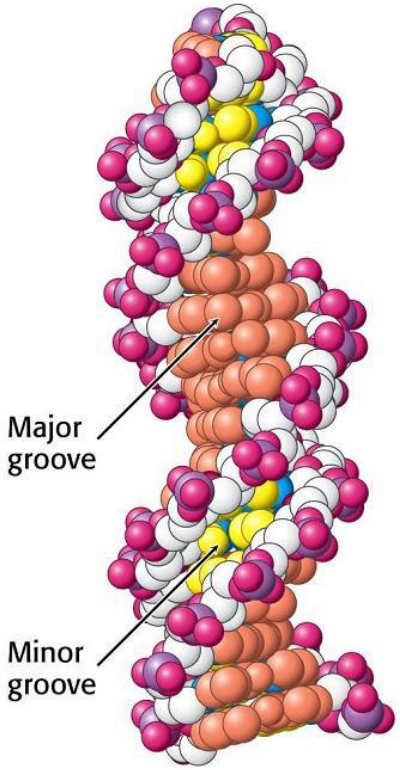
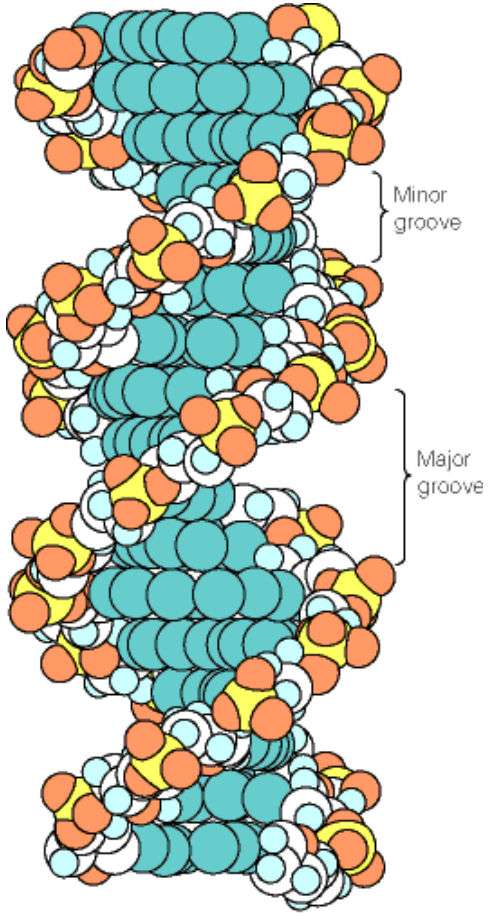


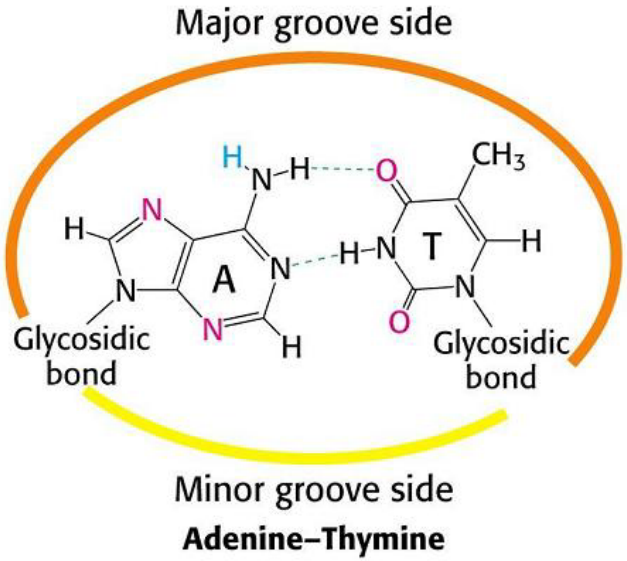
TABLE 5.1 **Base compositions experimentally determined for a variety of organisms**

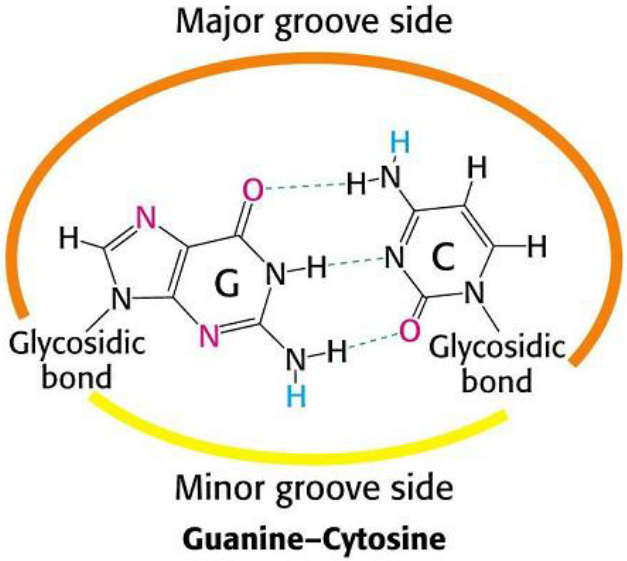
Species	A:T	G:C	A:G
Human being	1.00	1.00	1.56
Salmon	1.02	1.02	1.43
Wheat	1.00	0.97	1.22
Yeast	1.03	1.02	1.67
Escherichia coli	1.09	0.99	1.05
Serratia marcescens	0.95	0.86	0.70

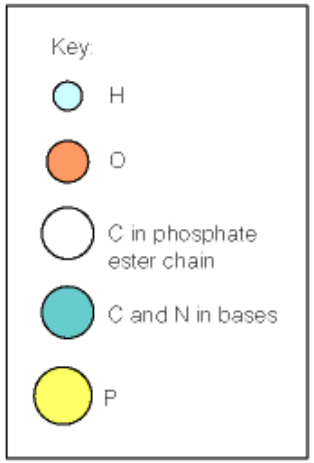
Presenta un surco mayor (donde las bases están más expuestas y pueden ser reconocidas) y otro menor.







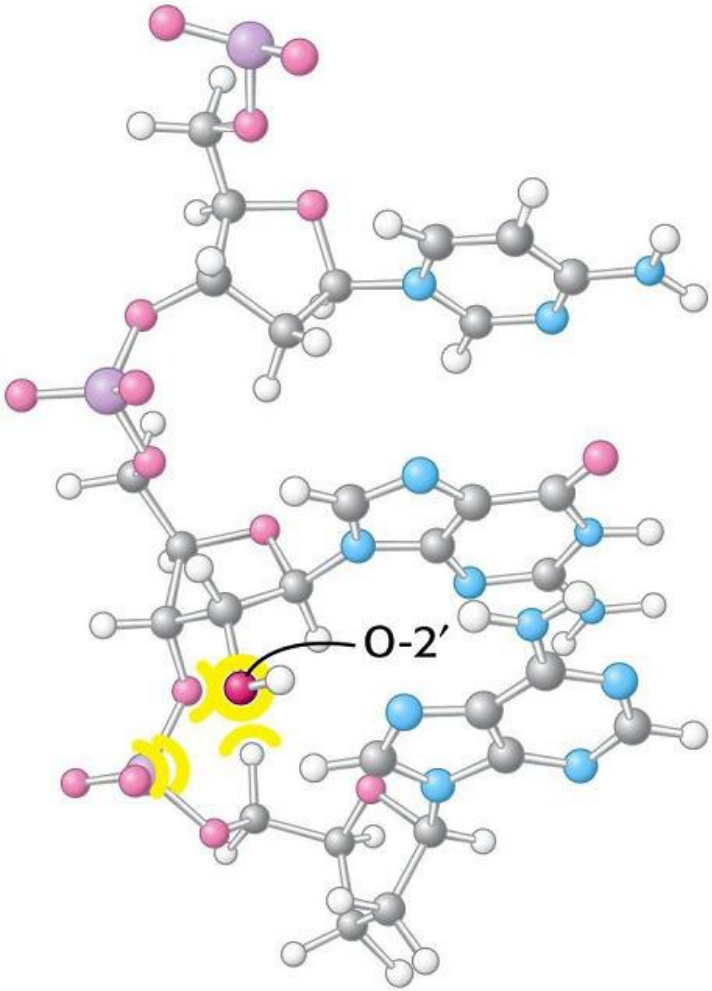




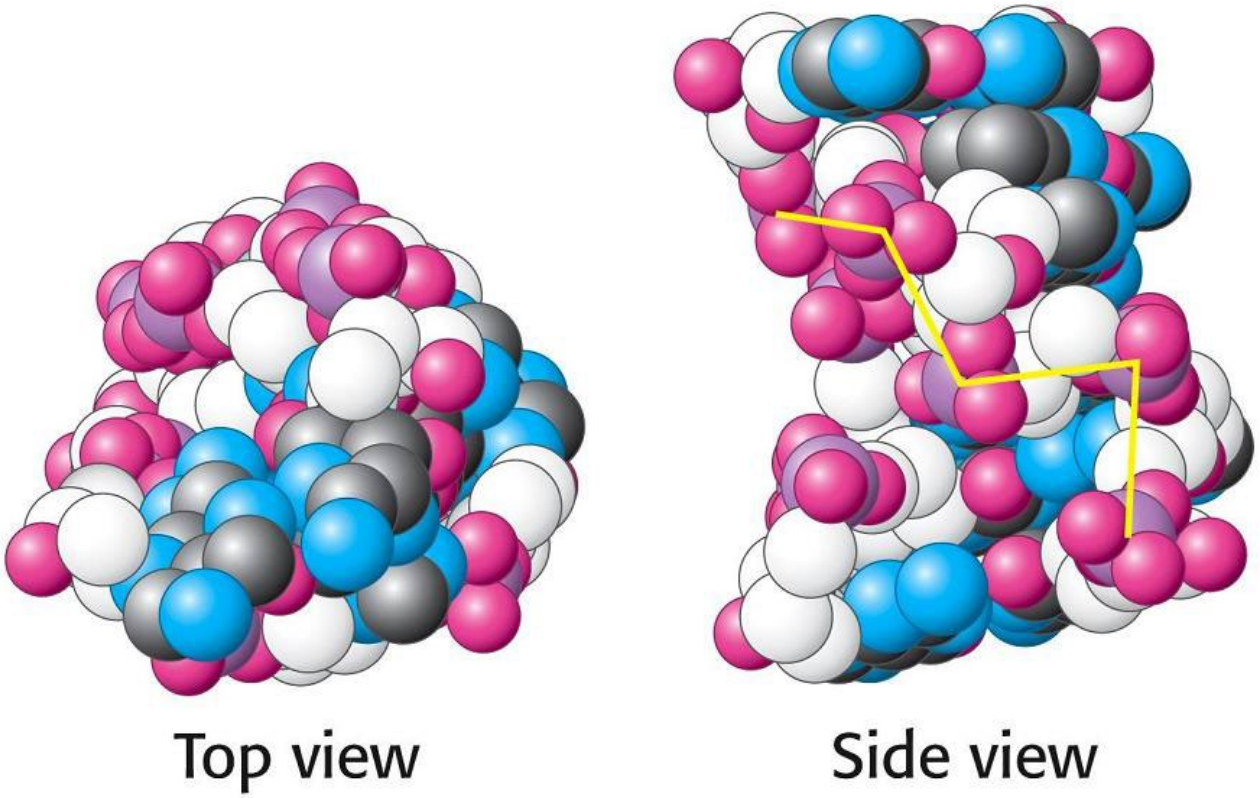
2.2. OTRAS FORMAS DE DOBLE HÉLICE: A-DNA Y Z-DNA

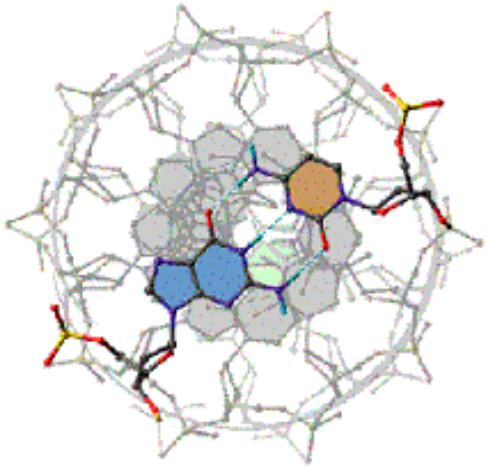
2.2.1. A-DNA.

Se forma a partir de la forma B por deshidratación. Es la forma predominante en regiones de RNA de doble hélice porque el OH en 2' impide la formación de la forma B. Es más gruesa y corta de modo que el surco mayor es más profundo y menos accesible y en el interior de la doble hélice cabe agua.

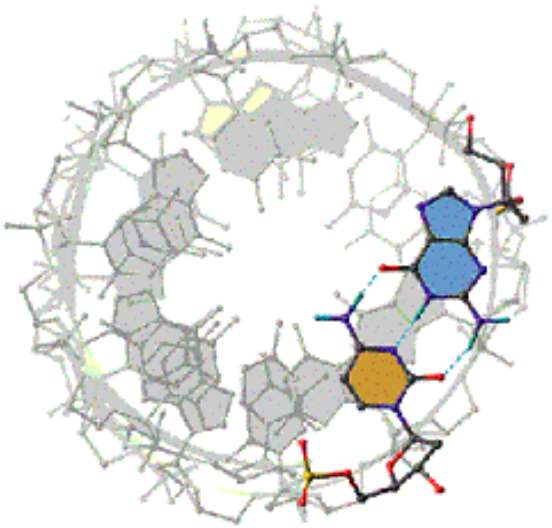


2.2.2. Z-DNA

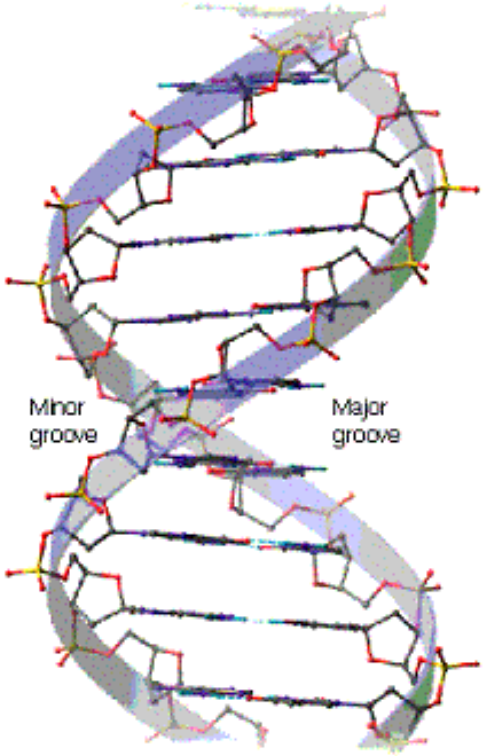




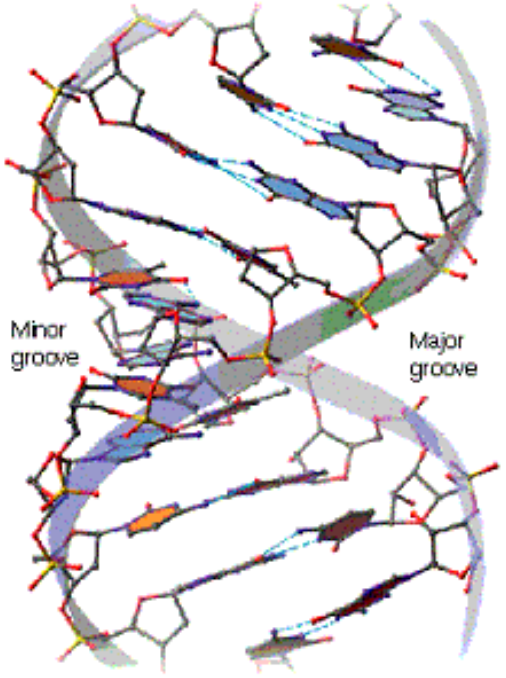
(a) B-DNA, end-on view



(c) A-DNA, end-on view



(b) B-DNA, side view



(d) A-DNA, side view

	A form	B form	Z form
Helical sense	Right handed	Right handed	Left handed
Diameter	~26 Å	~20 Å	~18 Å
Base pairs per helical turn	11	10.5	12
Helix rise per base pair	2.6 Å	3.4 Å	3.7 Å
Base tilt normal to the helix axis	20°	6°	7°
Sugar pucker conformation	C-3′ endo	C-2′ endo	C-2′ endo for pyrimidines; C-3′ endo for purines
Glycosyl bond conformation	Anti	Anti	Anti for pyrmidines;

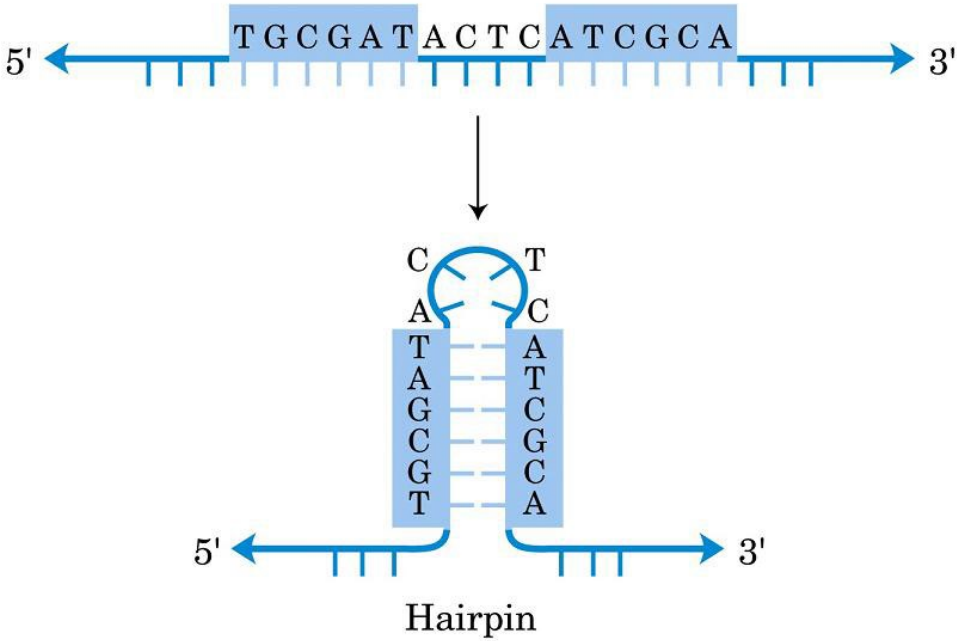
Datos de los tipos de hélices

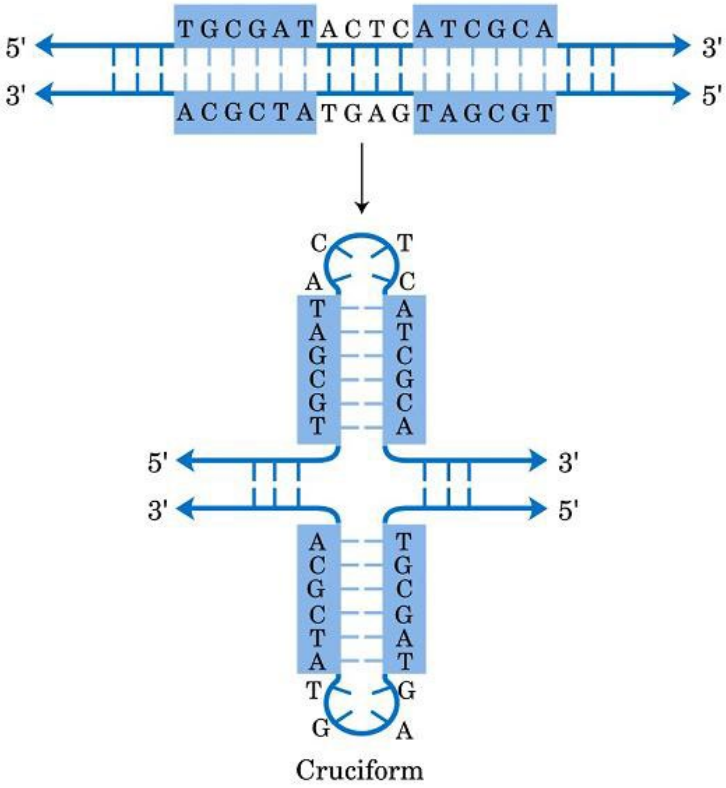
TABLE 27.1 Comparison of A-, B-, and Z-DNA

	Helix type		
	A	**B**	**Z**
Shape	Broadest	Intermediate	Narrowest
Rise per base pair	2.3 Å	3.4 Å	3.8 Å
Helix diameter	25.5 Å	23.7 Å	18.4 Å
Screw sense	Right-handed	Right-handed	Left-handed
Glycosidic bond	*anti*	*anti*	alternating *anti and syn*
Base pairs per turn of helix	11	10.4	12
Pitch per turn of helix	25.3 Å	35.4 Å	45.6 Å
Tilt of base pairs from normal to helix axis	19°	1°	9°
Major groove	Narrow and very deep	Wide and quite deep	Flat
Minor groove	Very broad and shallow	Narrow and quite deep	Very narrow and deep

2.3. ESTRUCTURAS INFRECUENTES DEL DNA

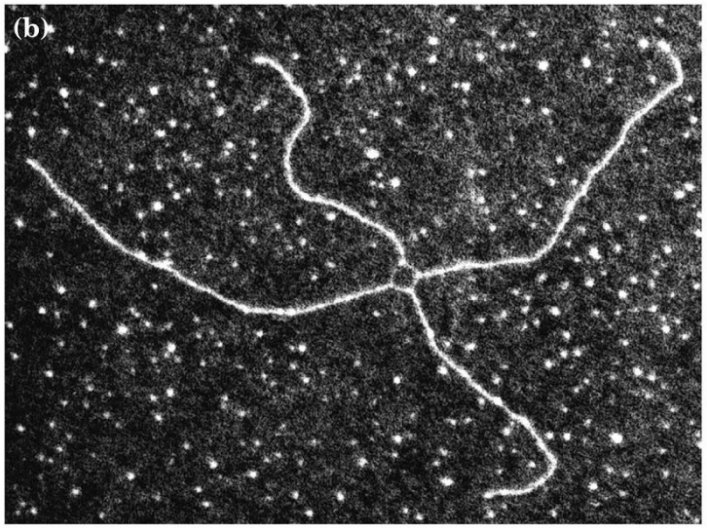
2.3.1. Horquillas y estructuras cruciformes.

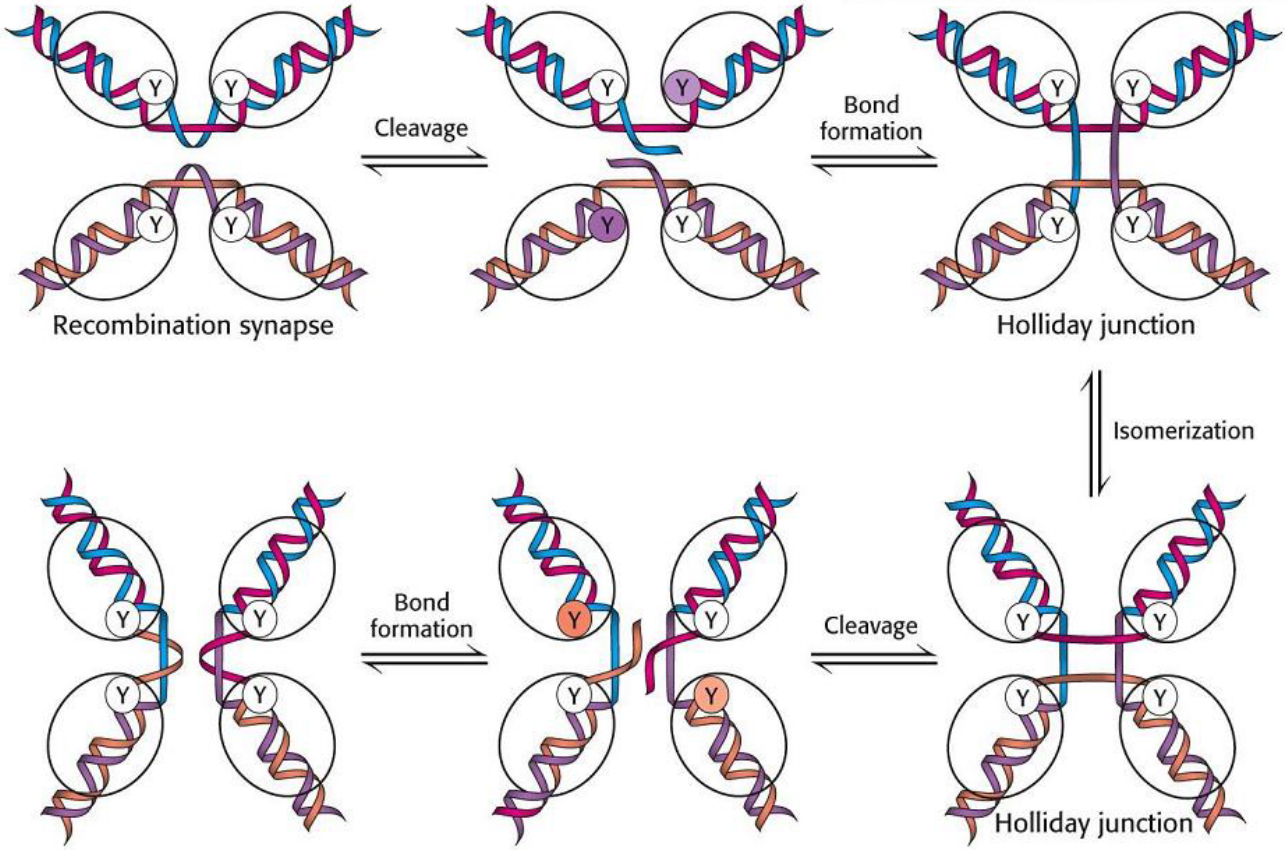




2.3. ESTRUCTURAS INFRECUENTES DEL DNA

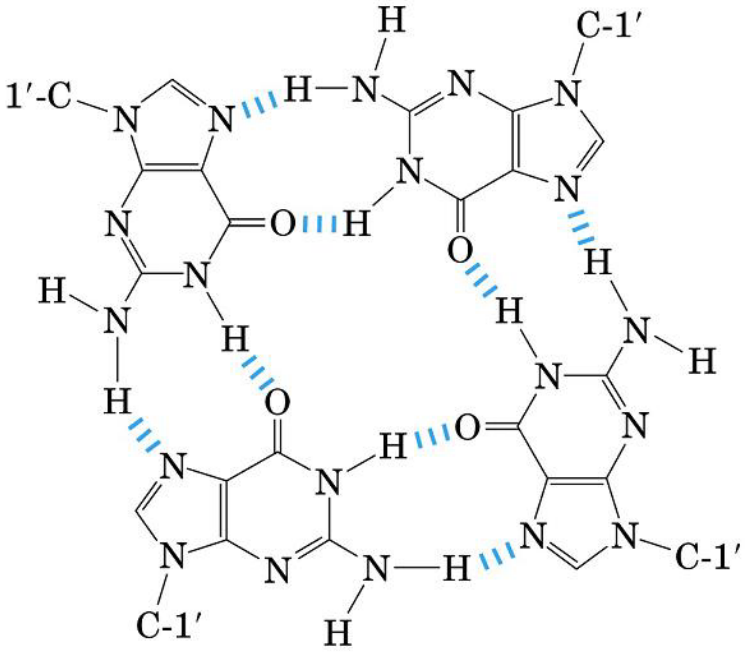
2.3.2. Intersecciones Holliday.



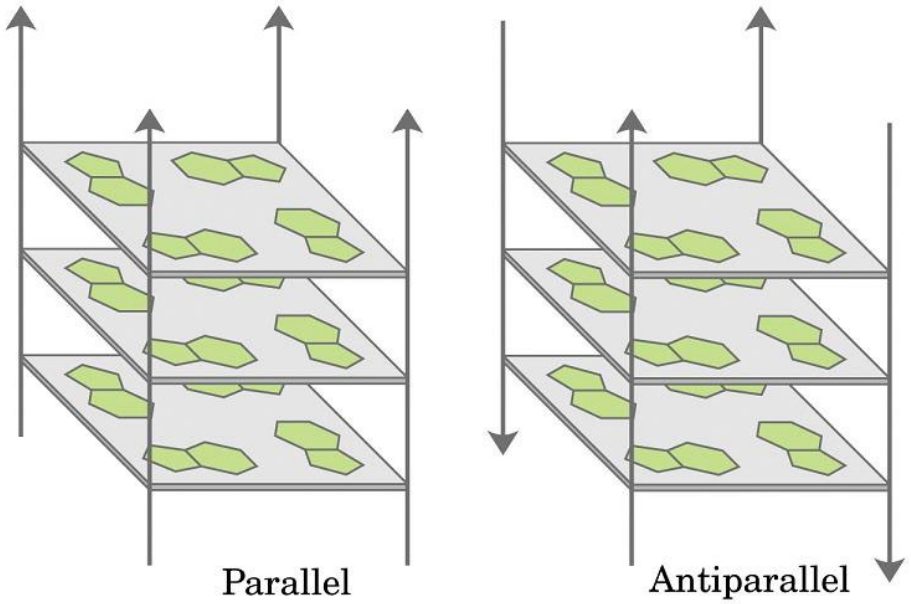


2.3. ESTRUCTURAS INFRECUENTES DEL DNA

2.3.3. Cadena G de los telómeros. Los telómeros aparecen en los extremos de los cromosomas eucariotas y contienen repeticiones de secuencias ricas en G (la cadena complementarias en C). La cadena rica en G puede formar una hélice cuádruple . Esta estructura podría evitar la degradación del cromosoma.

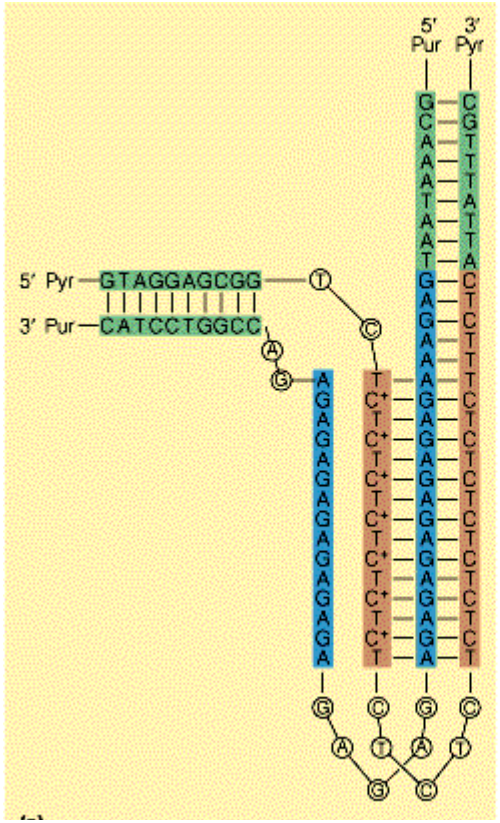


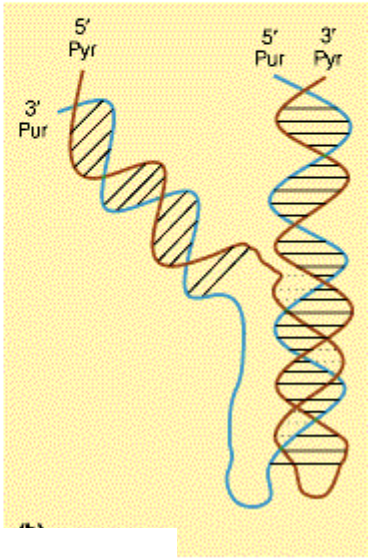
Guanosine tetraplex
(c)

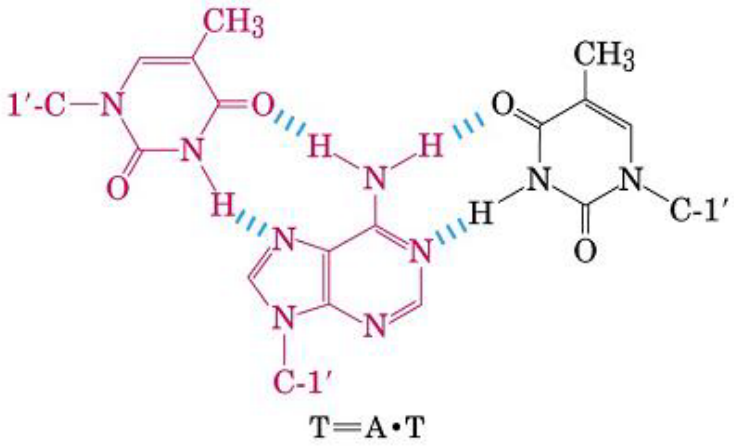


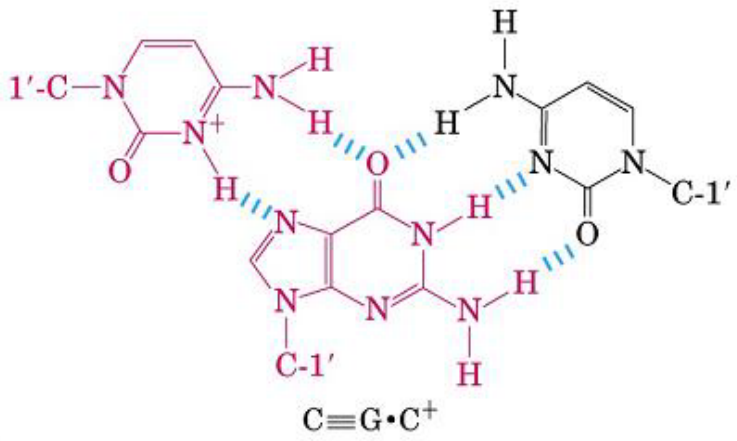
2.3. ESTRUCTURAS INFRECUENTES DEL DNA

2.3.4. Triples hélices (H-DNA). Aparece cuando una hebra sólo tiene purinas y la otra sólo pirimidinas. Se forman apareamientos no convencionales.





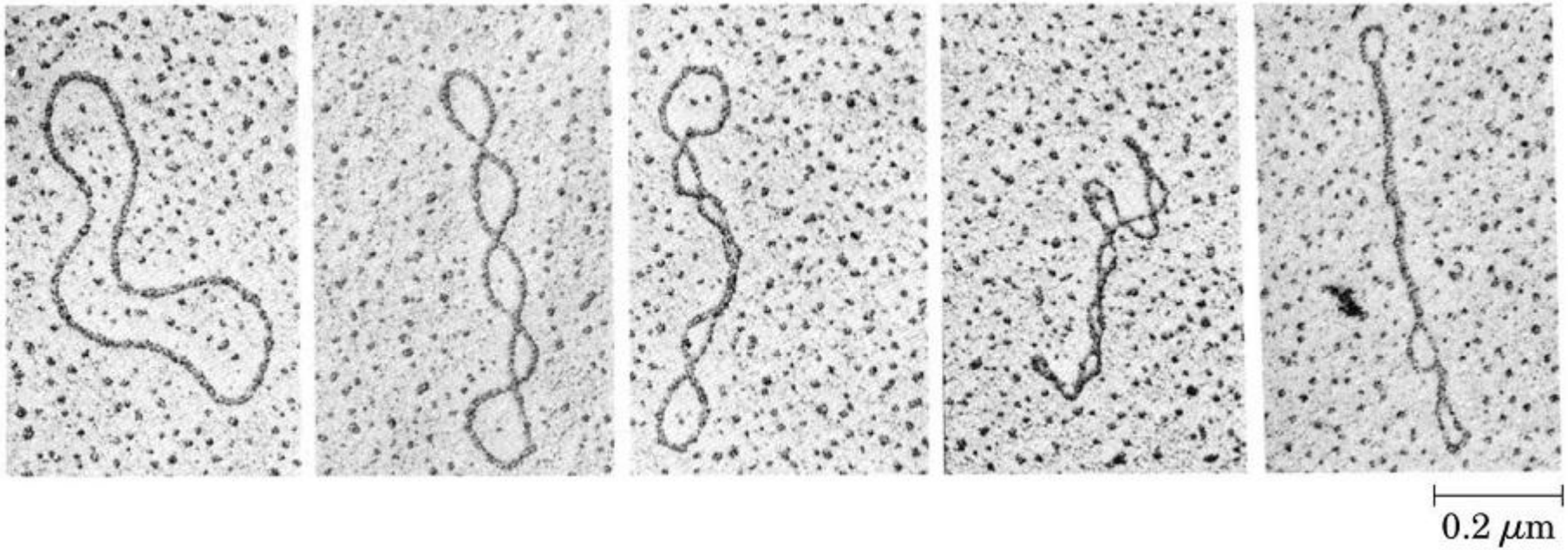




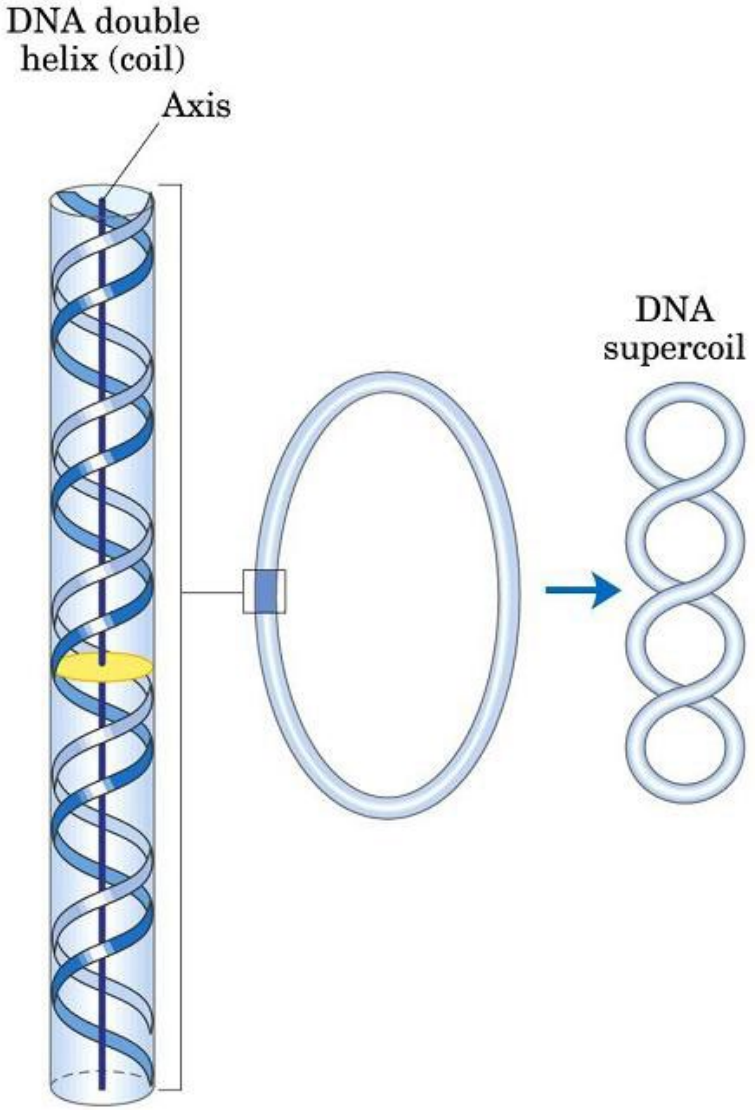
(a)

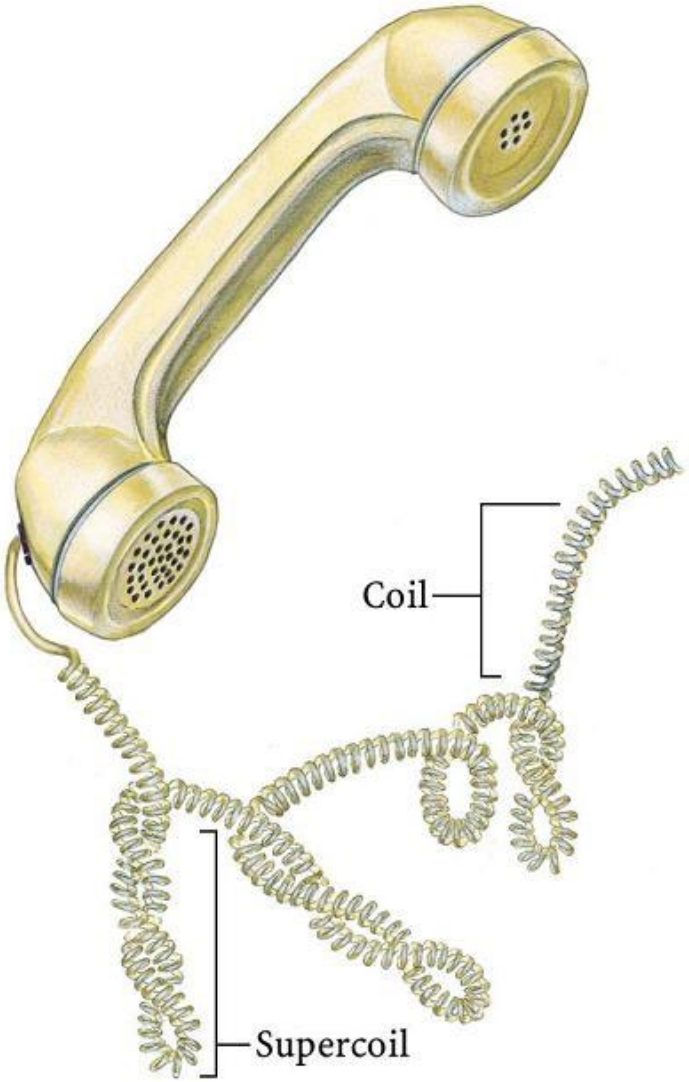
2.3. ESTRUCTURAS INFRECUENTES DEL DNA

2.3.5. Superenrollamiento. El DNA in vivo adopta estructuras cerradas: circulares (en bacterias) o bucles unidos por los extremos (en eucariotas).



Como la doble hélice ya consiste en un enrollamiento de dos cadenas, el superenrollamiento de la doble hélice puede ser: negativo: en sentido opuesto al de las cadenas, aflojándolas; positivo: en el mismo sentido, apretándolas.

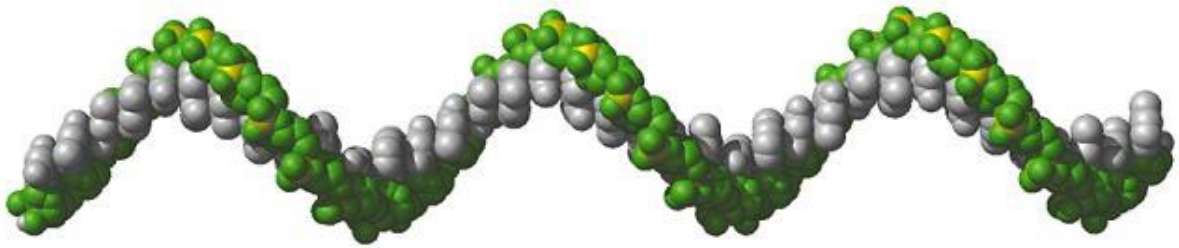




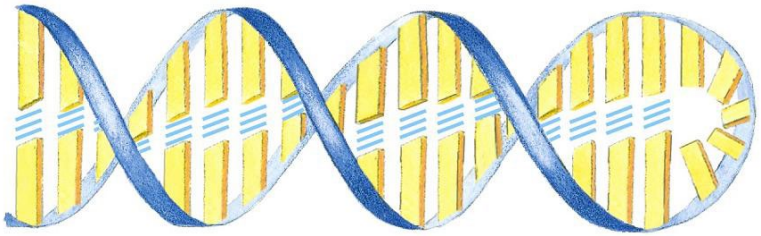
3. RNA

3.1. LA DOBLE HÉLICE TIPO A DEL RNA

El RNA es habitualmente monocatenario



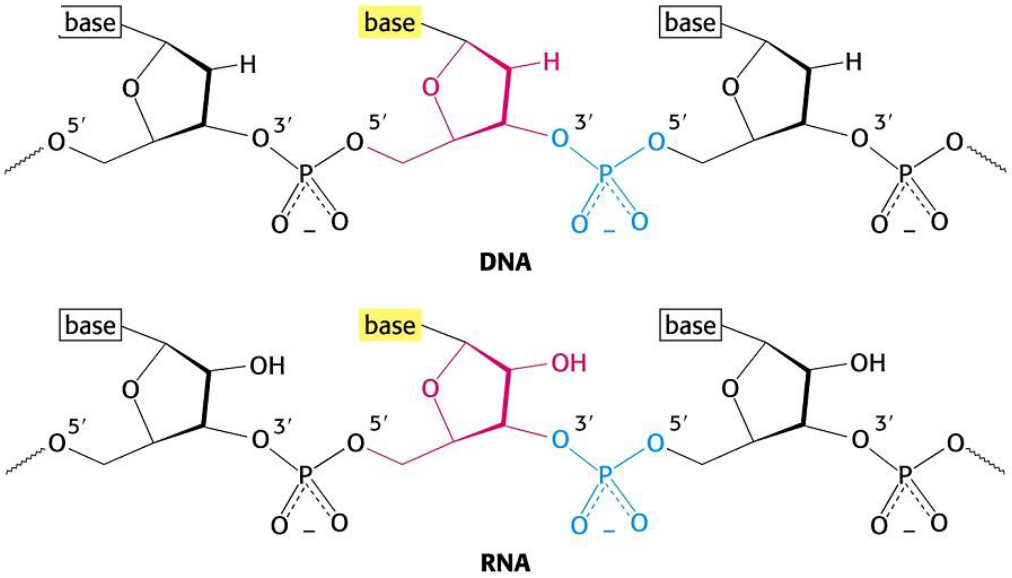
pero los segmentos de secuencia complementarios se aparean formando horquillas.



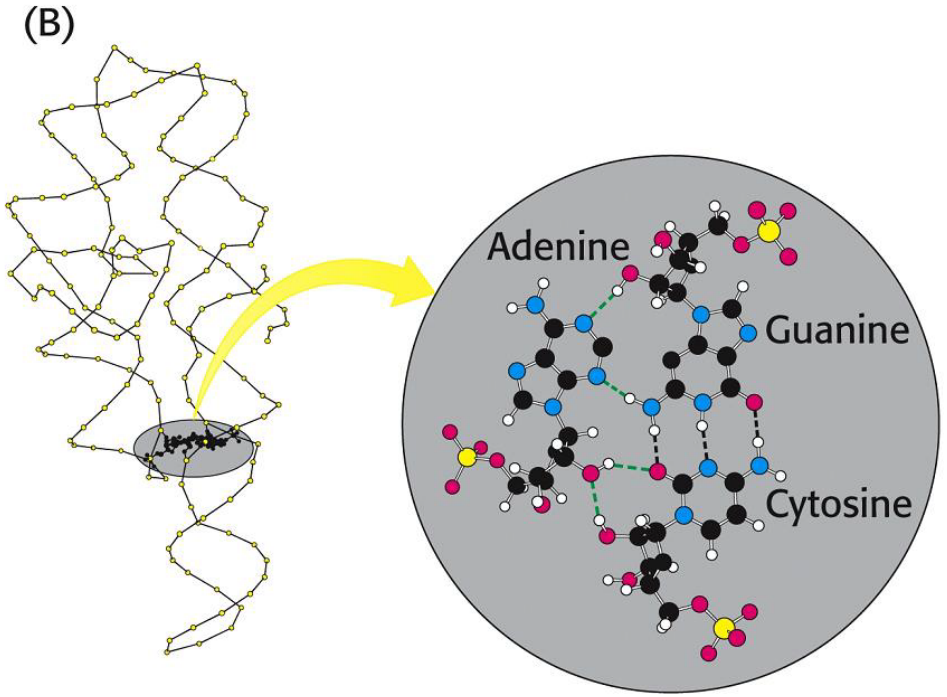
Hairpin double helix
(b)

3.2. EL RNA DE TRANSFERENCIA

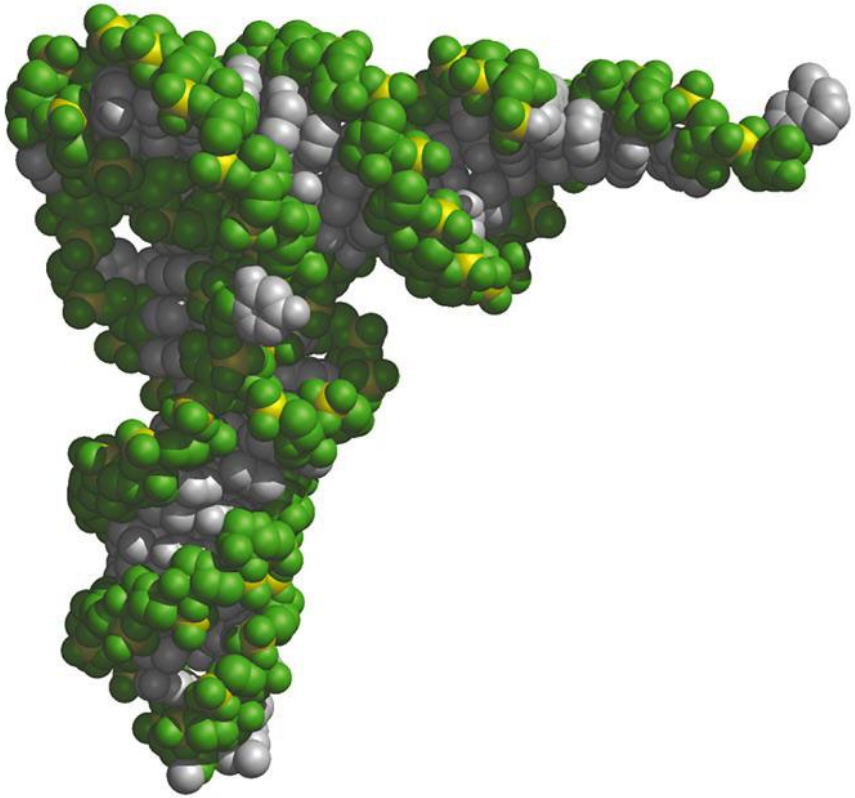
En las hélices de RNA hay bucles con bases sin aparear. Además, la ribosa posee un OH en 2' del que carece el DNA.

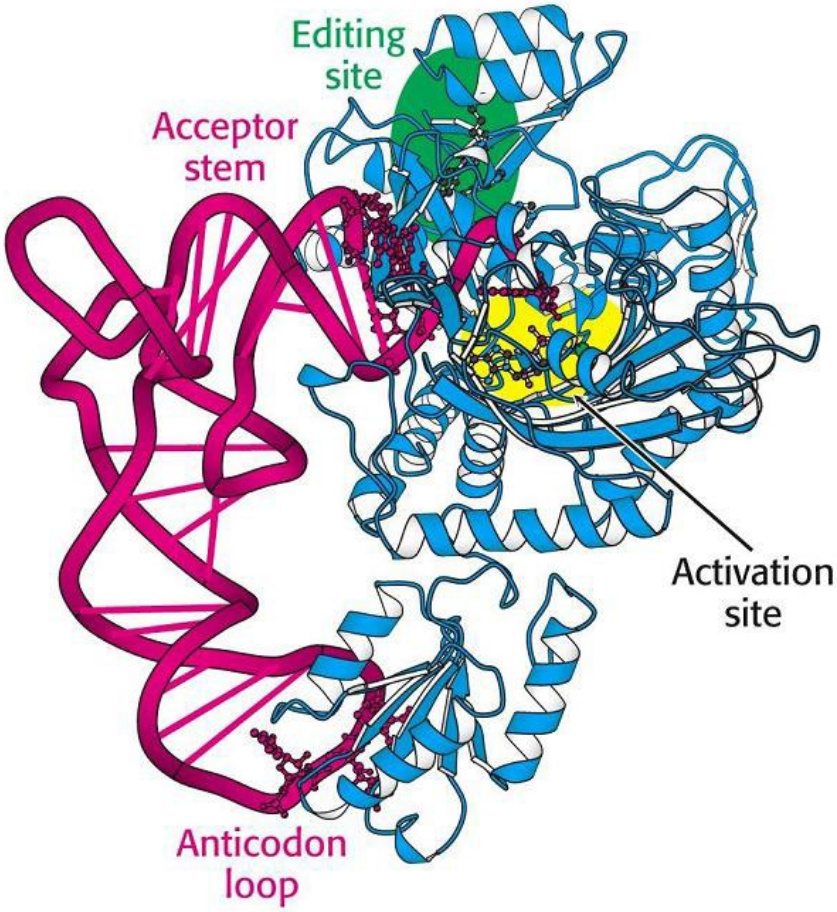


Al poseer grupos formadores de puentes de hidrógeno sin aparear y bases hidrofóbicas sin empaquetar algunas moleculas de RNA se pliegan y adoptan conformaciones compactas en las que esos grupos encuentran pareja.

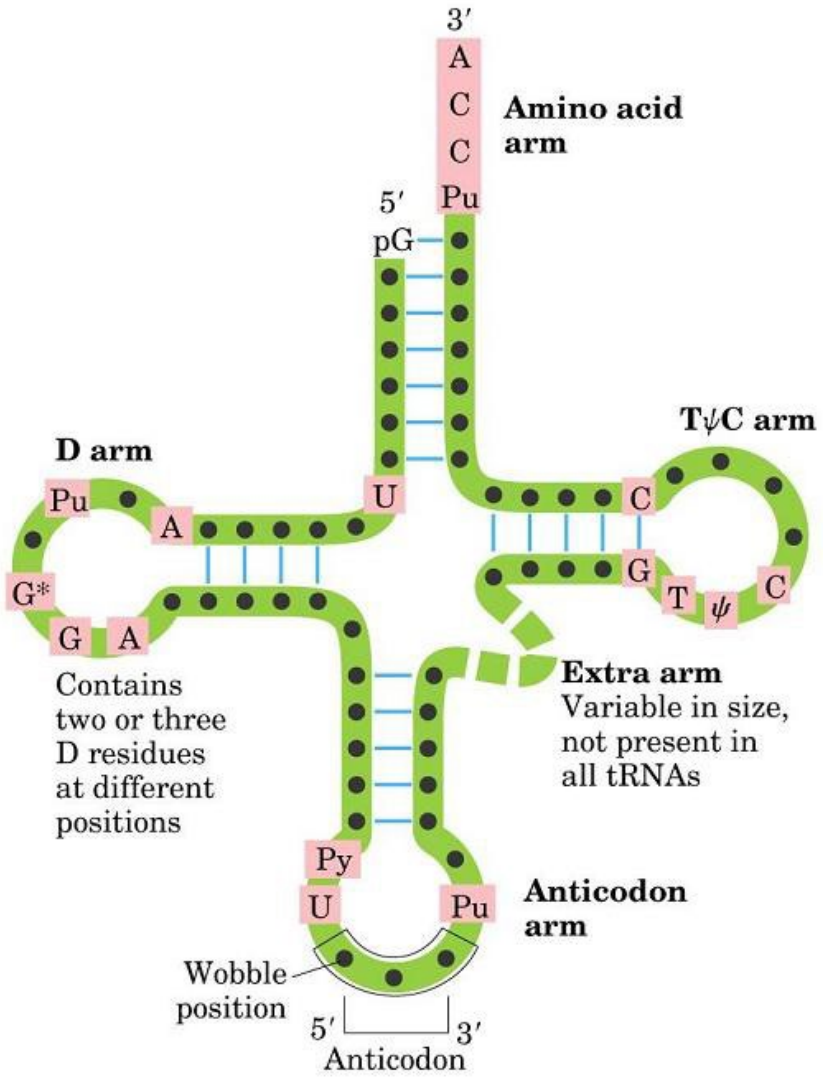


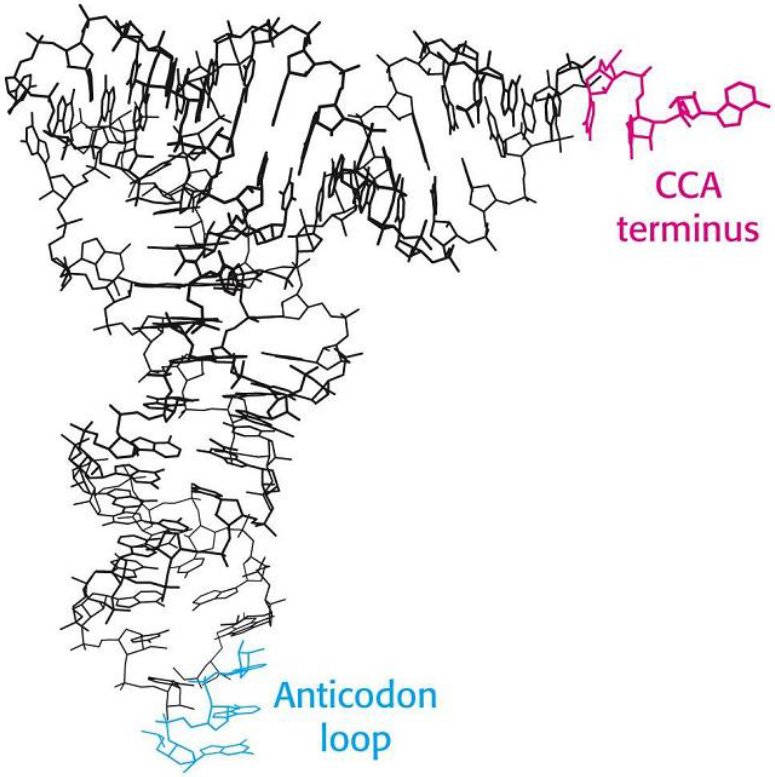
En su conformación plegada, estas moléculas de RNA presentan superficies complejas semejantes a las de las proteínas. Ejemplos de moléculas de RNA compactas son los RNAs de transferencia que deben interactuar específicamente con las aminoacil-t-RNA sintetasas, con el m-RNA y con los ribosomas.





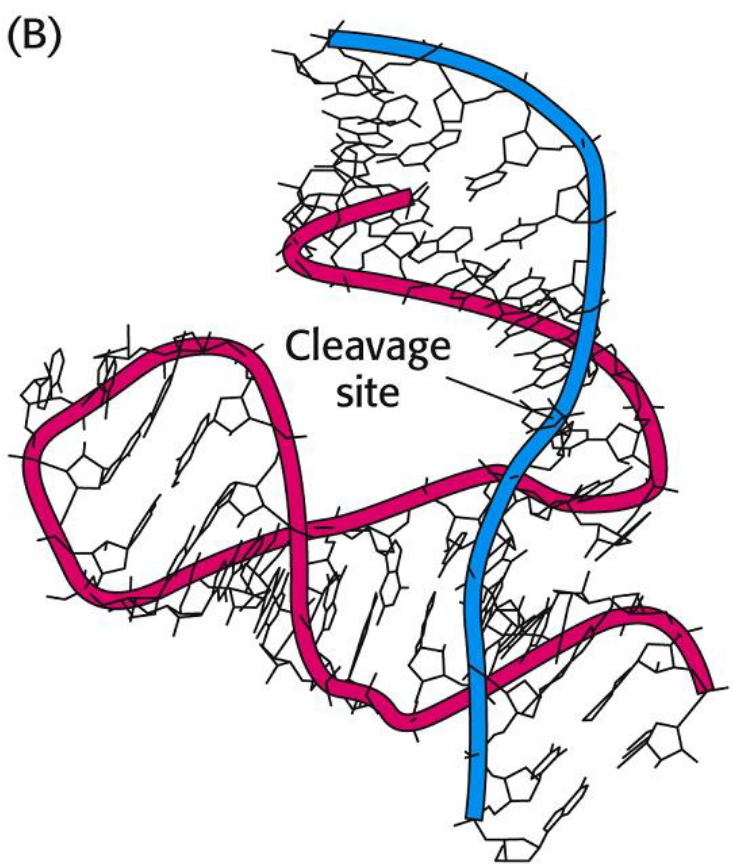
La estructura secundaria del t-RNA se asemeja a una hoja de trebol pero la tridimensional es como una "L" a uno de cuyos extremos se localiza el anticodón y al otro el aminoácido transportado.

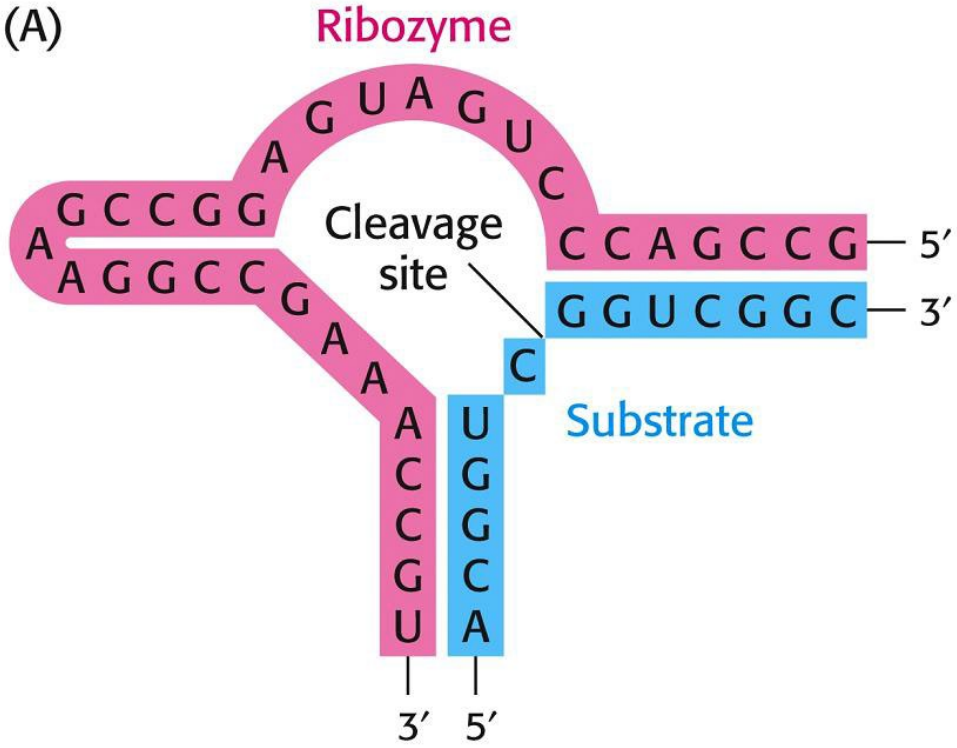




3.3. EL RIBOZIMA 'CABEZA DE MARTILLO'

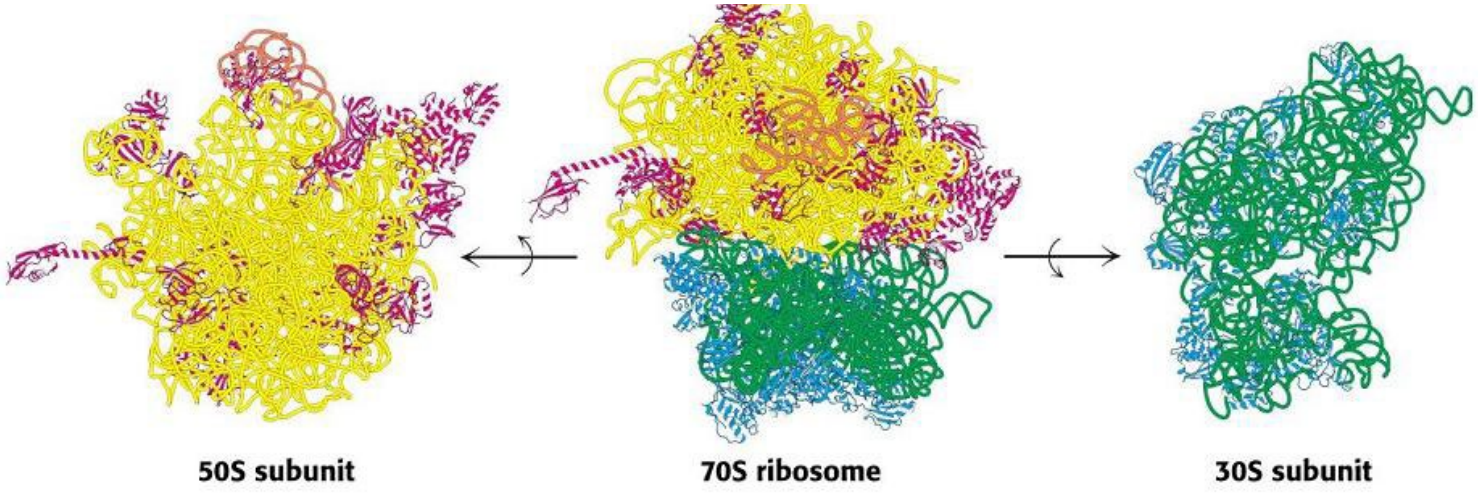
Es una molécula de RNA con actividad catalítica: se autohidroliza. Hay versiones sintéticas capaces de catalizar reacciones de hidrólisis de otras moléculas de RNA.

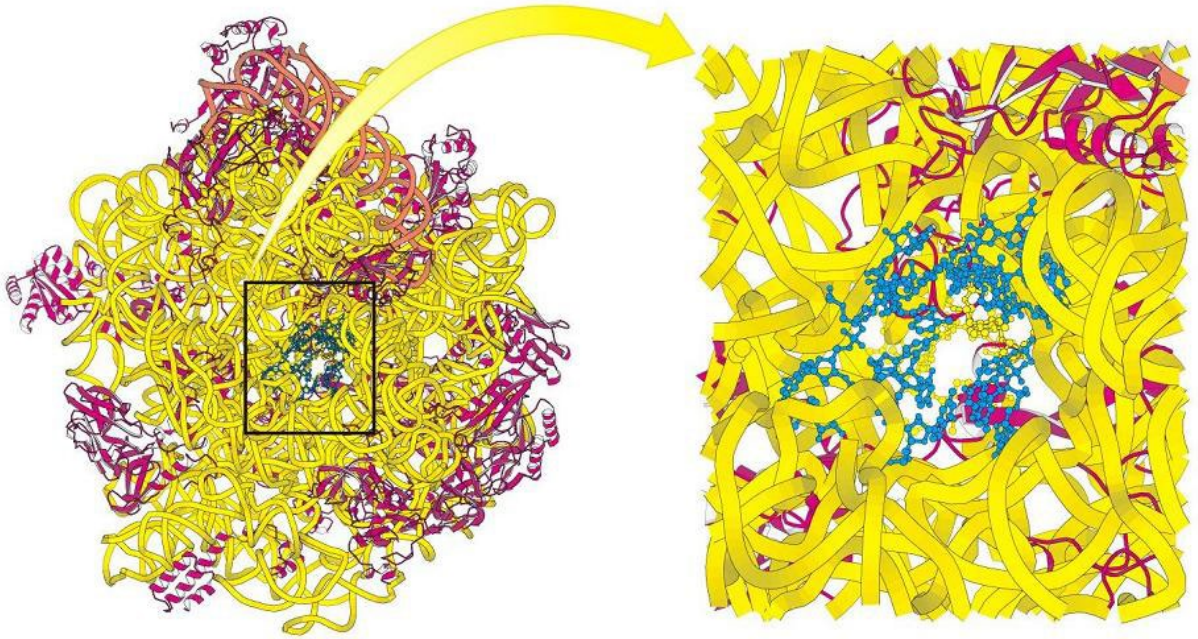




3.4. EL RIBOSOMA

El ribosoma es una ribonucleoproteína. Consta de dos subunidades con compleja estructura secundaria que se pliegan adoptando conformaciones compactas a las que se asocian proteínas.

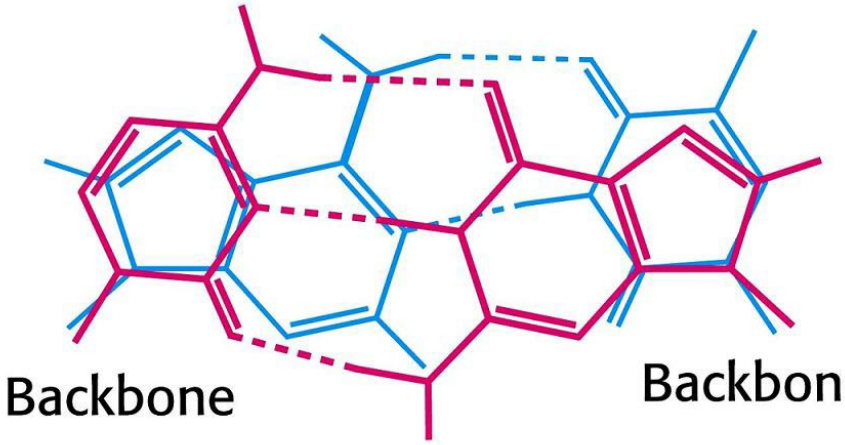


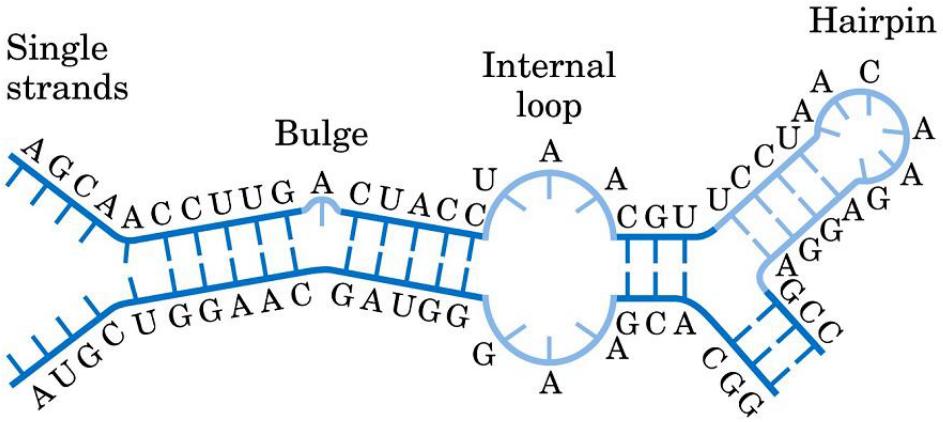


4. ESTABILIDAD Y PLEGAMIENTO DE ÁCIDOS NUCLEICOS

4.1. ESTABILIDAD DE LA DOBLE HÉLICE

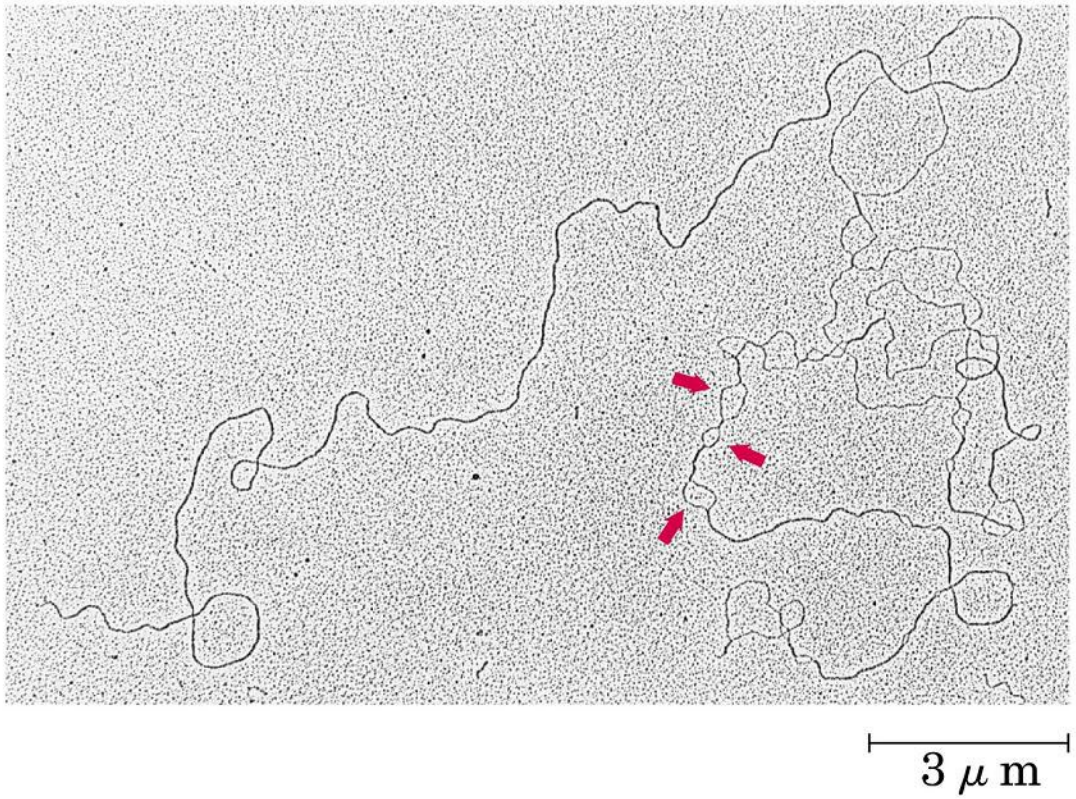
La estabilizan los puentes de hidrógeno (G---C y A--T) y el empaquetamiento entre las bases (efecto hidrofóbico, van der Waals, interacción aromático/aromático descentrada).



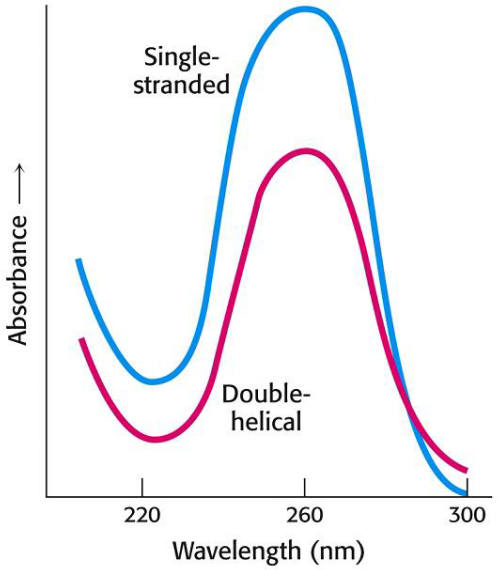


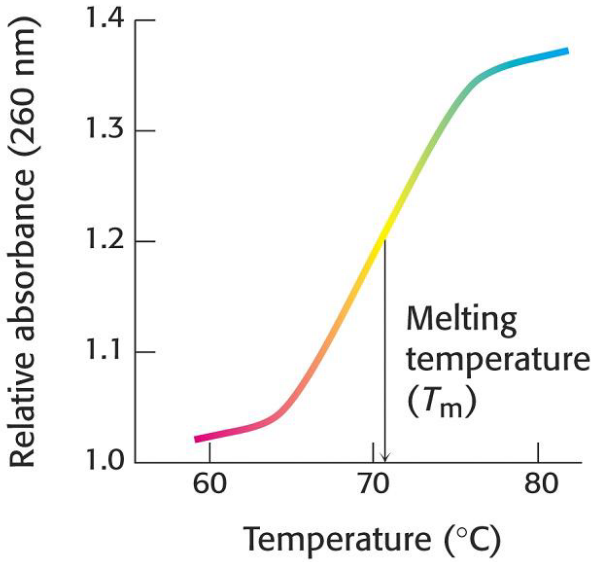
4.2. DESNATURALIZACIÓN TÉRMICA DE LA DOBLE HÉLICE

Al calentar, la doble hélice se desnaturaliza y las dos cadenas se separan

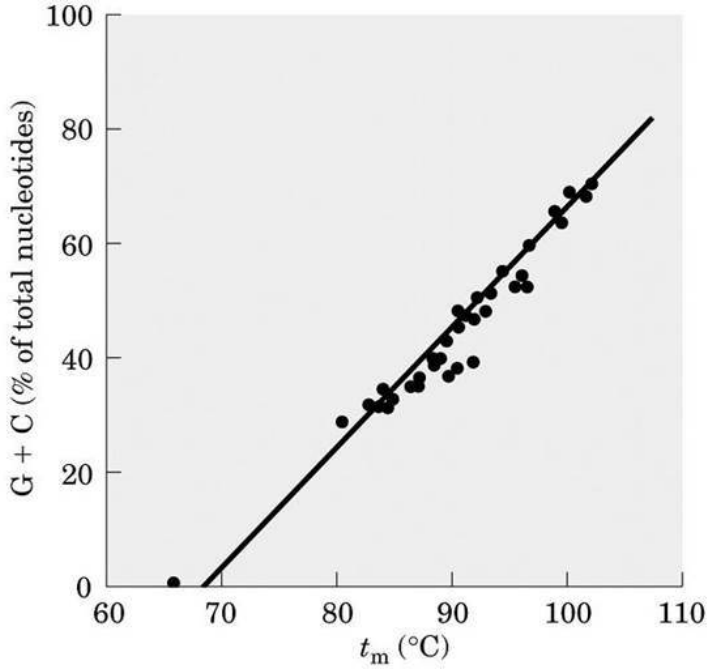


Como las bases tienen mayor coeficiente de extinción en el DNA monocatenario, la absorbancia en el UV aumenta. Midiendo la absorbancia en función de la temperatura se puede calcular la T_m.





La T_m del DNA es función lineal del % de bases GyC.



4.3. ESTABILIDAD DE MOLÉCULAS DE RNA PLEGADAS

En la desnaturalización térmica se observan dos transiciones. A baja temperatura se pierde la estructura terciaria (etapa sensible a la concentración de Mg^{2+}). A temperaturas superiores se pierde la estructura secundaria deshaciéndose las regiones de doble hélice (esta etapa es insensible a la concentración de Mg^{2+}). La contribución de las interacciones "terciarias" a la estabilidad de las moléculas de RNA se puede estudiar por mutagénesis dirigida.

.

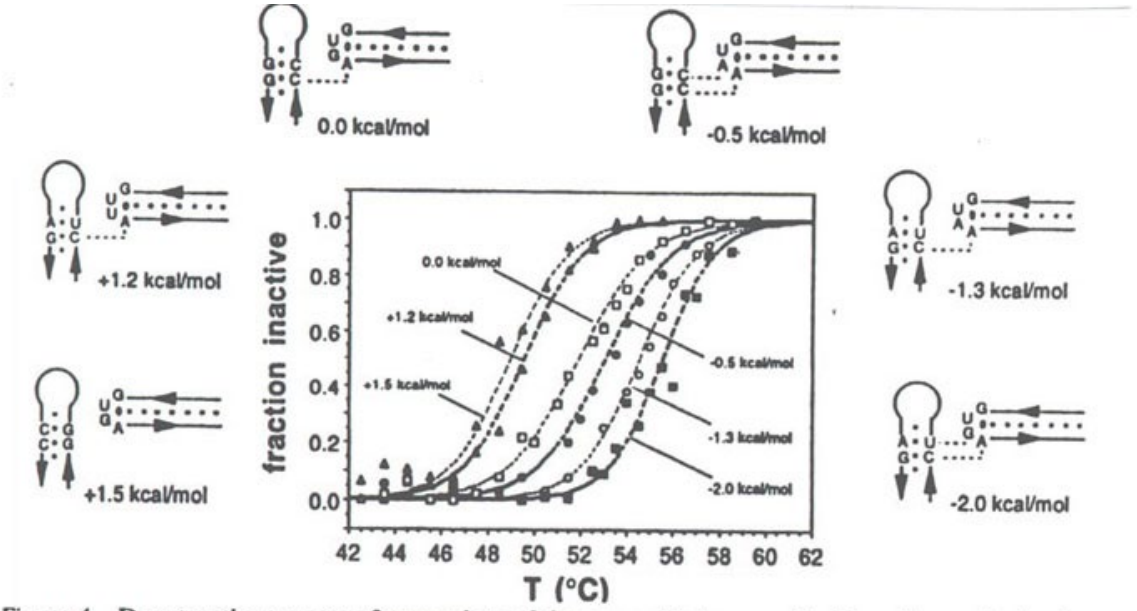
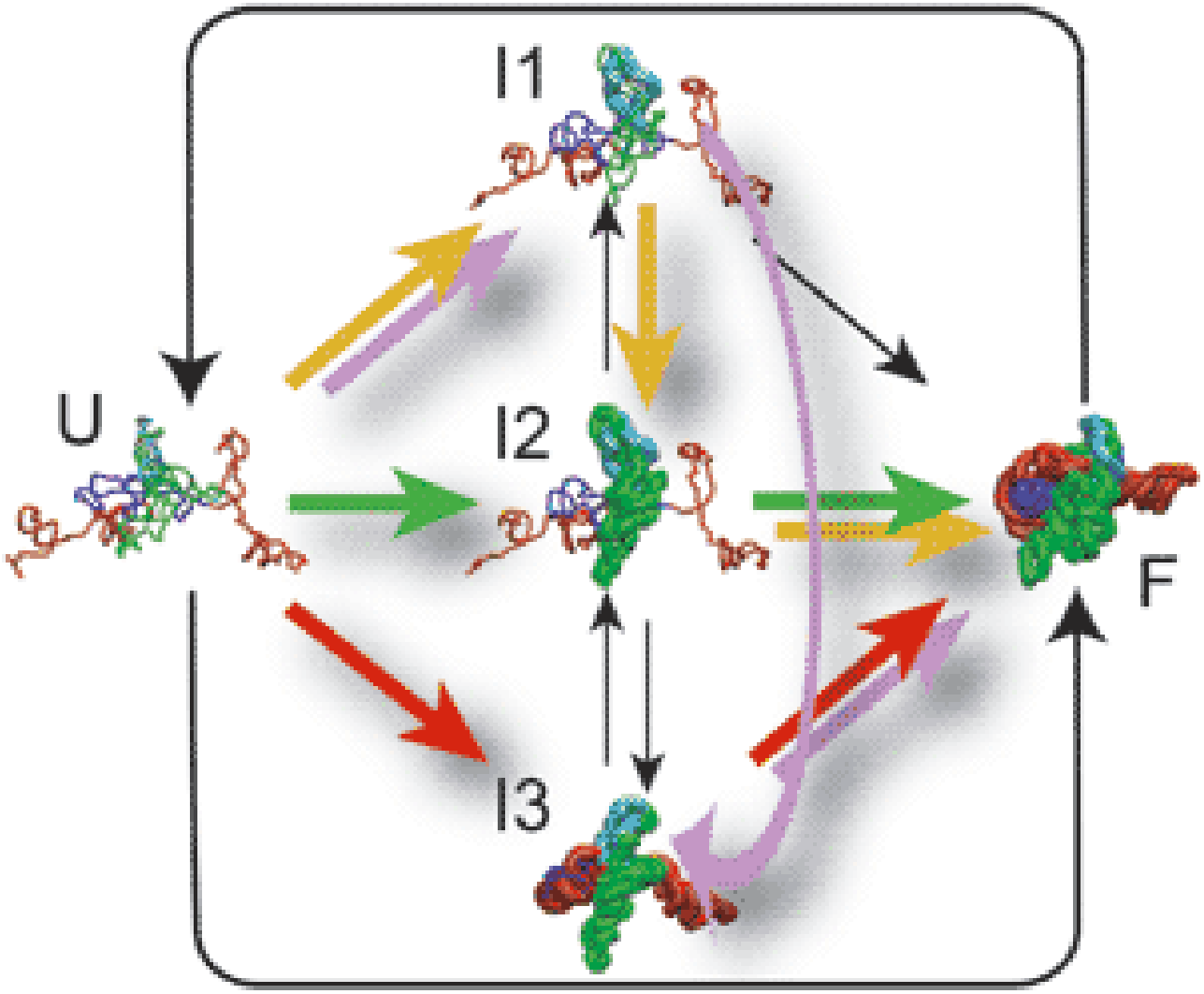


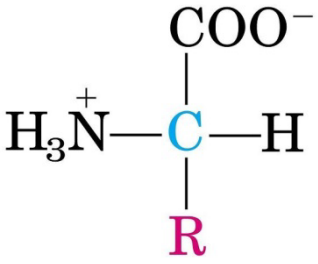
Figure 4 Denaturation curves of transcripts of the group I intron *sunY* with various substitutions in the L9 GNRA tetraloop and P5 helix (see Figure 3). Ordinates are fractions of molecules inactive at each temperature, determined as described by Jaeger et al (45). Broken and dotted curves correspond to theoretical two-state transitions (generated by fitting the data to straight lines in a van't Hoff representation). The corresponding free energies (ΔG's), indicated on the curves and below the corresponding 2D structures, were calculated at 52°C. The contacts between bases in the GNRA loop and the shallow groove of helical base pairs are indicated by dotted lines in the 2-D motifs drawn around the denaturation curves. The motif at the top right is not the lowest energy one because of the presence of a third G-C base pair above the couple of G-C base pairs in the helix, a situation which introduces an ambiguity. The ωt sequence is at bottom right. (Adapted from 45.)

4.4. PLEGAMIENTO DE ÁCIDOS NUCLEICOS

La formación de la doble hélice de DNA sigue un mecanismo de nucleación/propagación. La nucleación es lenta y, tras ella, la propagación es rápida.

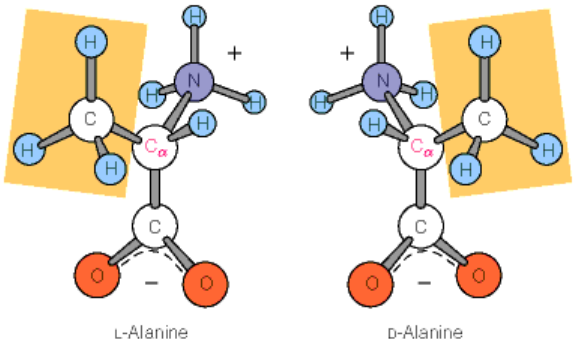


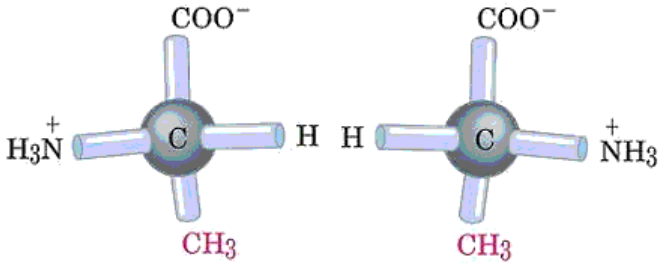
1. AMINOÁCIDOS Y PÉPTIDOS

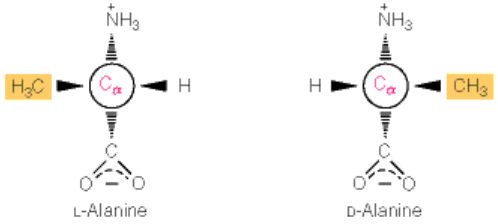


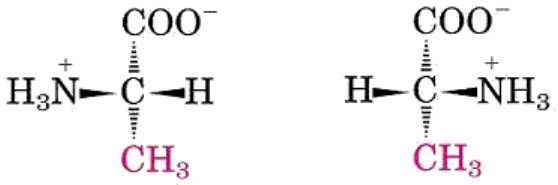
1.1. FÓRMULA GENERAL Y ESTEREOQUÍMICA DE LOS AMINOÁCIDOS

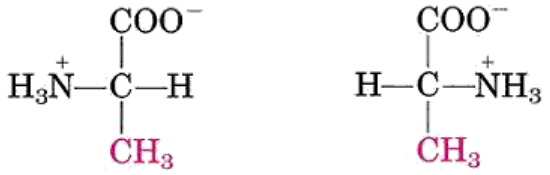
Los aminoácidos de las proteínas son alfa aminoácidos: tanto el grupo ácido (carboxílico) como el grupo amino están unidos al mismo carbono (carbono alfa).











1.2. ESTRUCTURA Y CLASIFICACIÓN DE LOS 20 AMINOÁCIDOS CODIFICADOS GENÉTICAMENTE

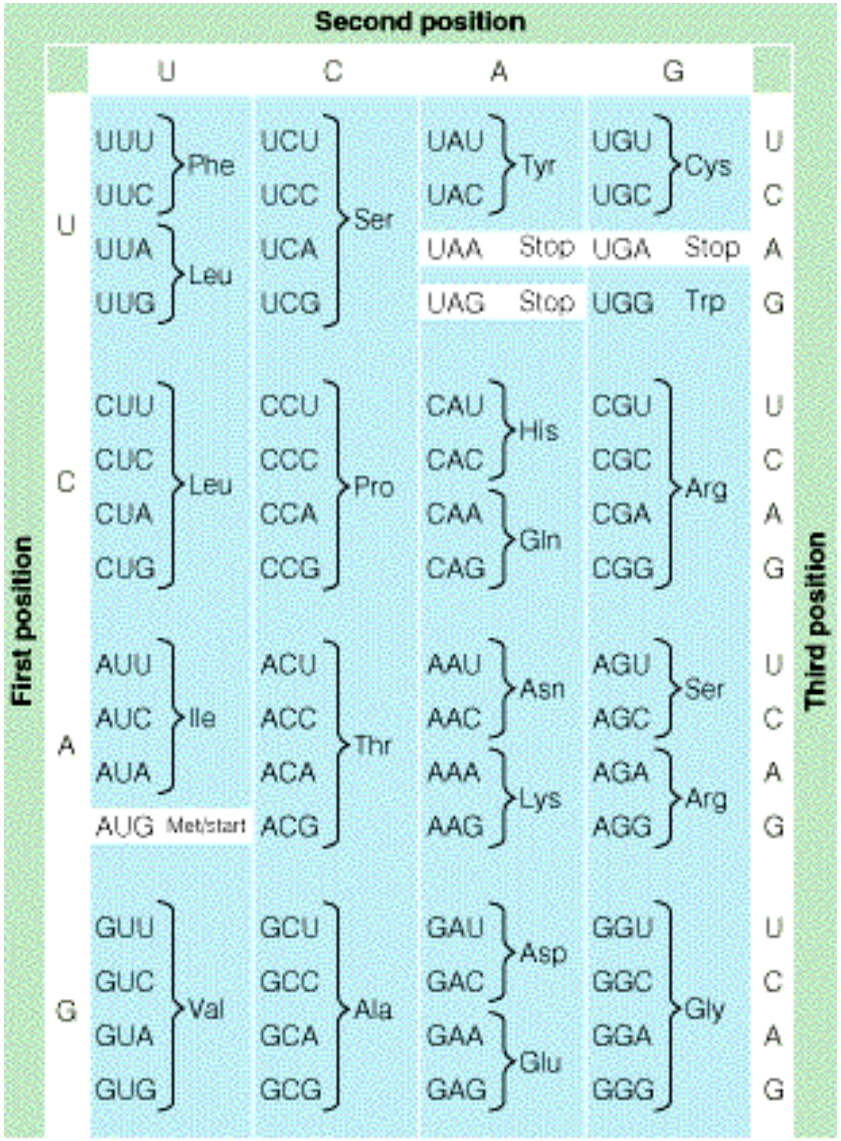
El código genético codifica 20 aminoácidos distintos con los que se sintetizan las proteínas en los ribosomas.

Los aminoácidos se clasifican en grupos según las propiedades de su cadena lateral.
Se suelen agrupar en:

Apolares: distinguiendo entre alifáticos (glicina, alanina, valina, leucina, isoleucina, prolina, metionina) y aromáticos (fenilalanina, tirosina, triptófano)

Polares neutros (serina, treonina, cisteína, asparragina, glutamina)

Cargados: ácidos (aspártico, glutámico) y básicos (lisina, arginina, histidina).



Todas las clasificaciones son arbitrarias y lo importante es conocer y comprender las propiedades de los aminoácidos.

table 5–1

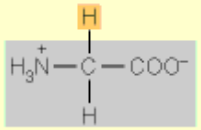
Properties and Conventions Associated with the Standard Amino Acids

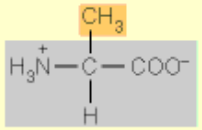
Amino acid	Abbreviated names		M_r	pK_a values: pK_1 (—COOH)	pK_2 ($—NH_3^+$)	pK_R (R group)	pI	Hydropathy index*	Occurrence in proteins (%)
Nonpolar, aliphatic R groups									
Glycine	Gly	G	75	2.34	9.60		5.97	−0.4	7.2
Alanine	Ala	A	89	2.34	9.69		6.01	1.8	7.8
Valine	Val	V	117	2.32	9.62		5.97	4.2	6.6
Leucine	Leu	L	131	2.36	9.60		5.98	3.8	9.1
Isoleucine	Ile	I	131	2.36	9.68		6.02	4.5	5.3
Methionine	Met	M	149	2.28	9.21		5.74	1.9	2.3
Aromatic R groups									
Phenylalanine	Phe	F	165	1.83	9.13		5.48	2.8	3.9
Tyrosine	Tyr	Y	181	2.20	9.11	10.07	5.66	−1.3	3.2
Tryptophan	Trp	W	204	2.38	9.39		5.89	−0.9	1.4
Polar, uncharged R groups									
Serine	Ser	S	105	2.21	9.15		5.68	−0.8	6.8
Proline	Pro	P	115	1.99	10.96		6.48	1.6	5.2
Threonine	Thr	T	119	2.11	9.62		5.87	−0.7	5.9
Cysteine	Cys	C	121	1.96	10.28	8.18	5.07	2.5	1.9
Asparagine	Asn	N	132	2.02	8.80		5.41	−3.5	4.3
Glutamine	Gln	Q	146	2.17	9.13		5.65	−3.5	4.2
Positively charged R groups									
Lysine	Lys	K	146	2.18	8.95	10.53	9.74	−3.9	5.9
Histidine	His	H	155	1.82	9.17	6.00	7.59	−3.2	2.3
Arginine	Arg	R	174	2.17	9.04	12.48	10.76	−4.5	5.1
Negatively charged R groups									
Aspartate	Asp	D	133	1.88	9.60	3.65	2.77	−3.5	5.3
Glutamate	Glu	E	147	2.19	9.67	4.25	3.22	−3.5	6.3

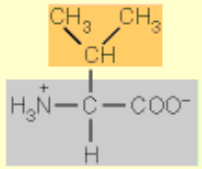
*A scale combining hydrophobicity and hydrophilicity of R groups; it can be used to measure the tendency of an amino acid to seek an aqueous environment (− values) or a hydrophobic environment (+ values). See Chapter 12. From Kyte, J. & Doolittle, R.F. (1982) *J. Mol. Biol.* **157,** 105 – 132.

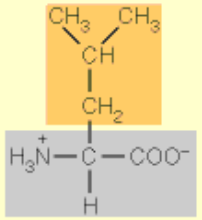
†Average occurrence in over 1150 proteins. From Doolittle, R.F. (1989) Redundancies in protein sequences. In *Prediction of Protein Structure and the Principles of Protein Conformation* (Fasman, G.D., ed) Plenum Press, NY, pp. 599–623.

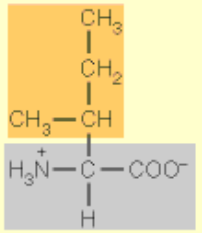
ALIPHATIC AMINO ACIDS


Glycine (Gly) G

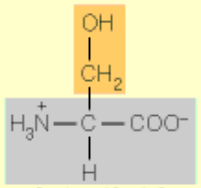

Alanine (Ala) A

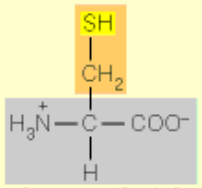

Valine (Val) V

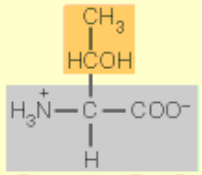

Leucine (Leu) L

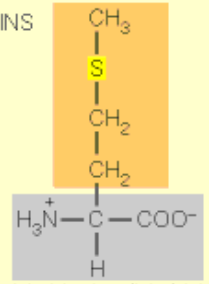

Isoleucine (Ile) I

AMINO ACIDS WITH HYDROXYL- OR SULFUR-CONTAINING SIDE CHAINS

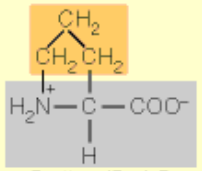

Serine (Ser) S


Cysteine (Cys) C

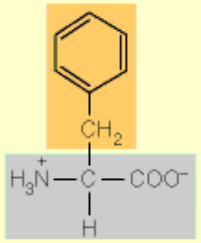

Threonine (Thr) T

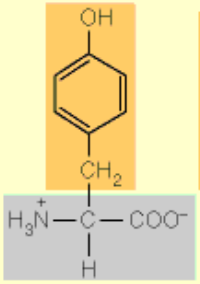

Methionine (Met) M

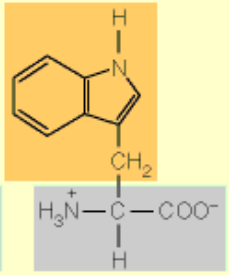
CYCLIC AMINO ACID


Proline (Pro) P

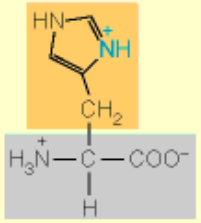
AROMATIC AMINO ACIDS

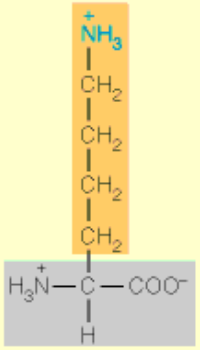

Phenylalanine (Phe) F

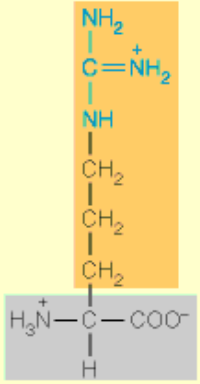

Tyrosine (Tyr) Y


Tryptophan (Trp) W

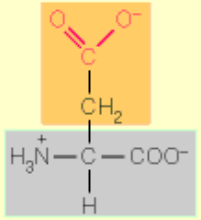
BASIC AMINO ACIDS

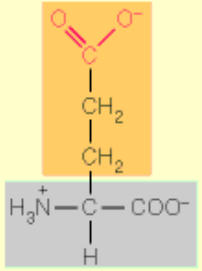

Histidine (His) H

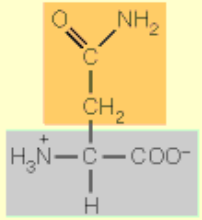

Lysine (Lys) K

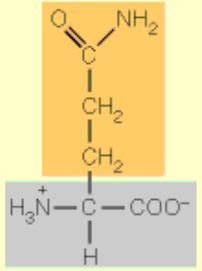

Arginine (Arg) R

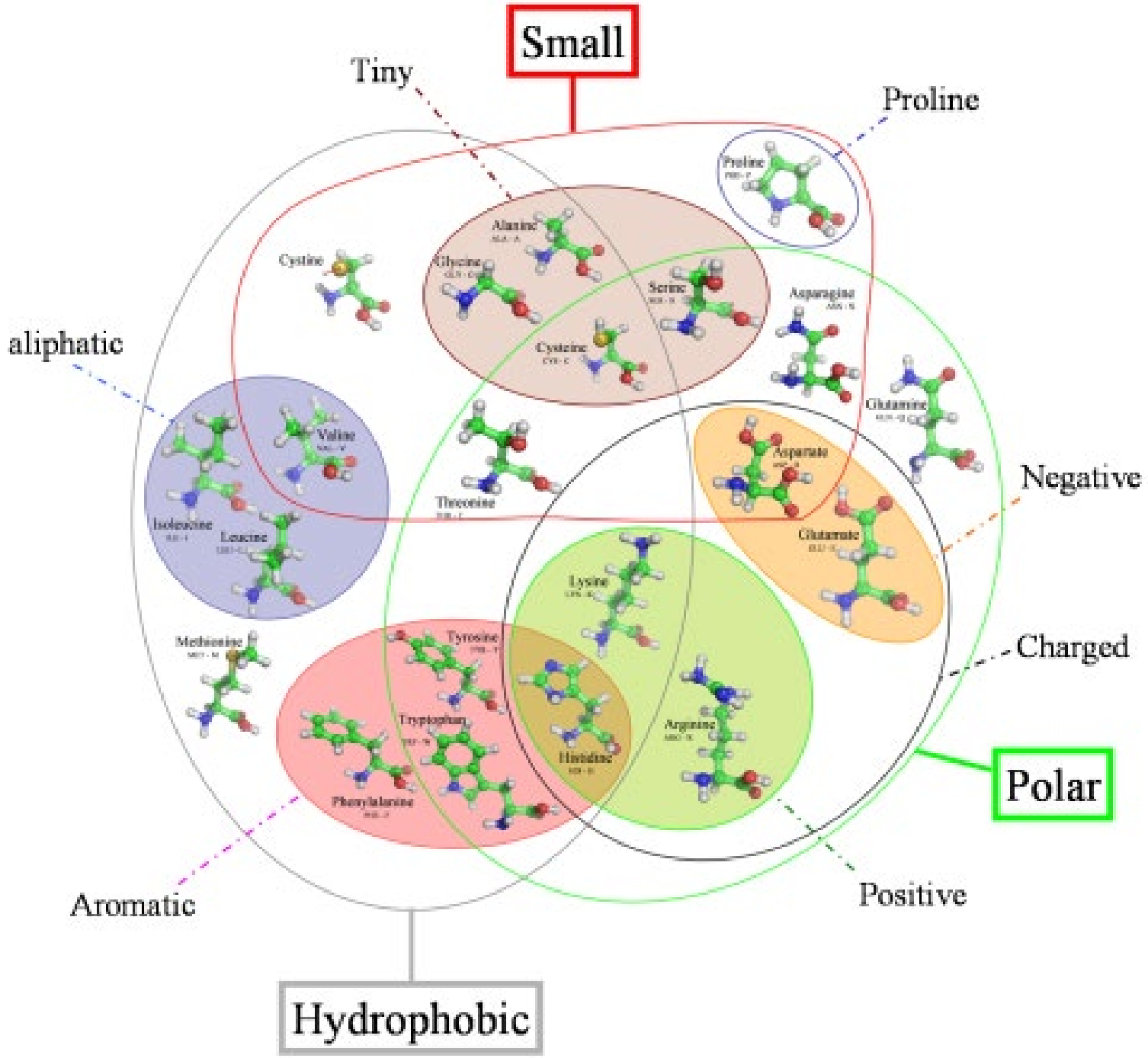
ACIDIC AMINO ACIDS AND THEIR AMIDES


Aspartic acid (Asp) D


Glutamic acid (Glu) E


Asparagine (Asn) N

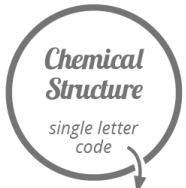

Glutamine (Gln) Q

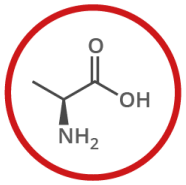

Small
Tiny
Proline
aliphatic
Negative
Charged
Polar
Aromatic
Positive
Hydrophobic
Proline
Alanine
Glycine
Serine
Cystine
Cysteine
Asparagine
Glutamine
Valine
Threonine
Isoleucine
Leucine
Aspartate
Glutamate
Lysine
Methionine
Tyrosine
Tryptophan
Arginine
Histidine
Phenylalanine

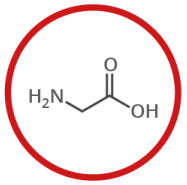
A GUIDE TO THE TWENTY COMMON AMINO ACIDS

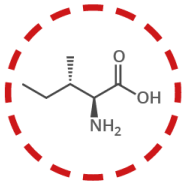
AMINO ACIDS ARE THE BUILDING BLOCKS OF PROTEINS IN LIVING ORGANISMS. THERE ARE OVER 500 AMINO ACIDS FOUND IN NATURE - HOWEVER, THE HUMAN GENETIC CODE ONLY DIRECTLY ENCODES 20. 'ESSENTIAL' AMINO ACIDS MUST BE OBTAINED FROM THE DIET, WHILST NON-ESSENTIAL AMINO ACIDS CAN BE SYNTHESISED IN THE BODY.

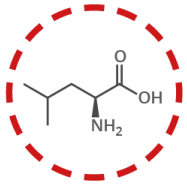
Chart Key: ALIPHATIC · AROMATIC · ACIDIC · BASIC · HYDROXYLIC · SULFUR-CONTAINING · AMIDIC · NON-ESSENTIAL · ESSENTIAL

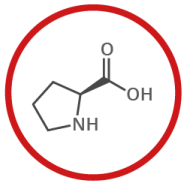


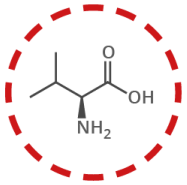

ALANINE A
ala

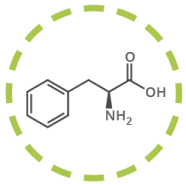

GLYCINE G
gly

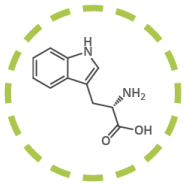

ISOLEUCINE I
ile

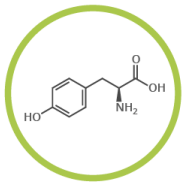

LEUCINE L
leu

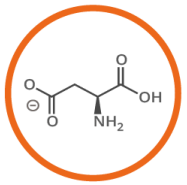

PROLINE P
pro

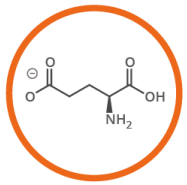

VALINE V
val

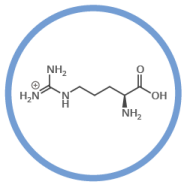

PHENYLALANINE F
phe

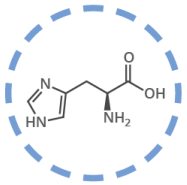

TRYPTOPHAN W
trp

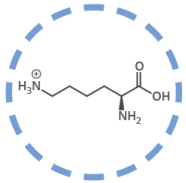

TYROSINE Y
tyr

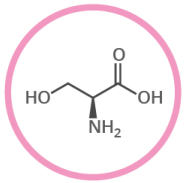

ASPARTIC ACID D
asp

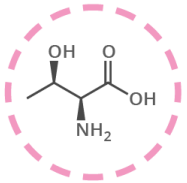

GLUTAMIC ACID E
glu

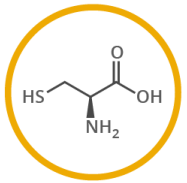

ARGININE R
arg

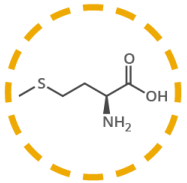

HISTIDINE H
his

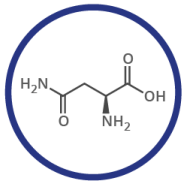

LYSINE K
lys

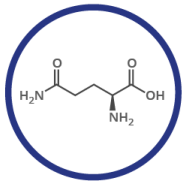

SERINE S
ser


THREONINE T
thr


CYSTEINE C
cys


METHIONINE M
met

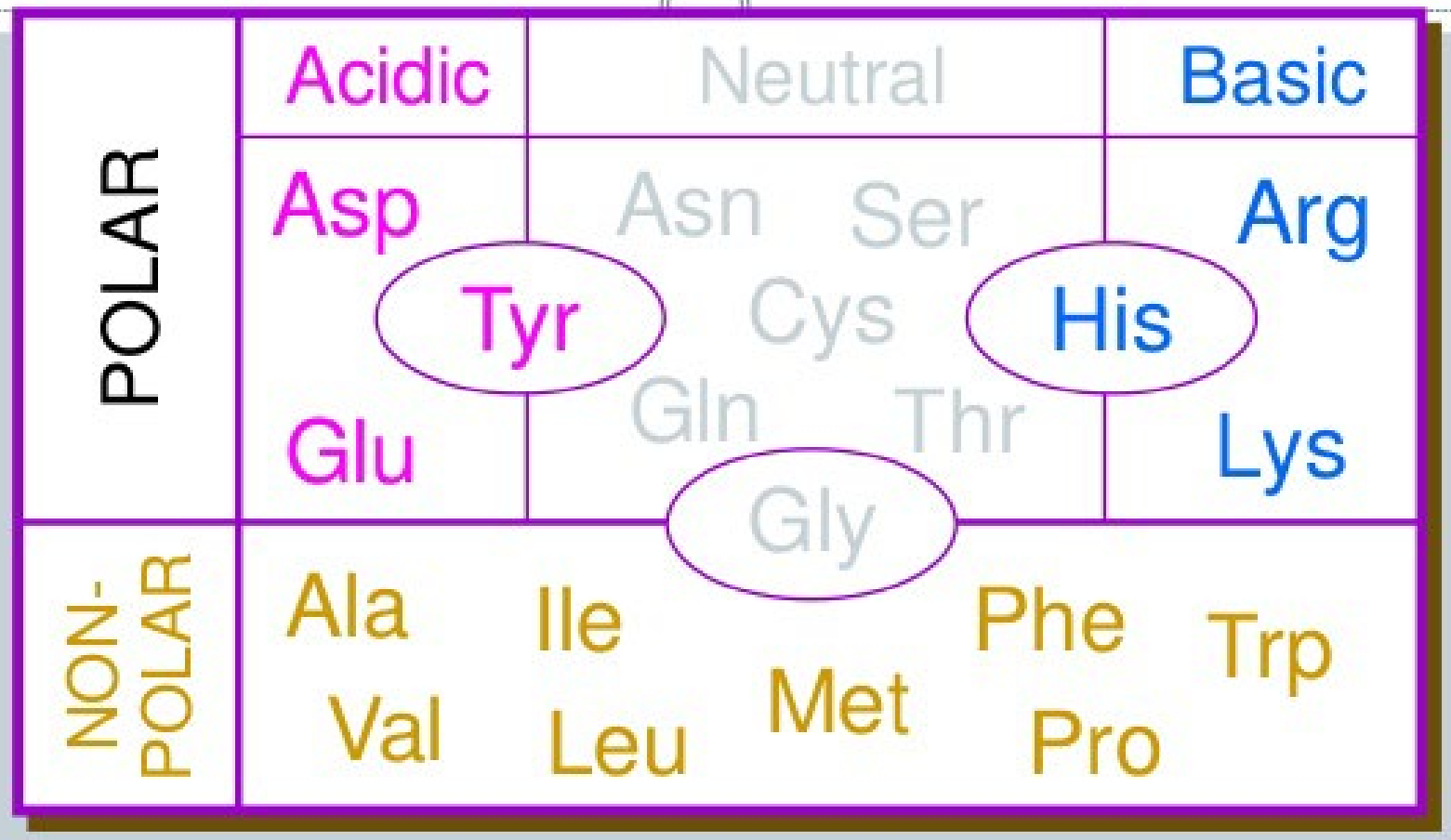

ASPARAGINE N
asn


GLUTAMINE Q
gln

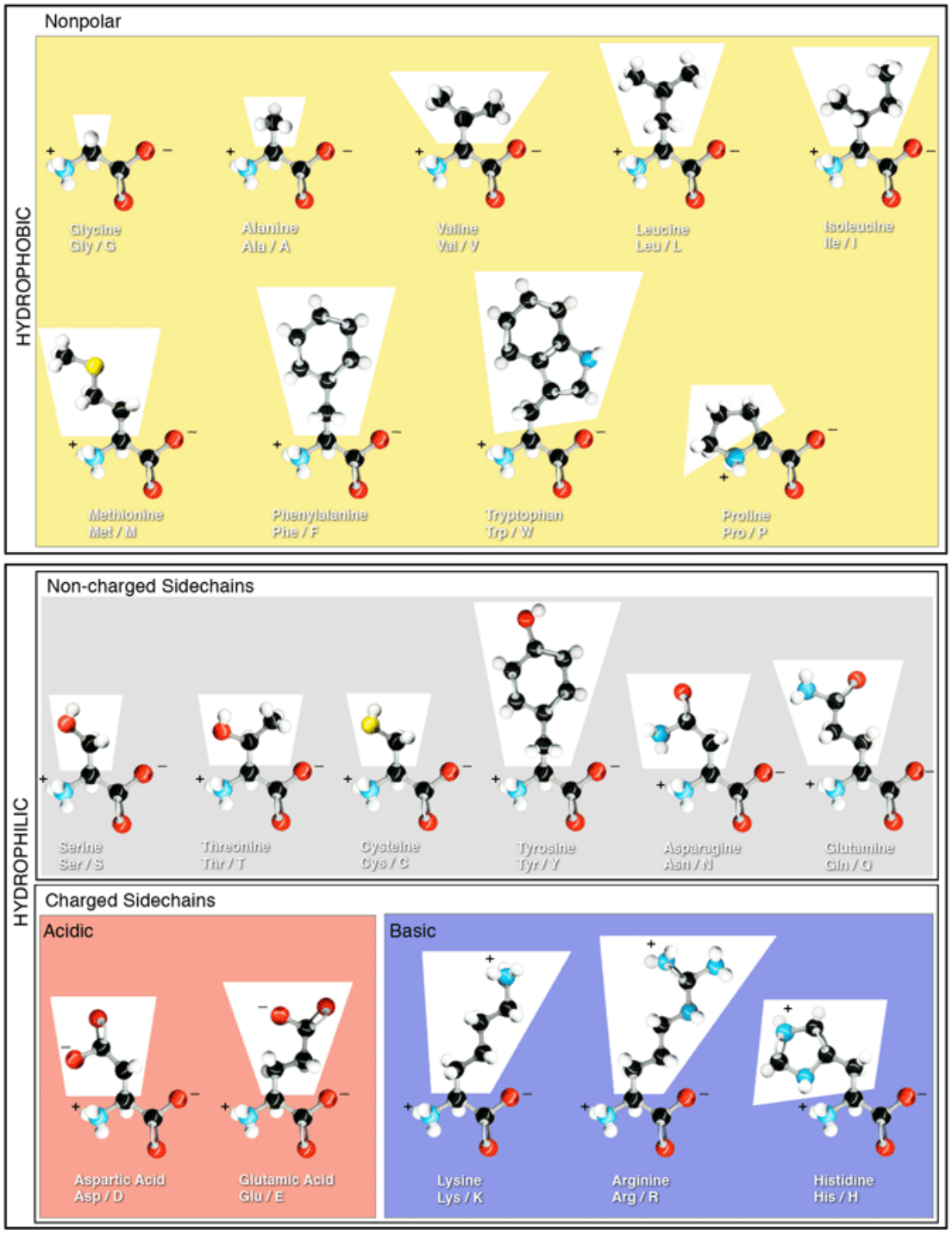
Note: This chart only shows those amino acids for which the human genetic code directly codes for. Selenocysteine is often referred to as the 21st amino acid, but is encoded in a special manner. In some cases, distinguishing between asparagine/aspartic acid and glutamine/glutamic acid is difficult. In these cases, the codes asx (B) and glx (Z) are respectively used.

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Classification of Amino Acids by Polarity

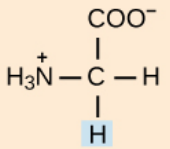


Polar or non-polar, it is the bases of the amino acid properties.

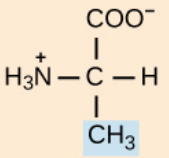

HYDROPHOBIC
Nonpolar
Glycine
Gly / G
Alanine
Ala / A
Valine
Val / V
Leucine
Leu / L
Isoleucine
Ile / I
Methionine
Met / M
Phenylalanine
Phe / F
Tryptophan
Trp / W
Proline
Pro / P
HYDROPHILIC
Non-charged Sidechains
Serine
Ser / S
Threonine
Thr / T
Cysteine
Cys / C
Tyrosine
Tyr / Y
Asparagine
Asn / N
Glutamine
Gln / Q
Charged Sidechains
Acidic
Aspartic Acid
Asp / D
Glutamic Acid
Glu / E
Basic
Lysine
Lys / K
Arginine
Arg / R
Histidine
His / H

PROTEIN AMINO ACIDS

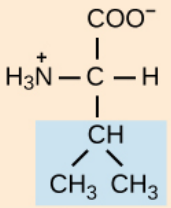
Nonpolar, aliphatic R groups



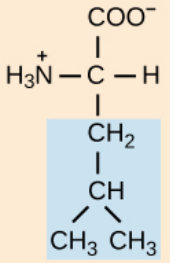
Glycine



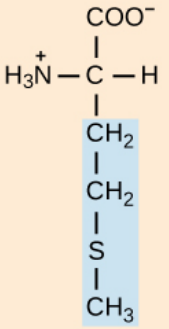
Alanine



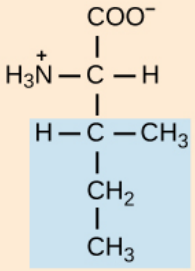
Valine



Leucine

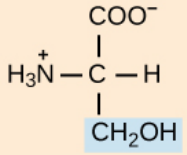


Methionine

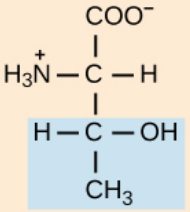


Isoleucine

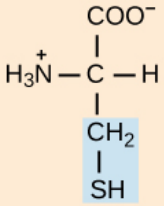
Polar, uncharged R groups



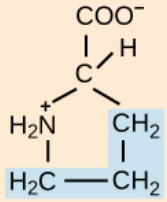
Serine



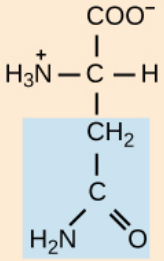
Threonine



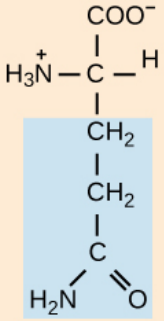
Cysteine



Proline



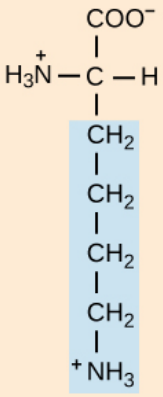
Asparagine



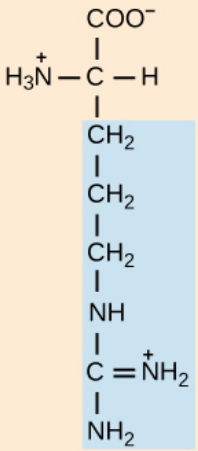
Glutamine

PROTEIN AMINO ACIDS

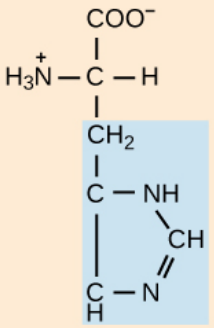
Positively charged R groups



Lysine

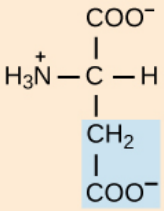


Arginine

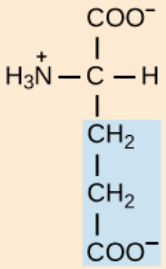


Histidine

Negatively charged R groups

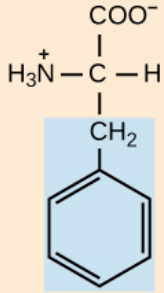


Aspartate

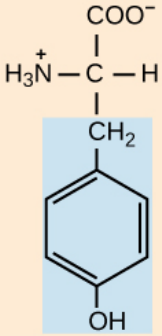


Glutamate

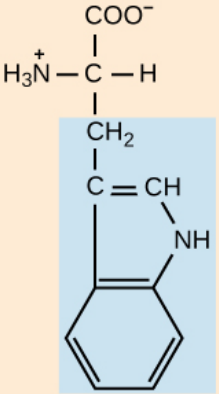
Nonpolar, aromatic R groups



Phenylalanine



Tyrosine



Tryptophan

Twenty-One Amino Acids

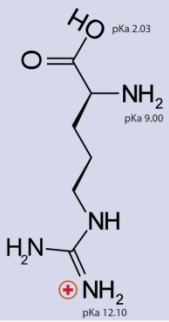
Positive Negative

• Side chain charge at physiological pH 7.4

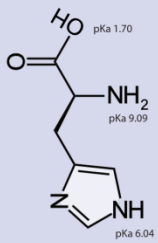
A. *Amino Acids with Electrically Charged Side Chains*

Positive

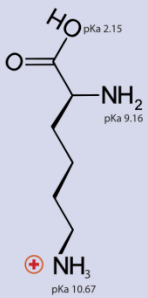
Arginine (Arg) R



Histidine (His) H

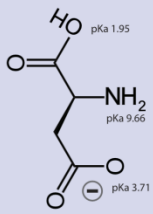


Lysine (Lys) K

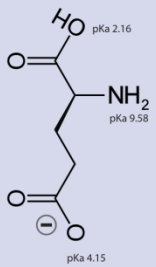


Negative

Aspartic Acid (Asp) D

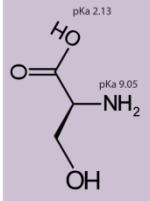


Glutamic Acid (Glu) E

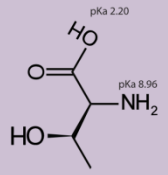


B. *Amino Acids with Polar Uncharged Side Chains*

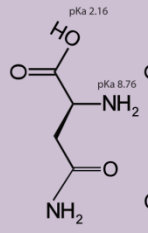
Serine (Ser) S



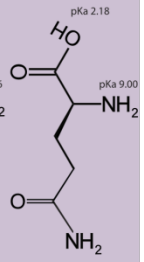
Threonine (Thr) T



Asparagine (Asn) N

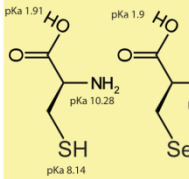


Glutamine (Gln) Q

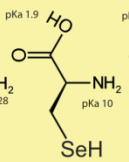


C. *Special Cases*

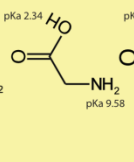
Cysteine (Cys) C



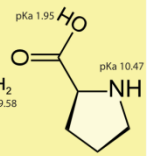
Selenocysteine (Sec) U



Glycine (Gly) G

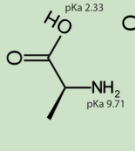


Proline (Pro) P

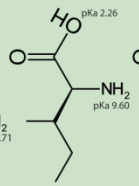


D. *Amino Acids with Hydrophobic Side Chain*

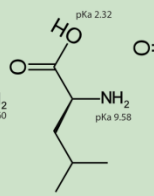
Alanine (Ala) A



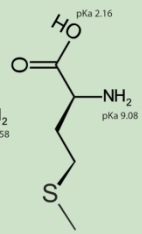
Isoleucine (Ile) I



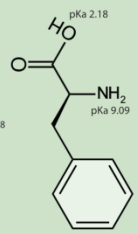
Leucine (Leu) L



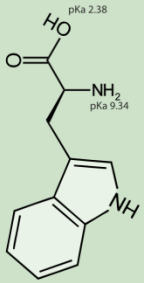
Methionine (Met) M



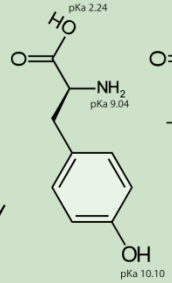
Phenylalanine (Phe) F



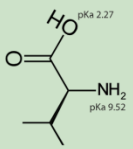
Tryptophan (Trp) W



Tyrosine (Tyr) Y



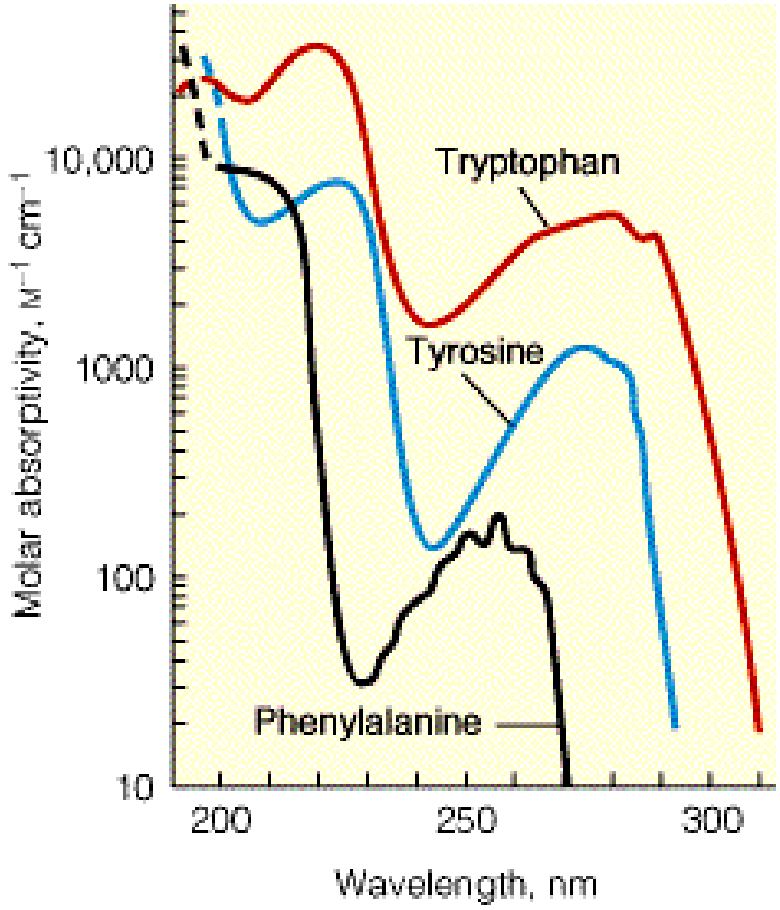
Valine (Val) V

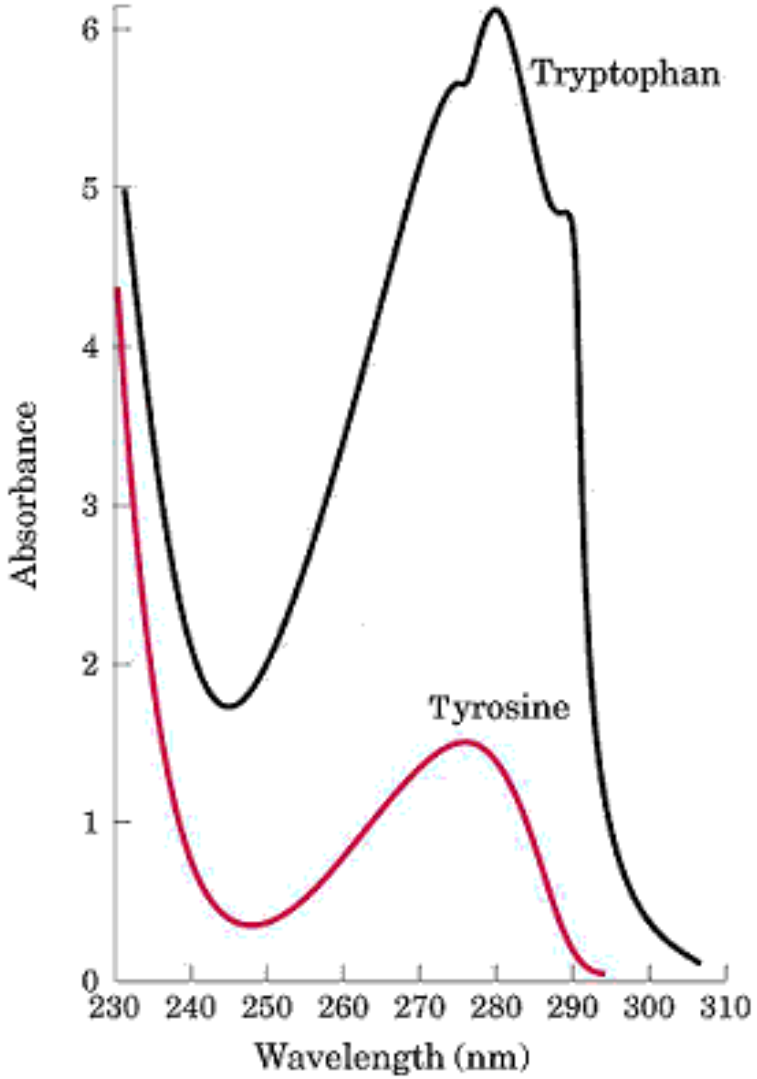


pKa Data: CRC Handbook of Chemistry, v.2010

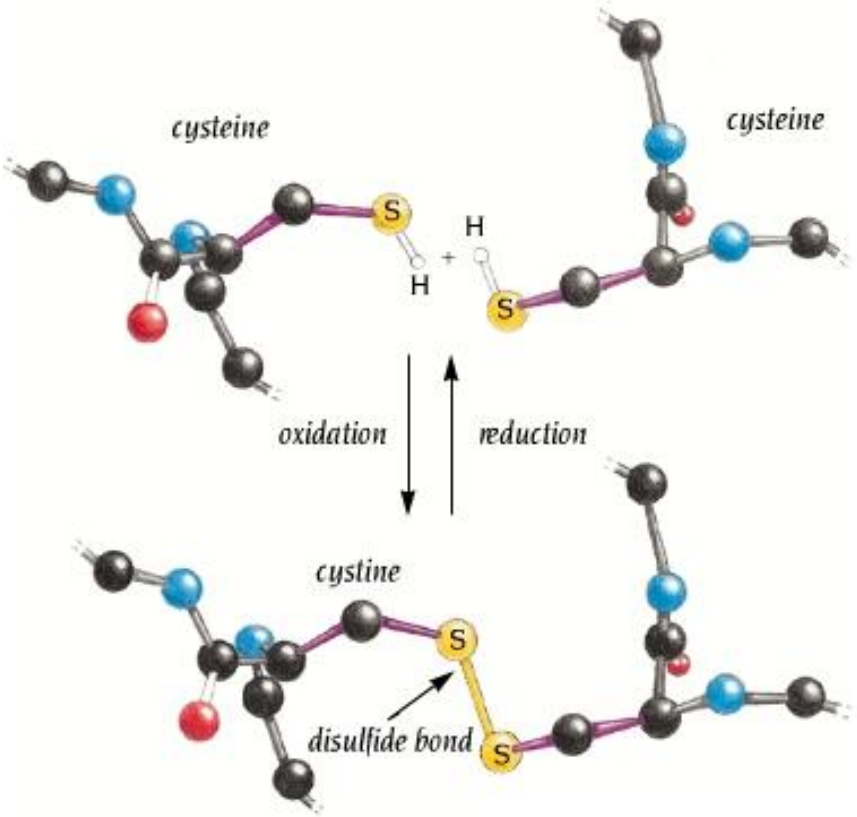
Dan Cojocari, Department of Medical Biophysics, **University of Toronto**, 2010

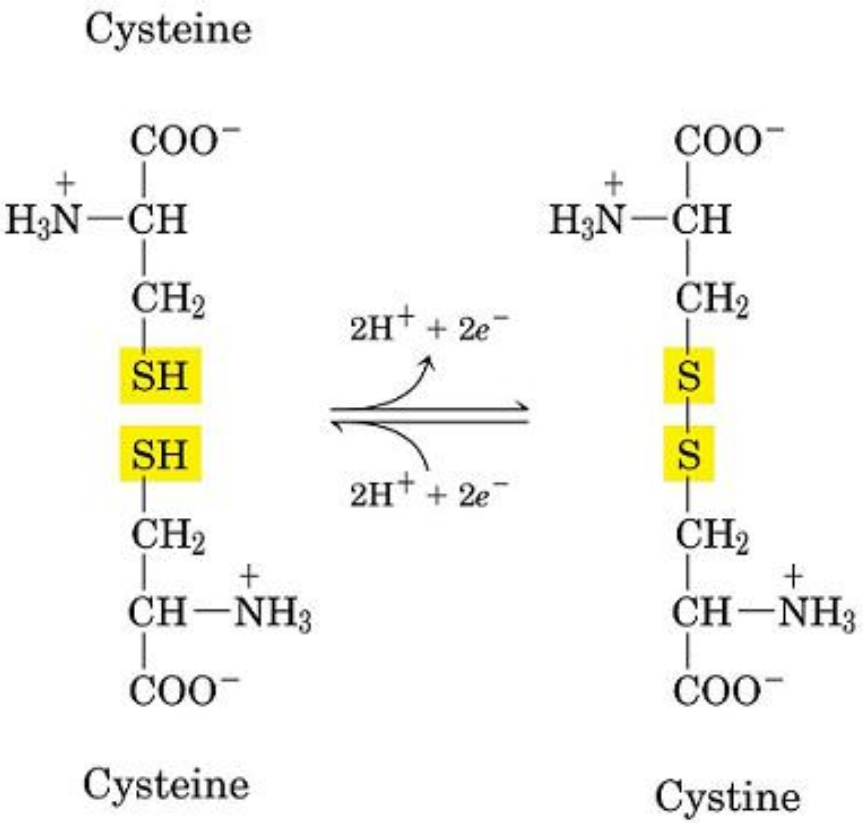
Los aminoácidos aromáticos son los responsables de la absorbancia a 280 nm típica de las proteínas.





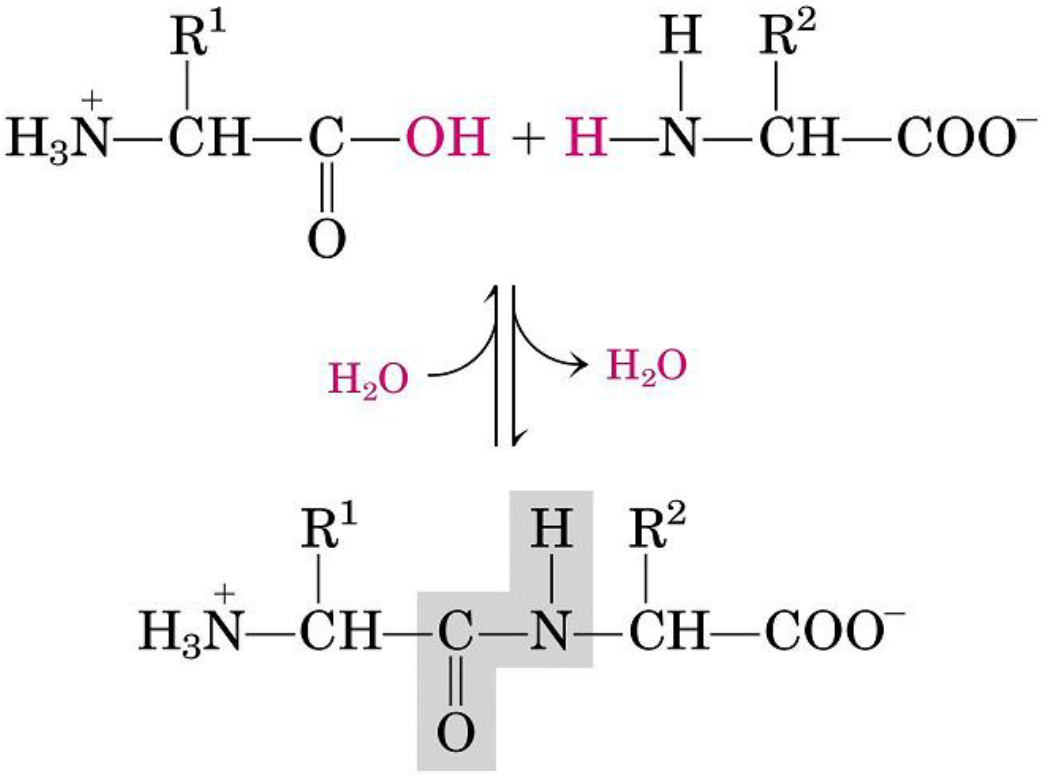
Las cisteínas se pueden oxidar formando puentes disulfuro entre distintas partes de una proteína.

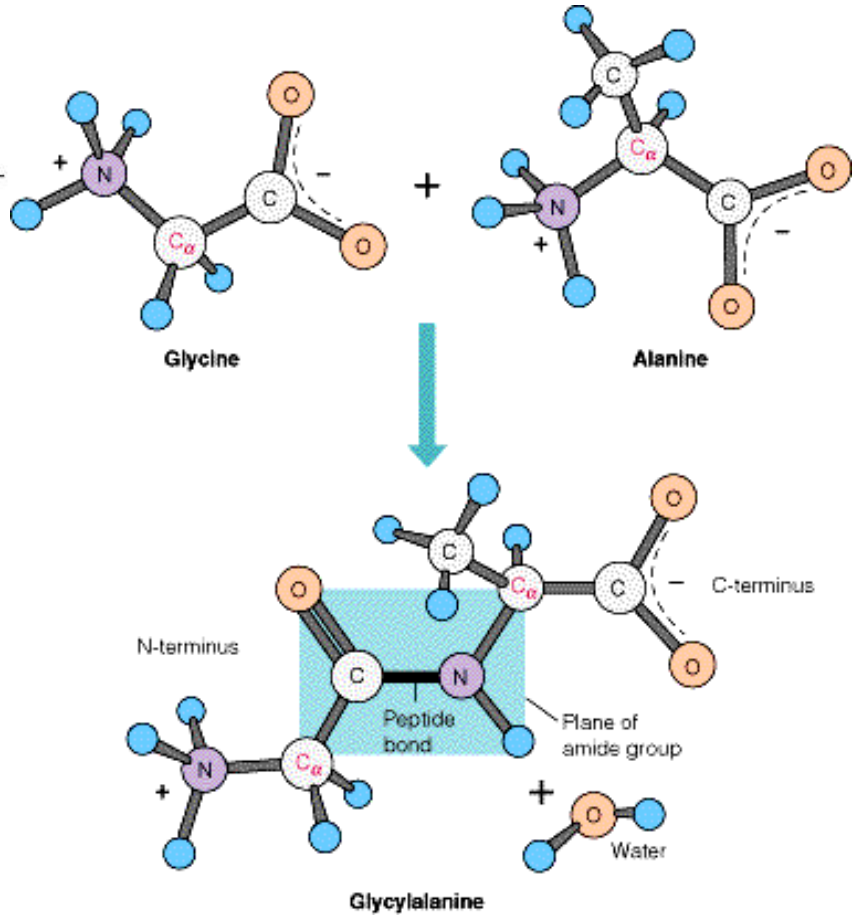




1.3. PEPTIDOS

Los péptidos son el resultado de la unión covalente de aminoácidos mediante un enlace amida (denominado enlace peptídico).



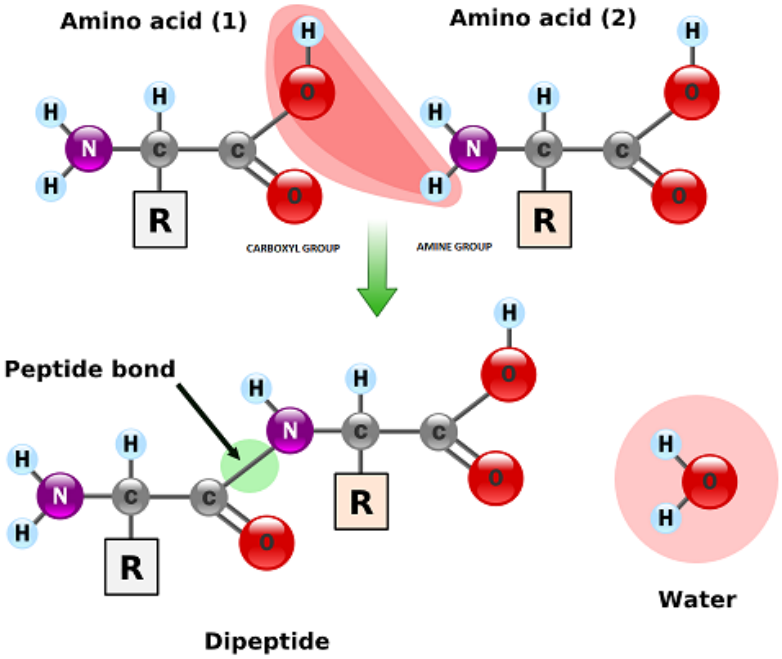


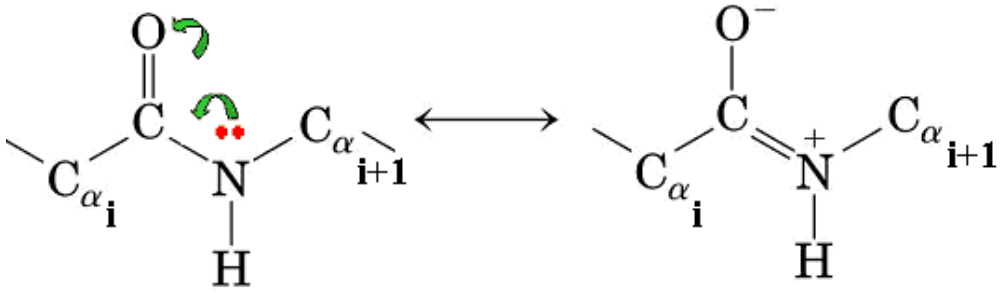
2. ENLACE PEPTÍDICO Y ESTRUCTURA SECUNDARIA

2.1. EL ENLACE PEPTÍDICO

El enlace peptídico es el resultado de la condensación del grupo carboxilo de un aminoácido y el grupo amino de otro, con liberación de una molécula de agua.

Es un enlace de tipo amida con resonancia entre dos formas y de ahí surge su carácter de enlace doble parcial.

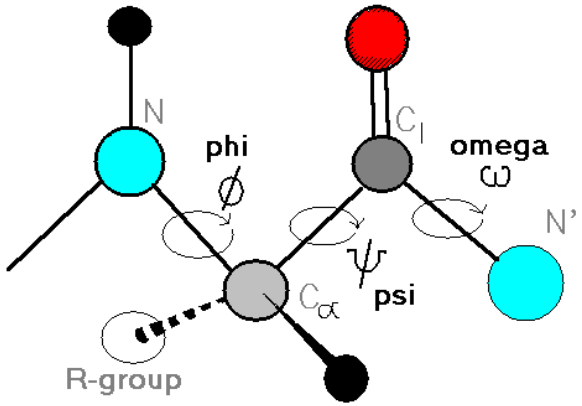




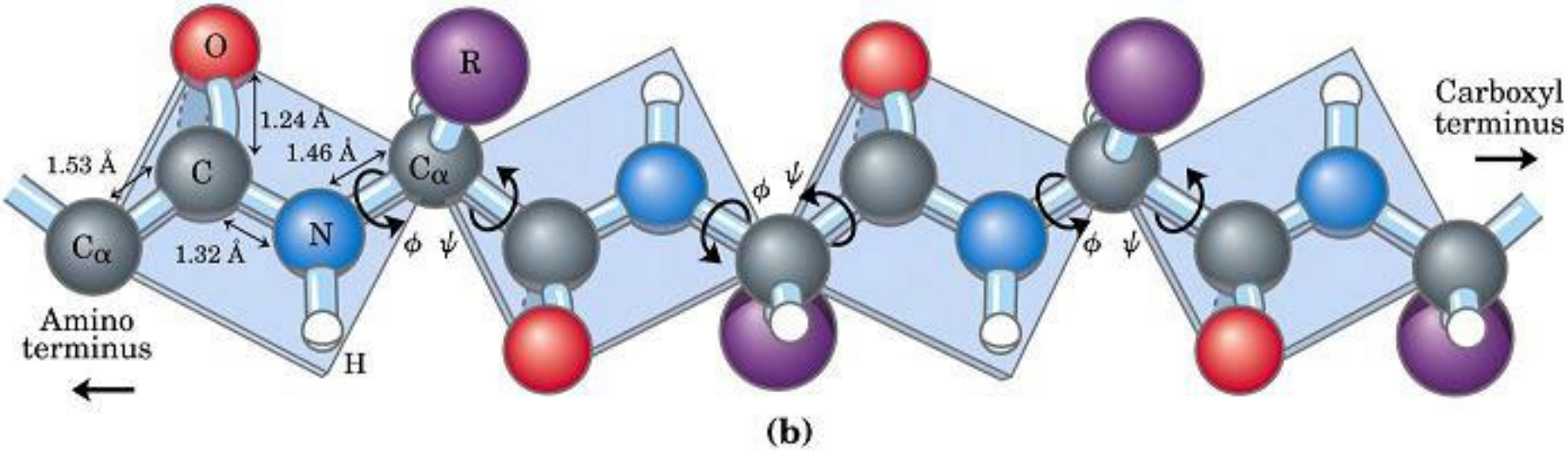
C-N	1.49 Å
C=N	1.27 Å
pept	1.32 Å

2.2. RESTRICCIÓNES CONFORMACIONALES DE LOS POLIPÉPTIDOS

2.2.1. Ángulo omega: planaridad de la unidad peptídica.



2.2.2. Ángulos phi y psi: diagrama de Ramachandran.



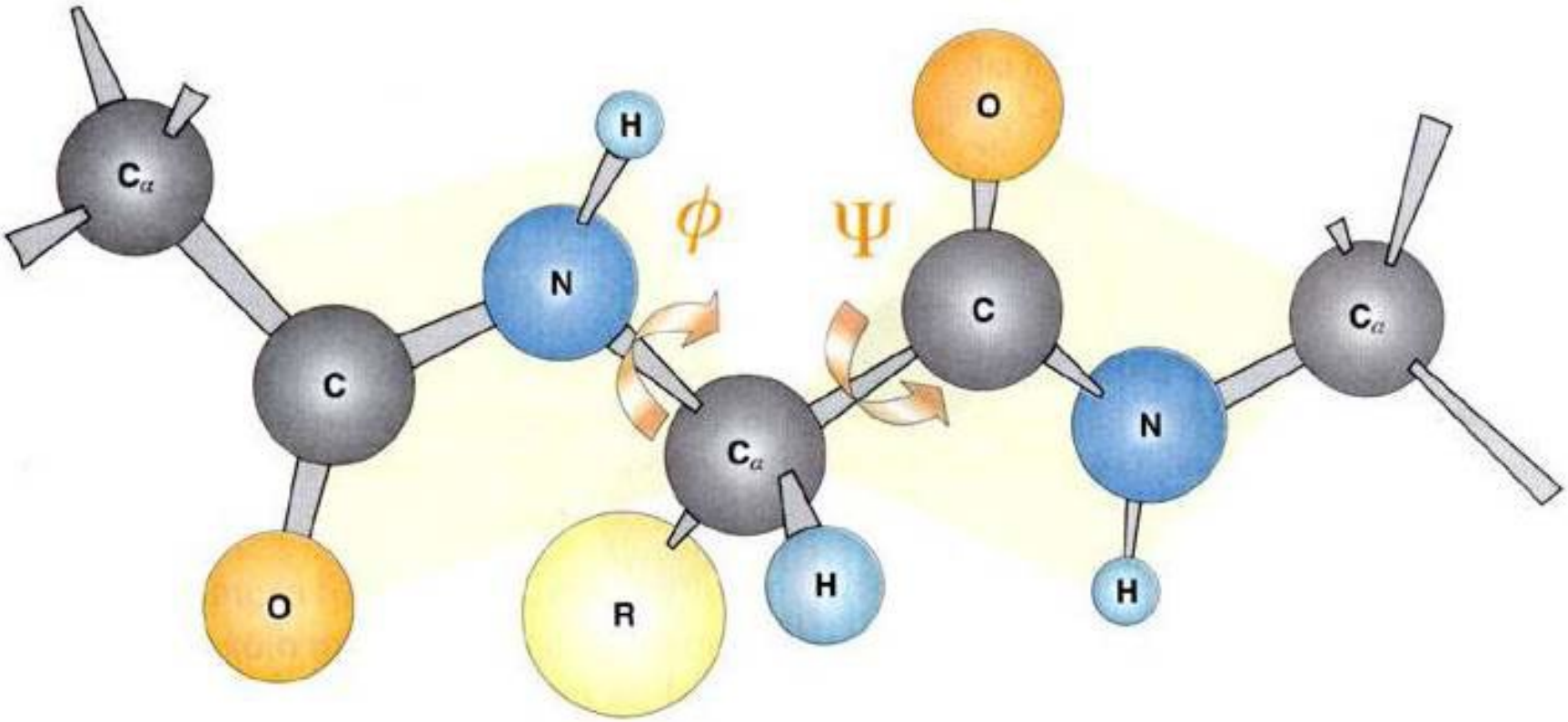
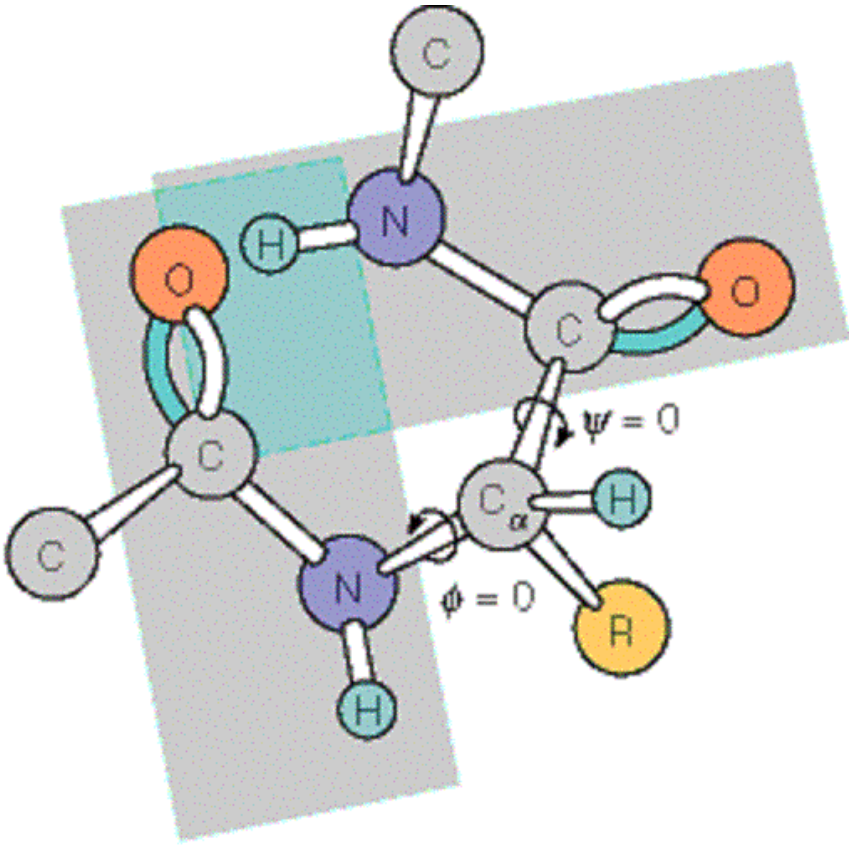
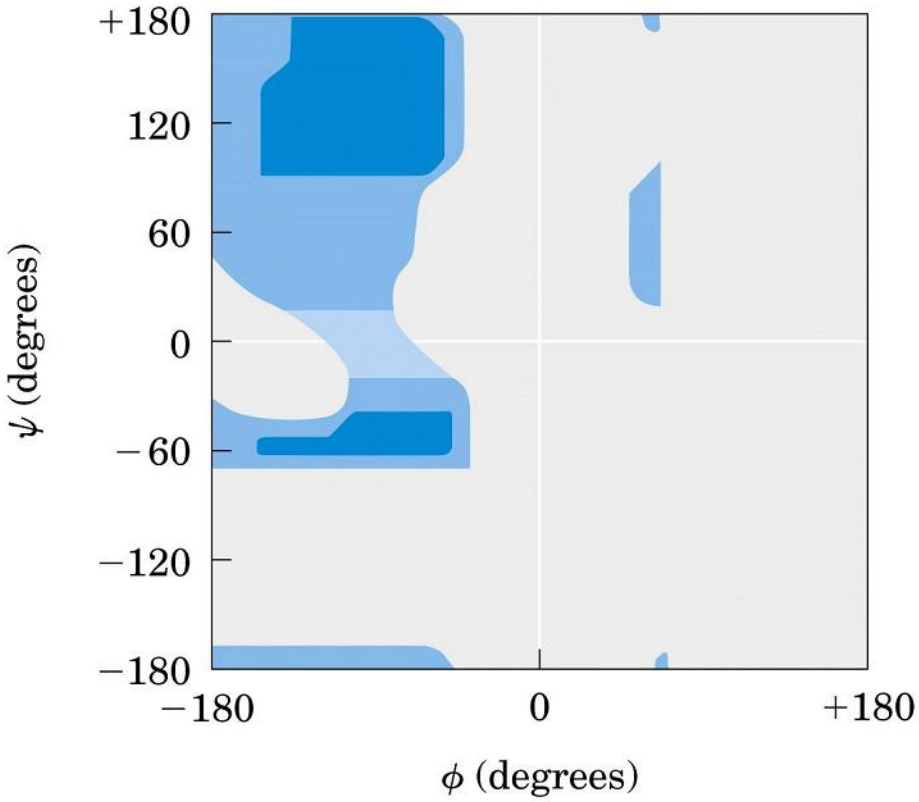
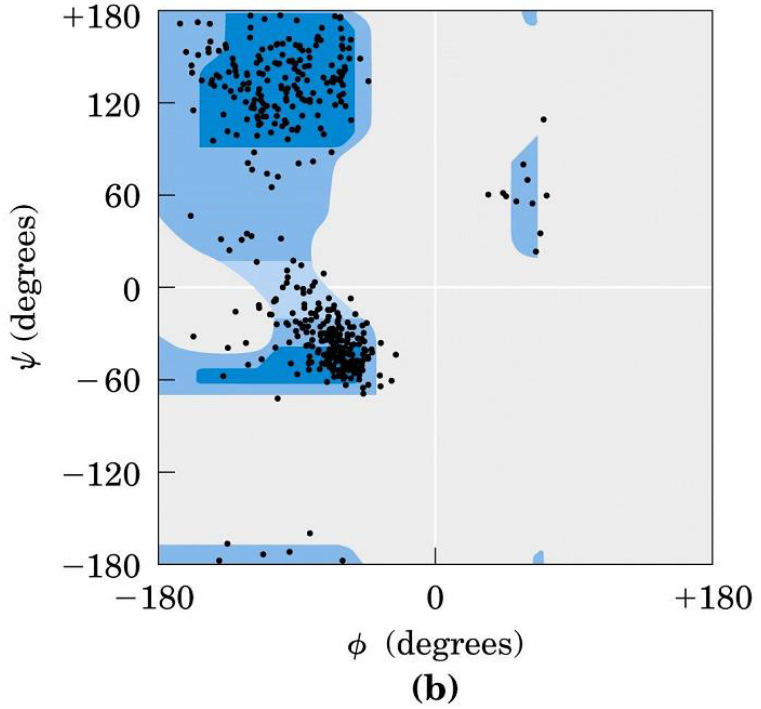


FIG. 2.18. *Ángulos ϕ y ψ de los residuos de las proteínas*. El ángulo ϕ mide el giro en torno al enlace que une el carbono α de un residuo *i* con el *N* (situado en el plano peptídico anterior) del mismo residuo. Su valor es de 0° cuando los átomos C (carbonílicos) de los residuos $i - 1$ e *i* quedan eclipsados en *cis*, y aumenta su valor cuando, mirando desde el *N* hacia el carbono α, el *C* del residuo *i* gira en el sentido de las agujas del reloj. El ángulo ψ mide el giro en torno al enlace que une el carbono α con el *C* del mismo residuo en el plano peptídico posterior (de forma semejante, vale cero cuando los *N* de los residuos *i* e $i + 1$ quedan eclipsados en *cis*) (http://www.chem.qmw.ac.uk/iupac/misc/biop.html).

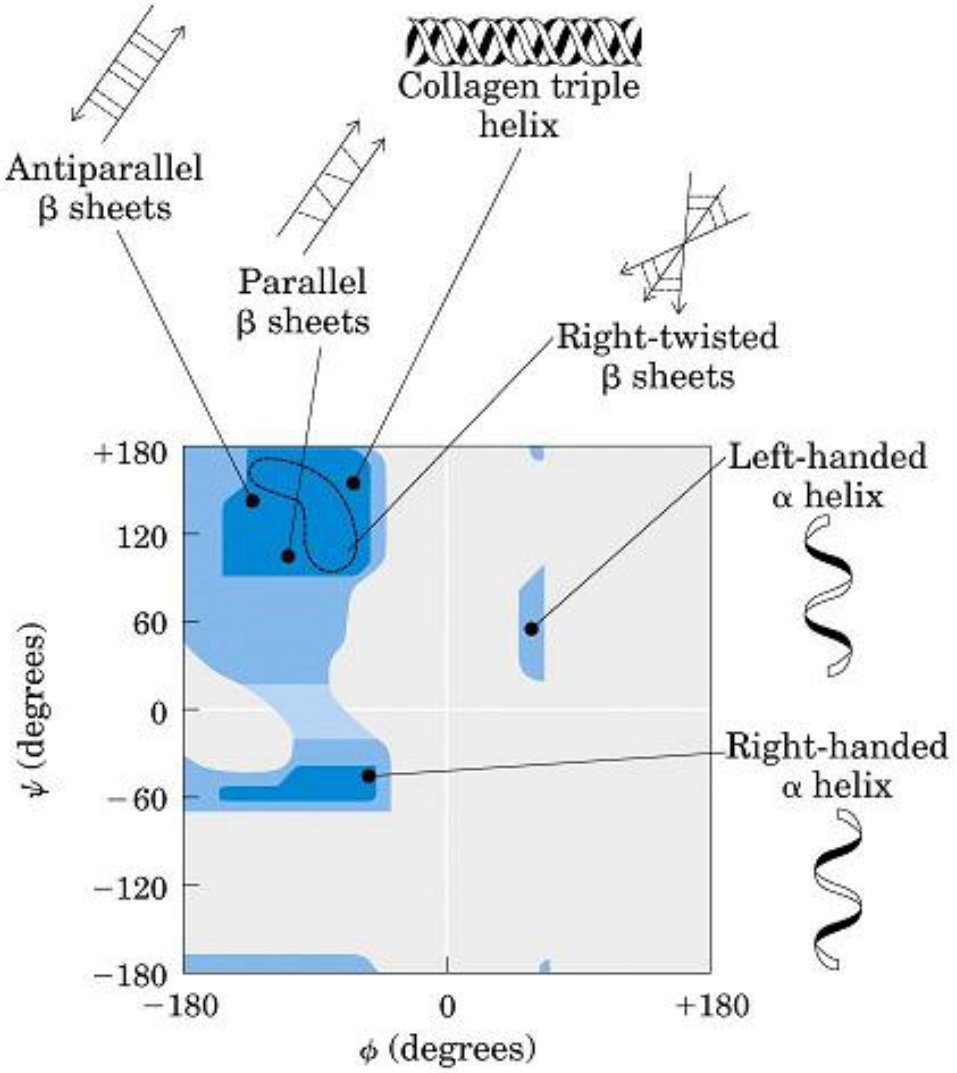

C
N
H
O
C
O
C
$\psi = 0$
C_α
H
C
N
$\phi = 0$
R
H


+180
120
60
0
−60
−120
−180
ψ (degrees)
−180
0
+180
ϕ (degrees)

La gran mayoría de los aminoácidos en las proteínas se sitúan en las regiones permitidas. Sólo las glicinas presentan regiones permitidas más extensas. Los elementos de estructura secundaria de las proteínas (hélices, láminas, giros) están constituidos por residuos de aminoácidos con ángulosphi y psi característicos.

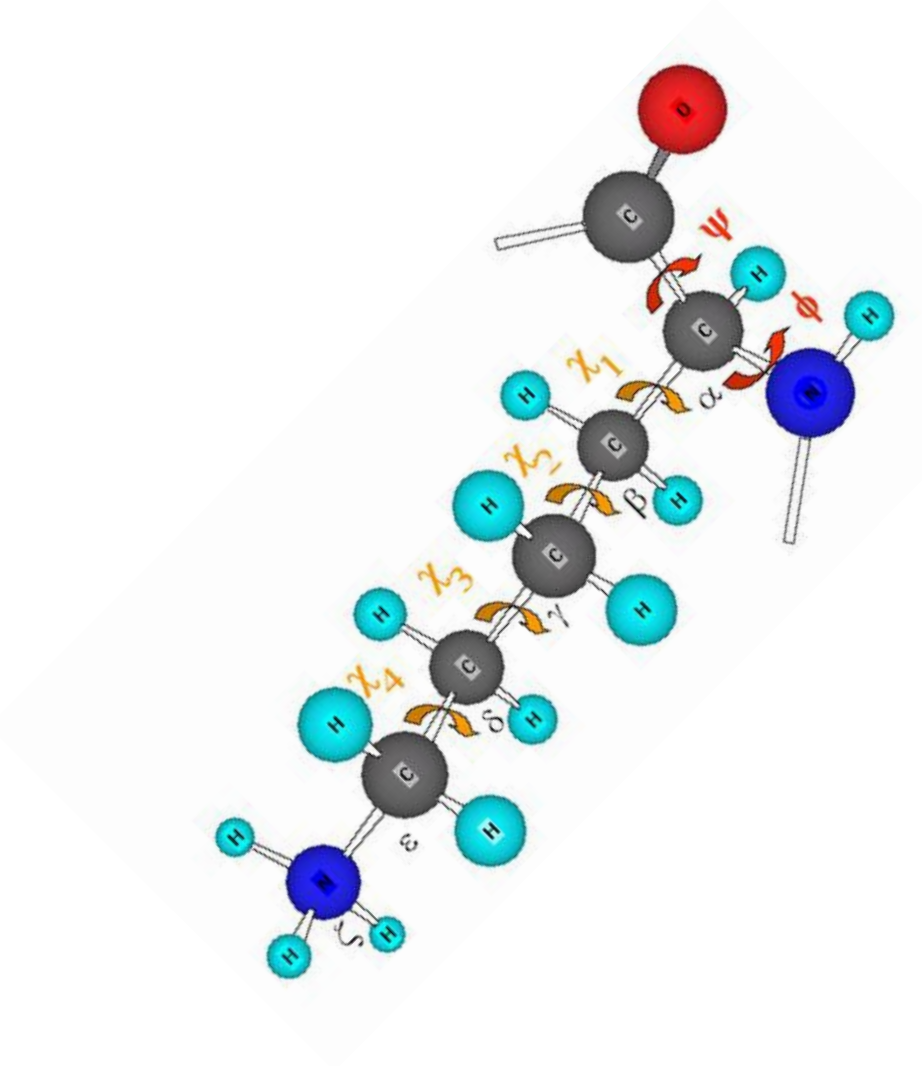


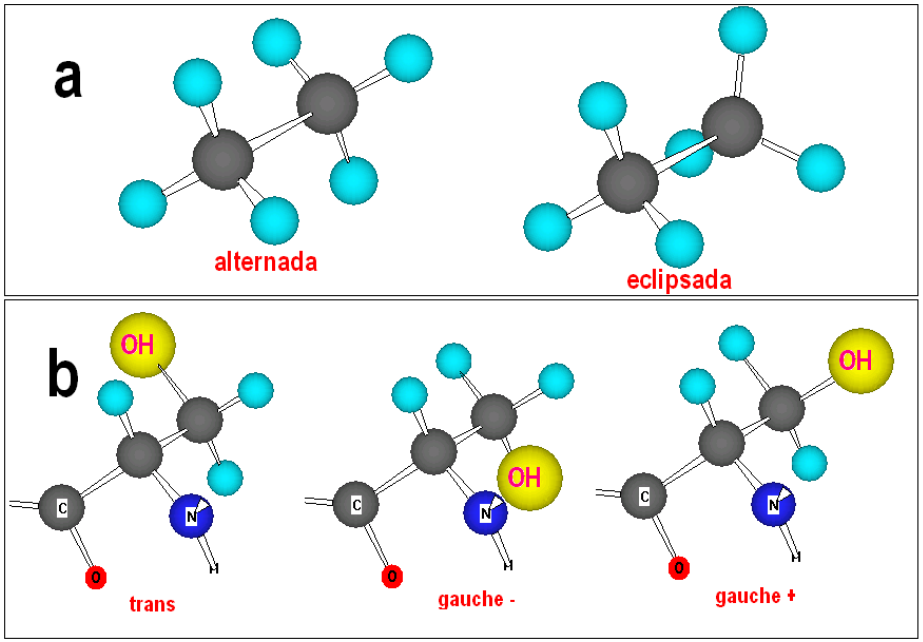
(b)



(a)

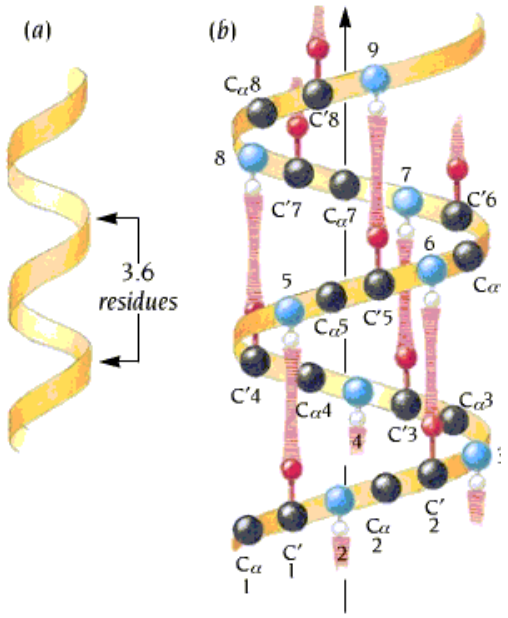
2.2.3. Ángulos chi: rotámeros más estables. El giro en torno al enlace que une el carbono alfa con el siguiente carbono de la cadena lateral se mide con el ángulo chi1.

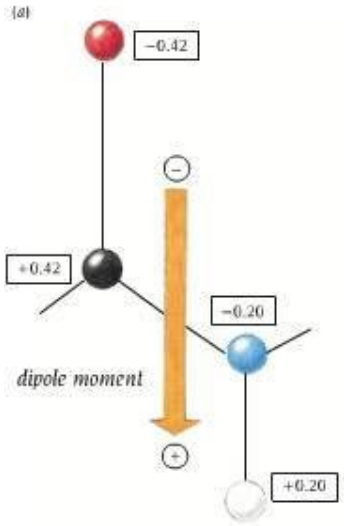


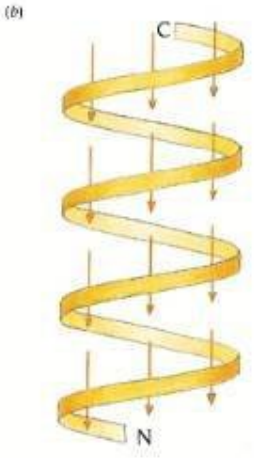


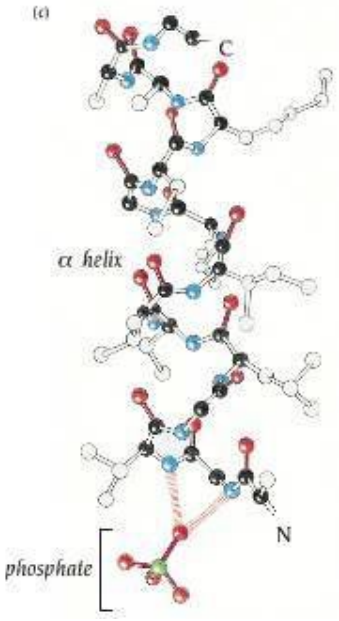
2.3. HÉLICES, LÁMINAS, GIROS Y BUCLES

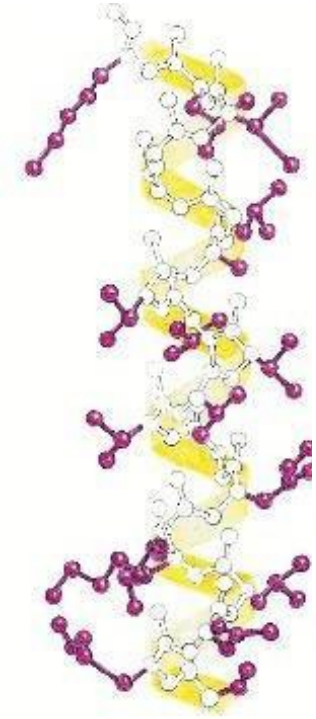
2.3.1. La hélice alfa. Cuando 5 o más aminoácidos consecutivos de una proteína adoptan ángulo phi de -60 y psi de -40 aparece una hélice alfa. En las hélices alfa el carbonilo (-CO) del residuo *i* forma un puente de hidrógeno con el NH del residuo *i+4* (salvo los 4 CO del extremo C de la hélice y los 4 NH del extremo N).

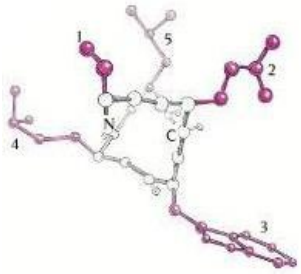


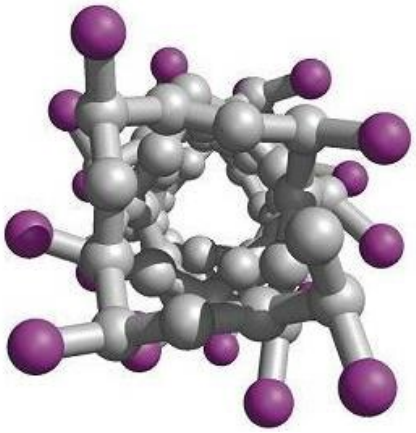




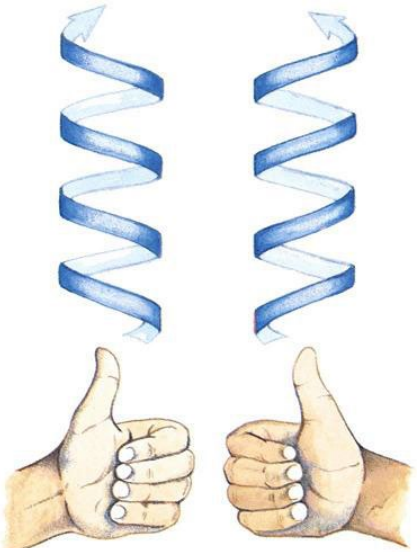




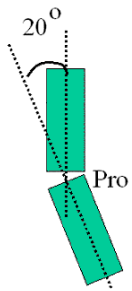




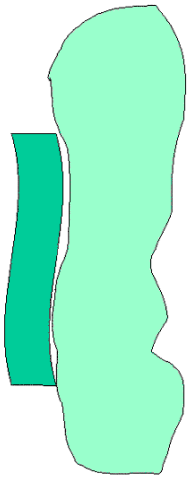
Las hélices alfa de las proteínas, formadas por L-aminoácidos, son R. Las hélices (no transmembrana) tienen una longitud media de 12 aminoácidos.



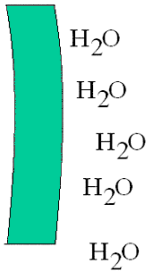
Las hélices alfa de las proteínas pueden presentan las siguientes distorsiones:



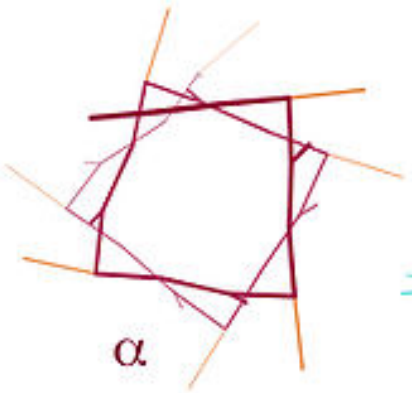
Prolina

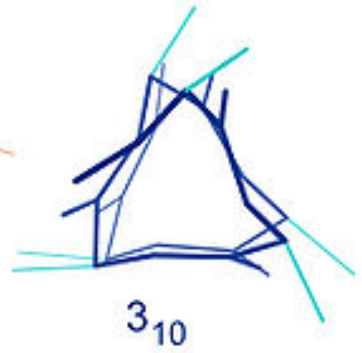


empaquetamiento

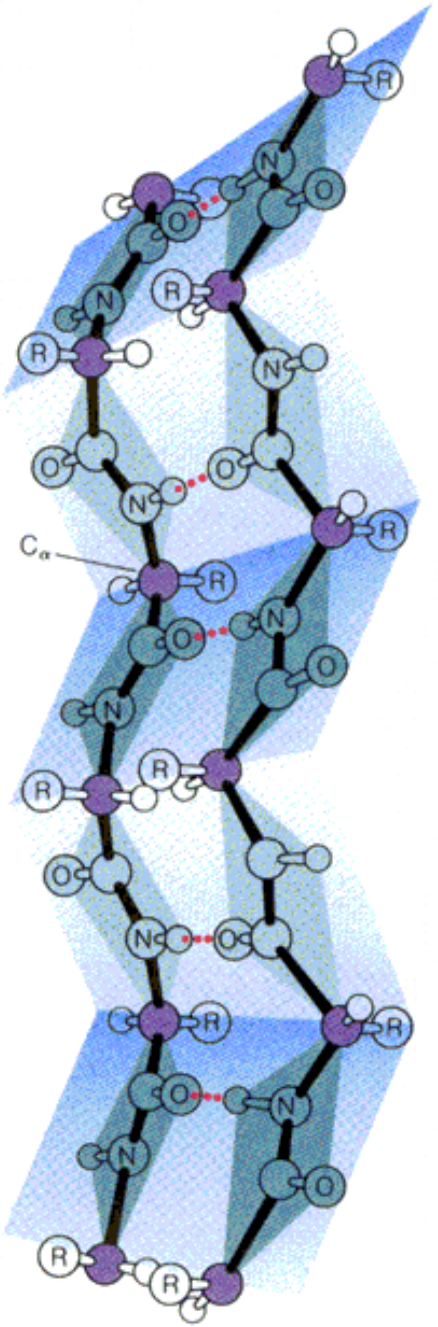


disolvente



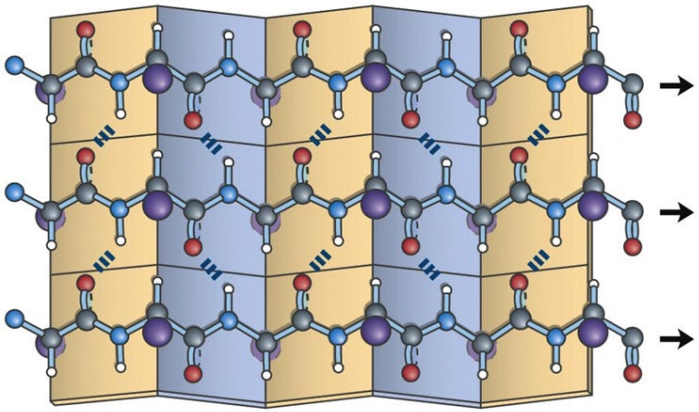


2.3.2. La lámina beta. Cuando 2 o más aminoácidos consecutivos de una proteína adoptan ángulos phi de ≈ -140 y psi de ≈+130, aparece una hebra beta (cada residuo avanza 3.5 Å a lo largo del eje de la hebra). Cuando 2 o más hebras beta se sitúan una al lado de la otra y establecen p de H entre ellas, aparece una lámina beta. Todos los CO y NH forman p de H con los de hebras adjacentes, salvo los que definen los bordes izdo y dcho de la lámina.

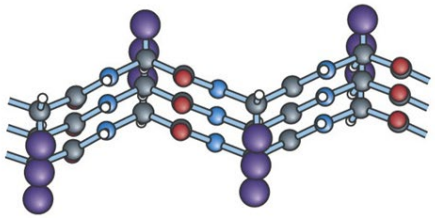


Parallel

Top view

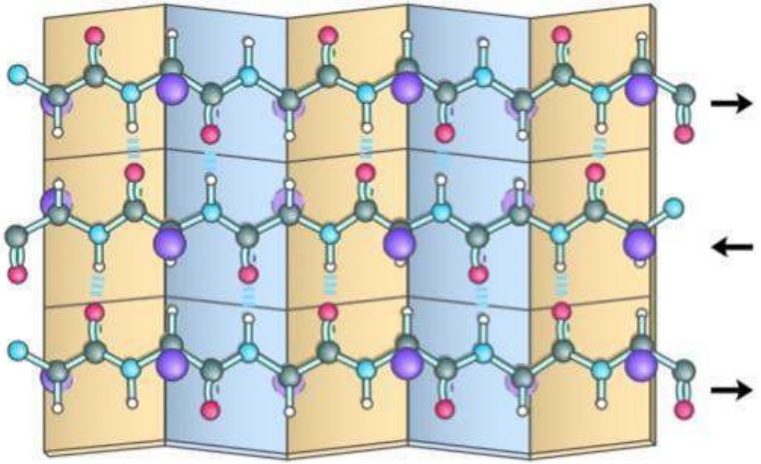


Side view

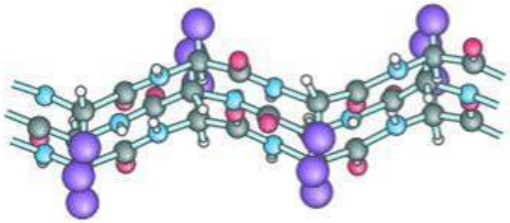


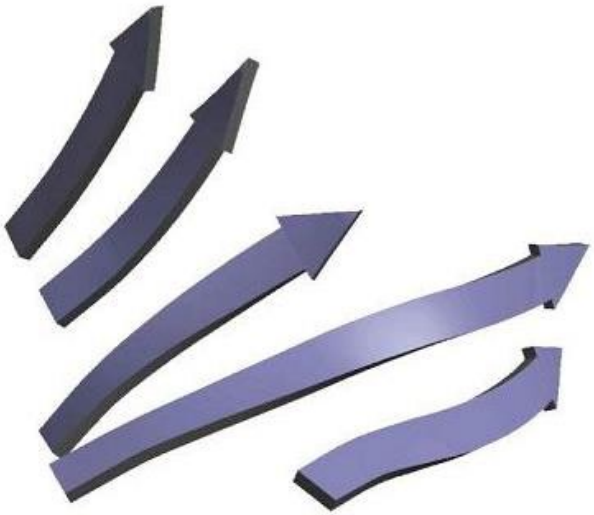
(a) Antiparallel

Top view



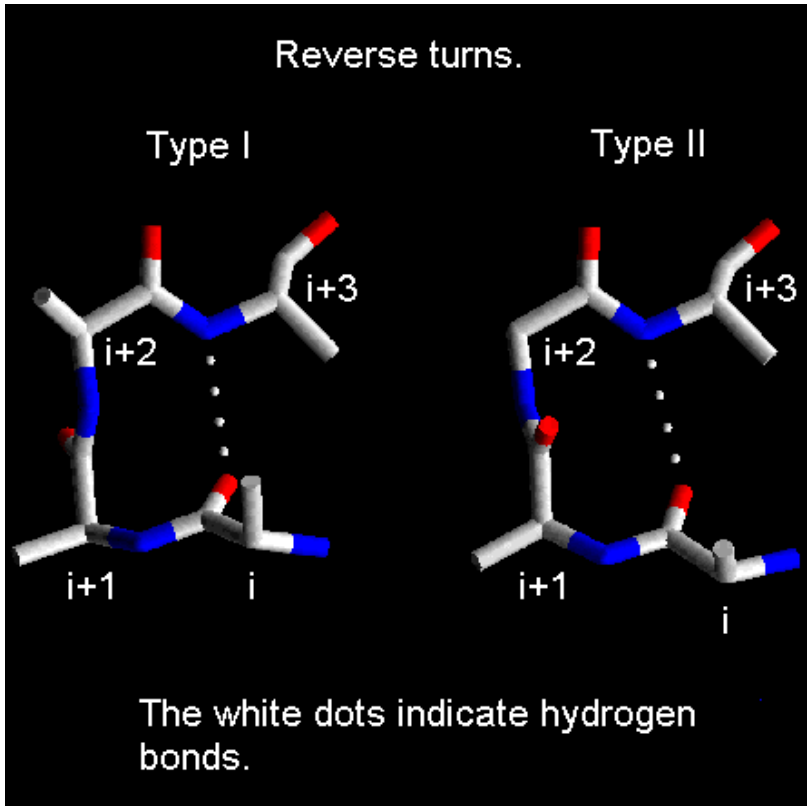
Side view

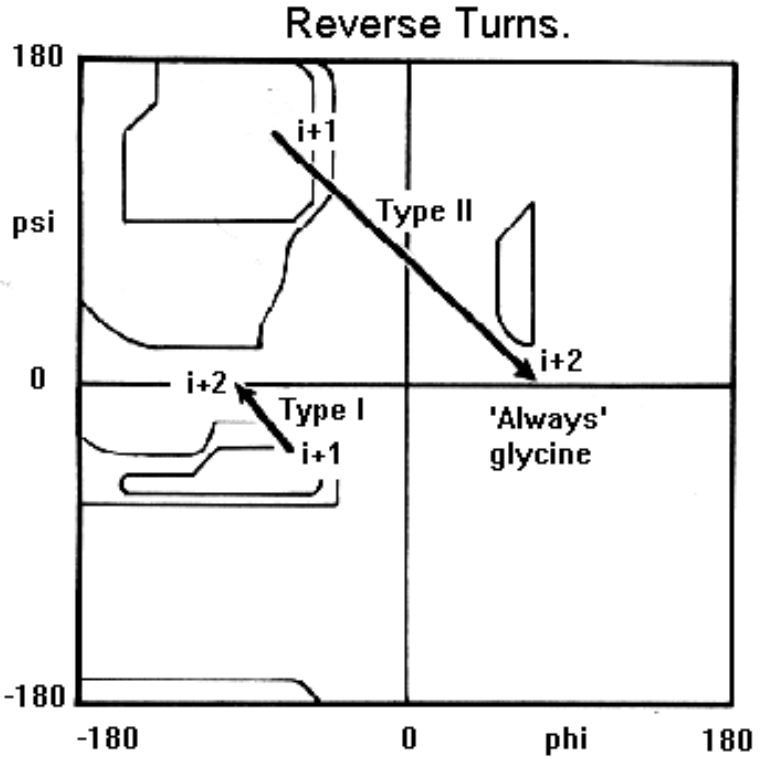


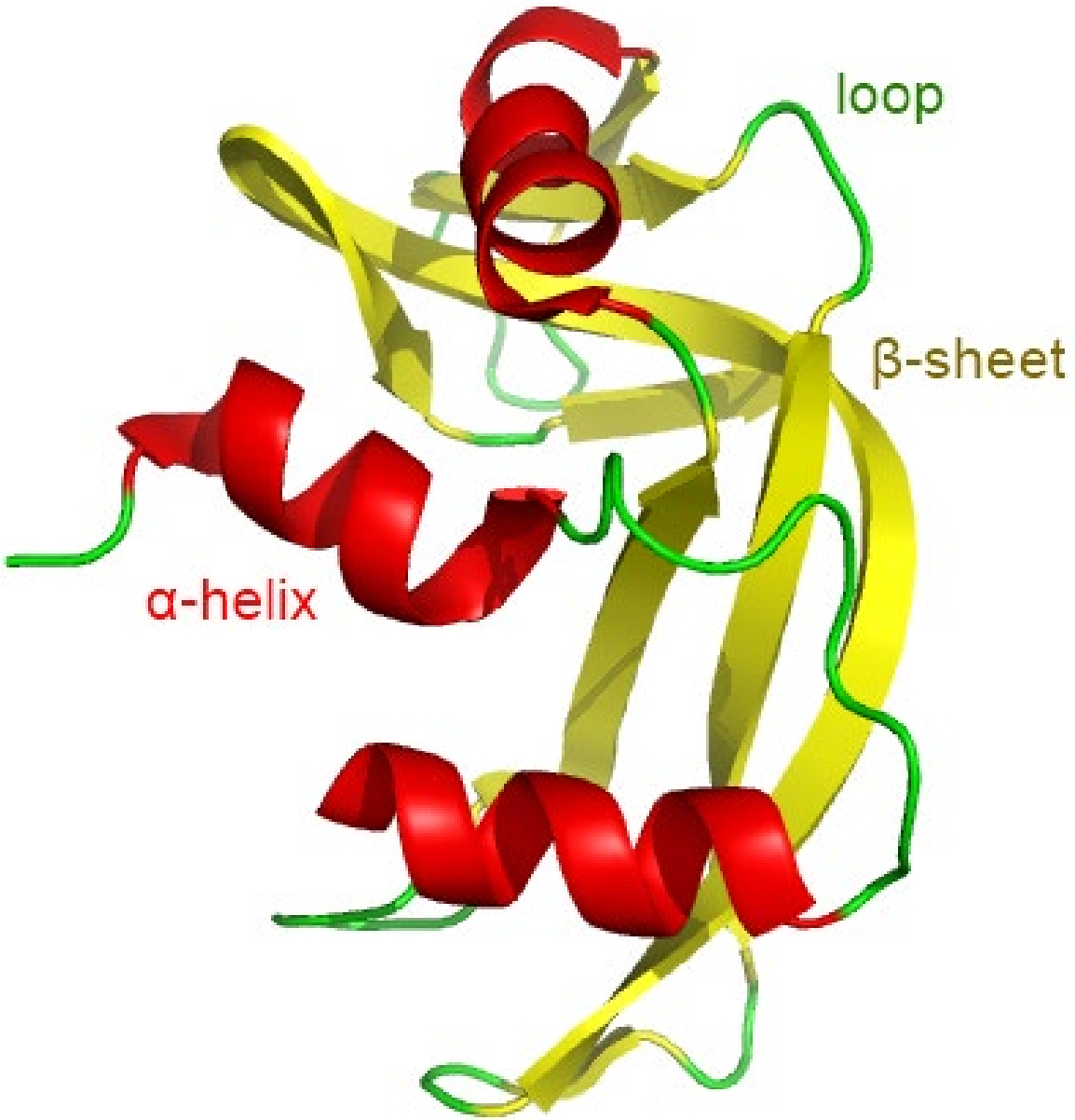


Twisted β sheet

2.3.3. Giros y bucles. Los giros reversos son regiones del polipéptido con un p de H entre el CO (i) y el NH (i+3) (se excluyen las hélices 3_{10}). Hay dos tipos principales (I y II) que se diferencian en los ángulos phi y psi de los residuos i+1 e i+2. Los giros que conectan hebras de una lámina beta antiparalela tienen ángulos distintos y se dominan giros beta de tipo I' y II' . En los giros abundan las Gly.






loop
β-sheet
α-helix

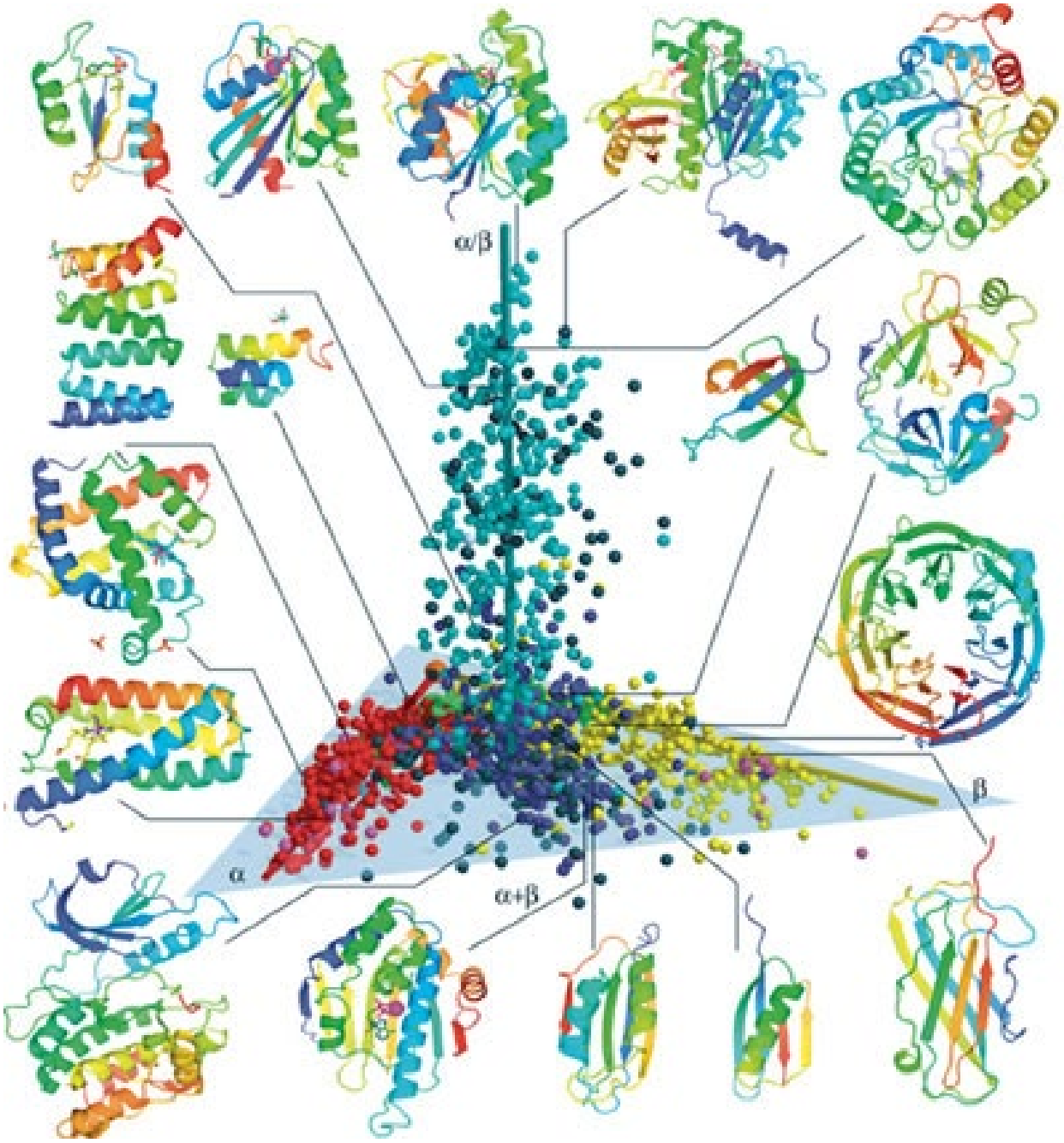
3. PROTEÍNAS SOLUBLES EN AGUA

3.1. CLASES DE PROTEÍNAS GLOBULARES SEGÚN LA ESTRUCTURA SECUNDARIA PREDOMINANTE

- Proteínas alfa (25 % del total de dominios conocidos)
- proteínas beta (20 %)
- proteínas alfa/beta (principalmente beta paralelas, unidades beta-alfa-beta)(30 %)
- proteínas alfa+beta (principalmente beta antiparalelas, regiones alfa y beta segregadas)(25 %).

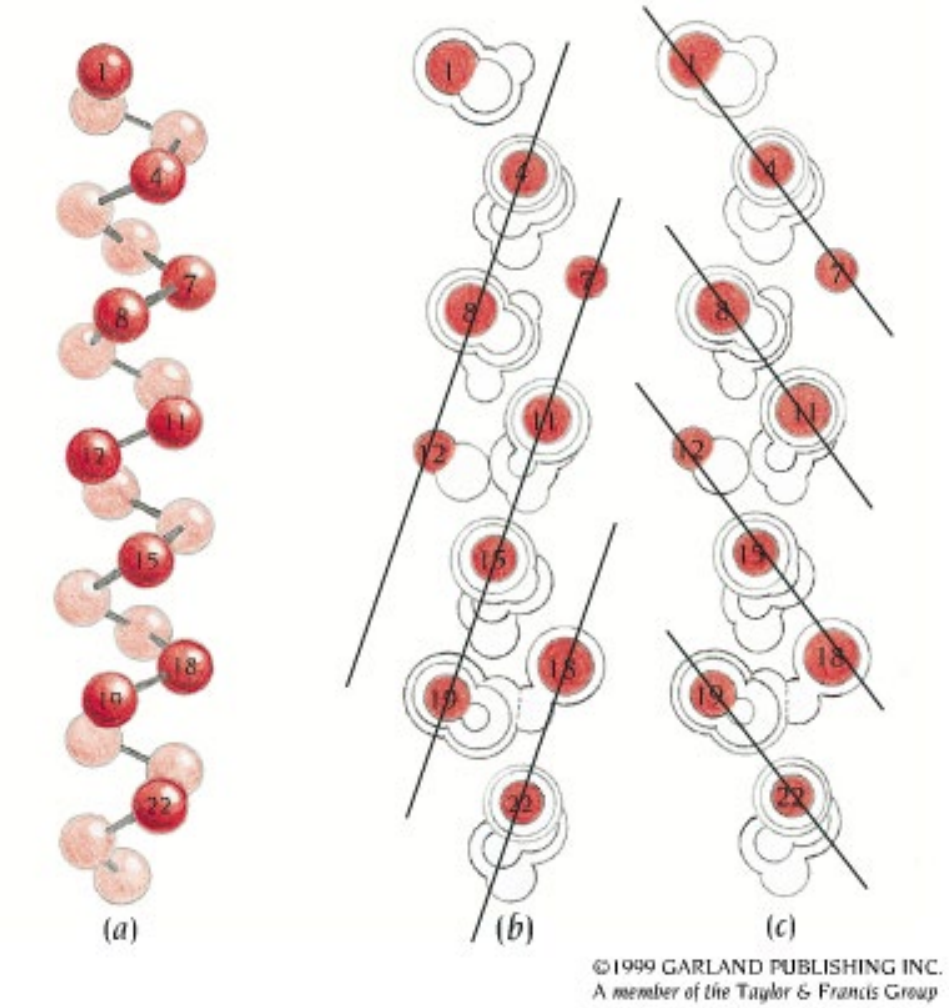
Existen, además, proteínas con poca estructura secundaria.

In this 3-D protein structure space map (SSM), each of the known 1,898 unique protein structures is represented by a sphere, color-coded according to the widely used Structural Classification of Proteins (SCOP) system. The spheres are distributed along three elongated regions. Spheres that are close together will usually share similar structure and function, as well as evolutionary history, making this map an excellent predictive tool for exploring the protein universe

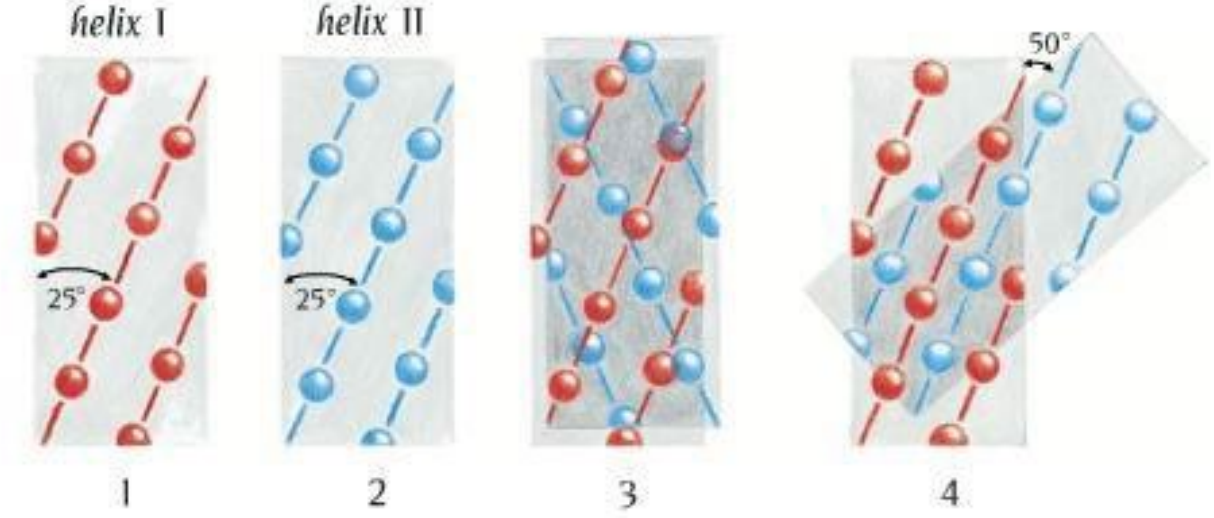


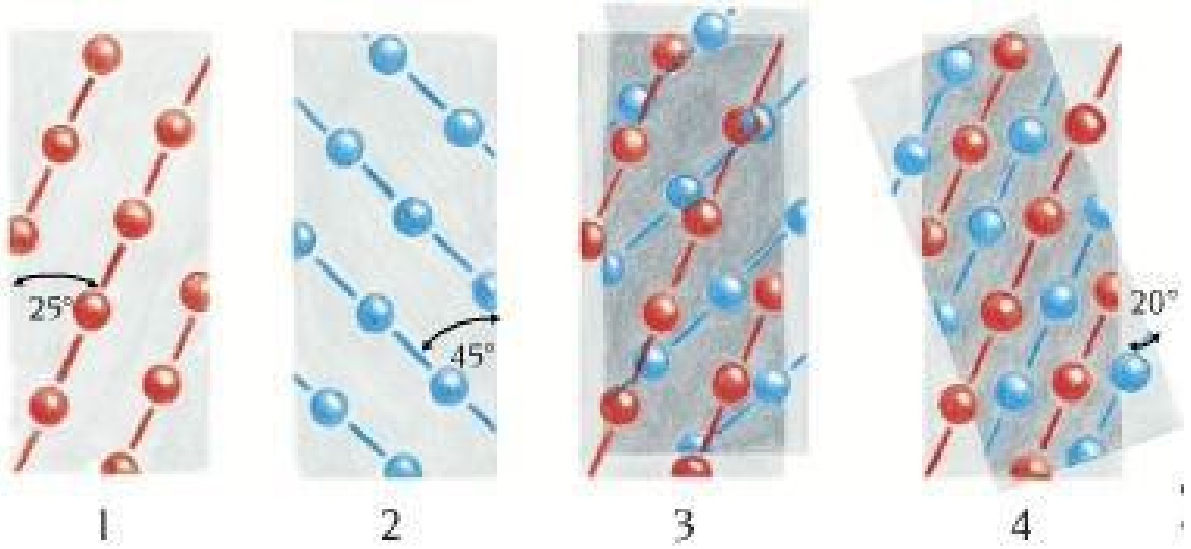
3.2. PROTEÍNAS ALFA

3.2.1. Cómo empaquetar dos hélices alfa. Las cadenas laterales de los aminoácidos a i+3 (y a i+4) en hélices alfa están alineadas. Cada una de esas líneas de cadenas laterales es como la cresta de un tornillo y entre línea y línea aparece un 'surco'.

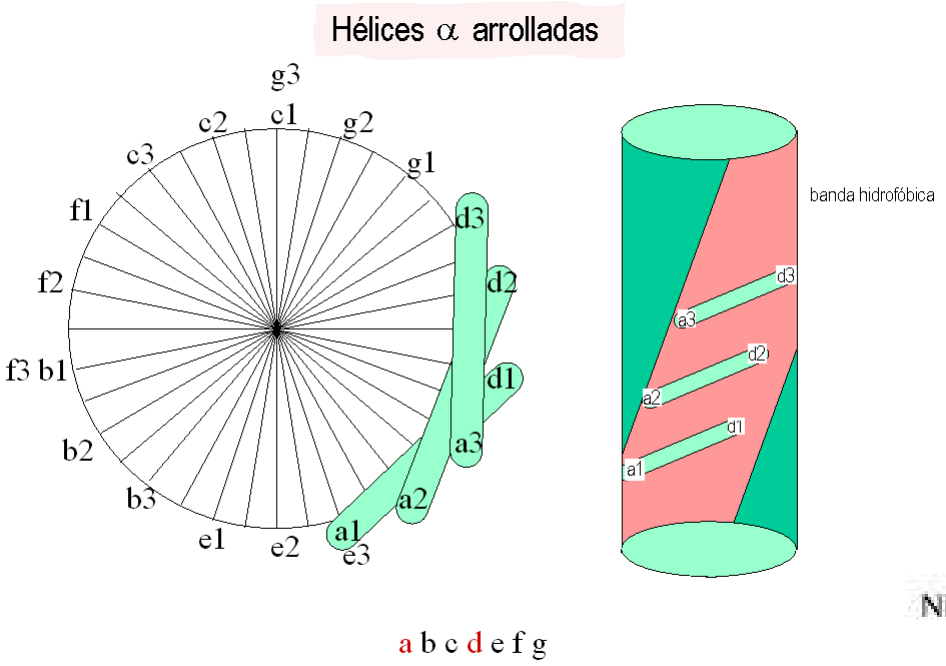


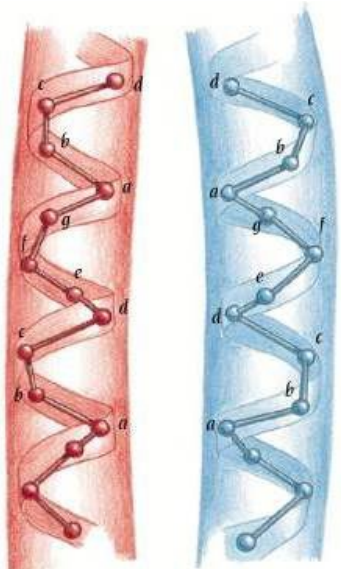
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helix I
helix II
25°
25°
50°
1
2
3
4


25°
45°
20°
1
2
3
4

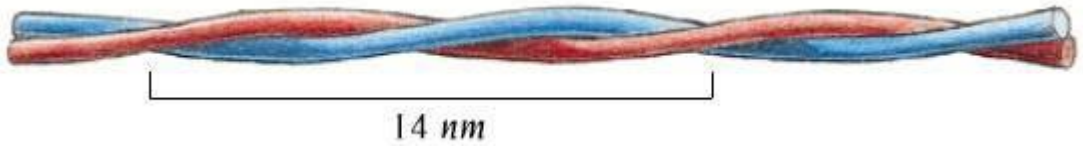
3.2.2. Arrollamientos de hélices ('coiled-coils'). Dos hélices muy largas con secuencias (absdef)n, donde *a* y *d* son hidrofóbos, se disponen con ángulos de ≈ 20° juntando sus zonas hidrófobas.





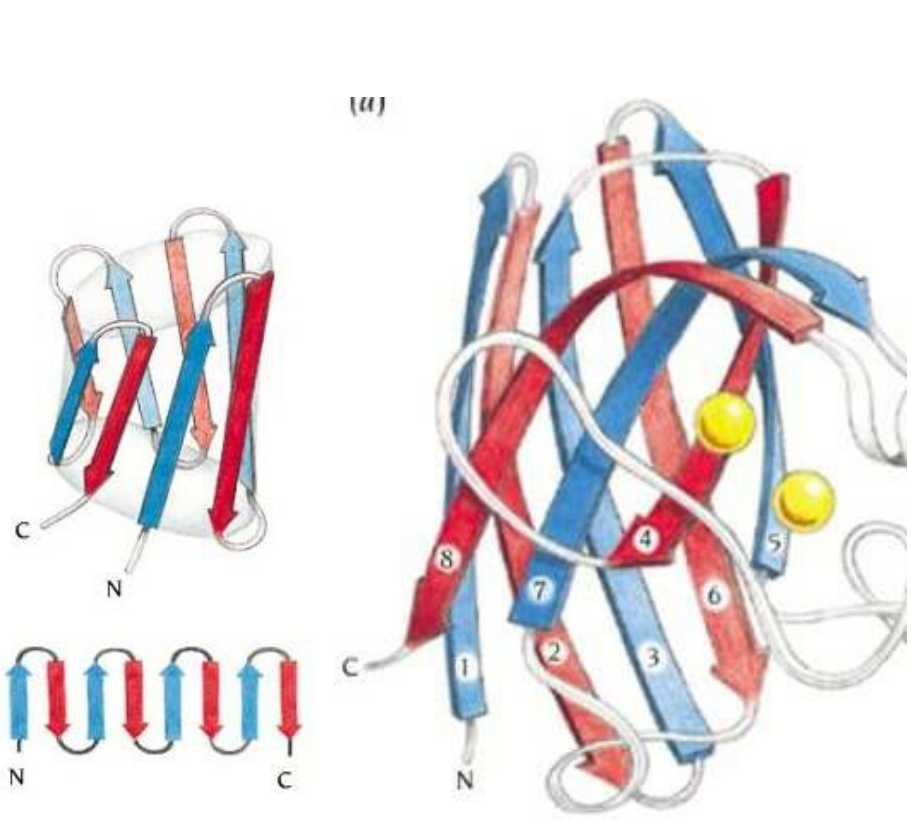
	a	*b*	*c*	*d*	*e*	*f*	*g*	
NH_2 -	Met	- Lys	- Gln	- **Leu**	- Glu	- Asp	- Lys	-
	Val	- Glu	- Glu	- **Leu**	- Leu	- Ser	- Lys	-
	Asn	- Tyr	- His	- **Leu**	- Glu	- Asn	- Glu	-
	Val	- Ala	- Arg	- **Leu**	- Lys	- Lys	- Leu	- COOH

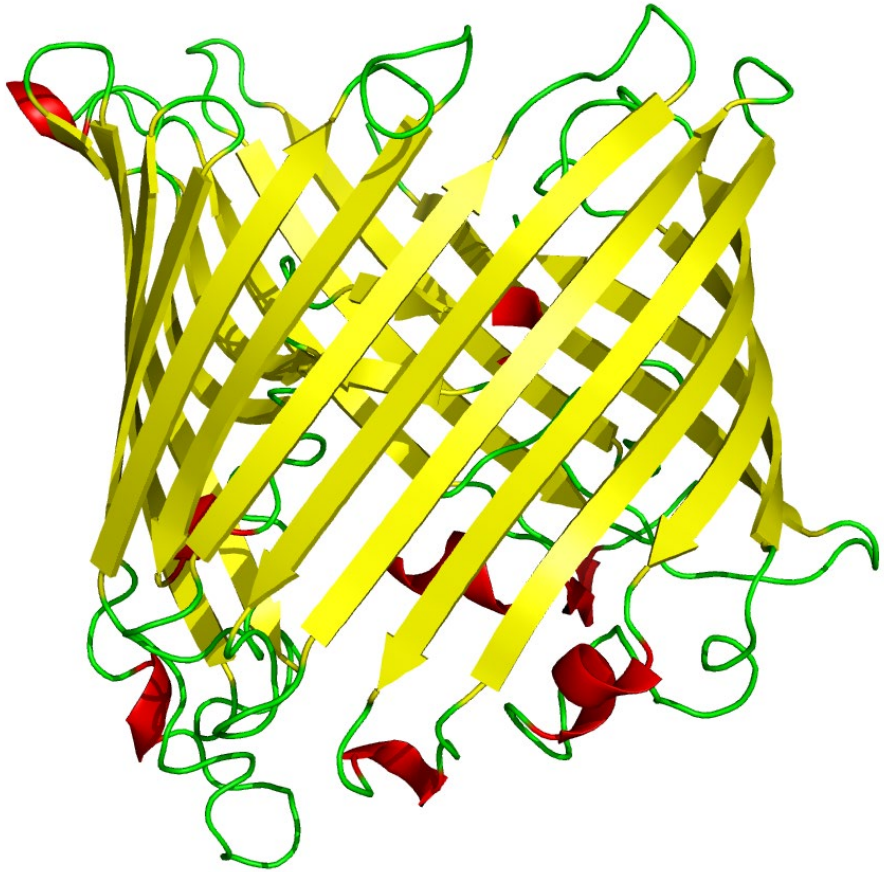
Los filamentos intermedios (proteínas fibrosas) tienen esta estructura.



3.3. PROTEÍNAS BETA

3.3.1. De una sóla lámina (barril beta). Suelen tener 7 u 8 hebras, la primera enlazada por p de H con la última. El orden de las hebras no tiene porque reflejar su orden en la secuencia. La secuencia de las hebras presenta una marcada alternancia de residuos polares y apolares.





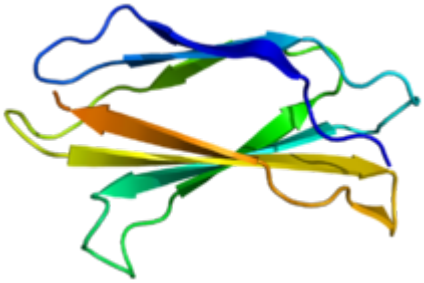
3.3.2. De varias láminas.

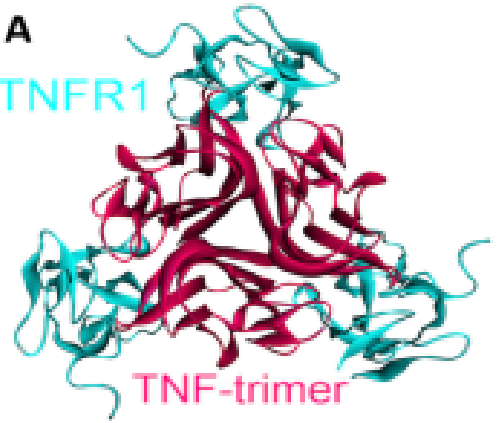
Emparedados beta(‘beta sandwiches')

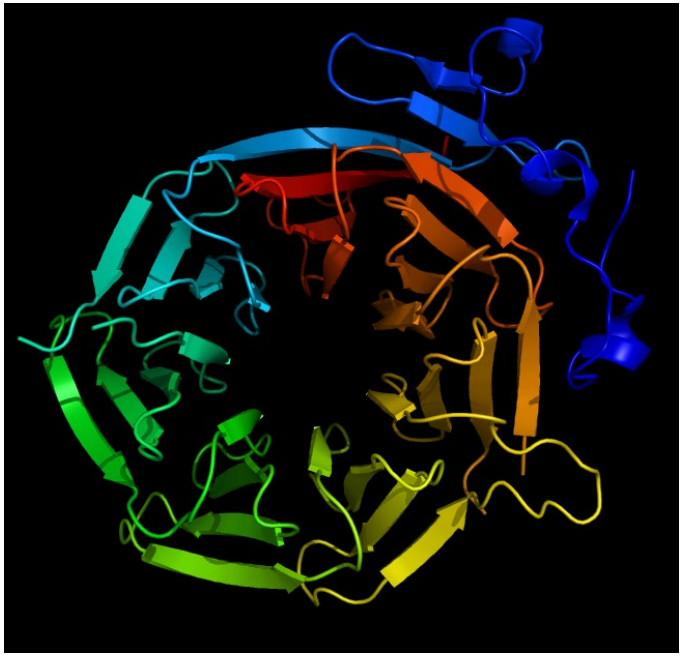
Prismas beta.

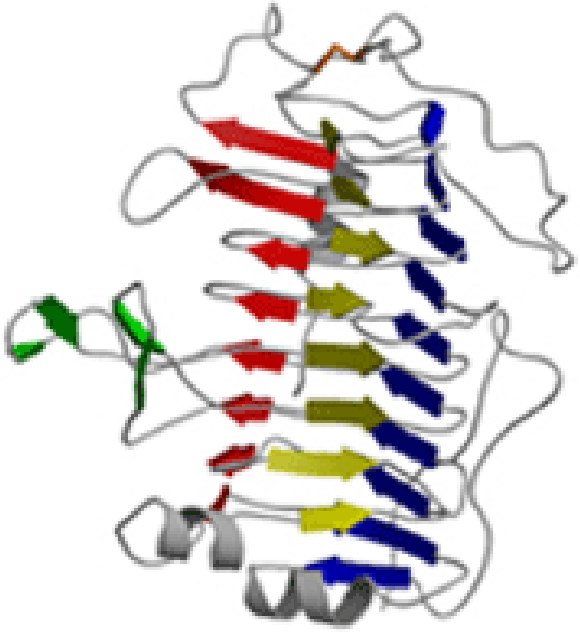
Solenoides

Turbinas





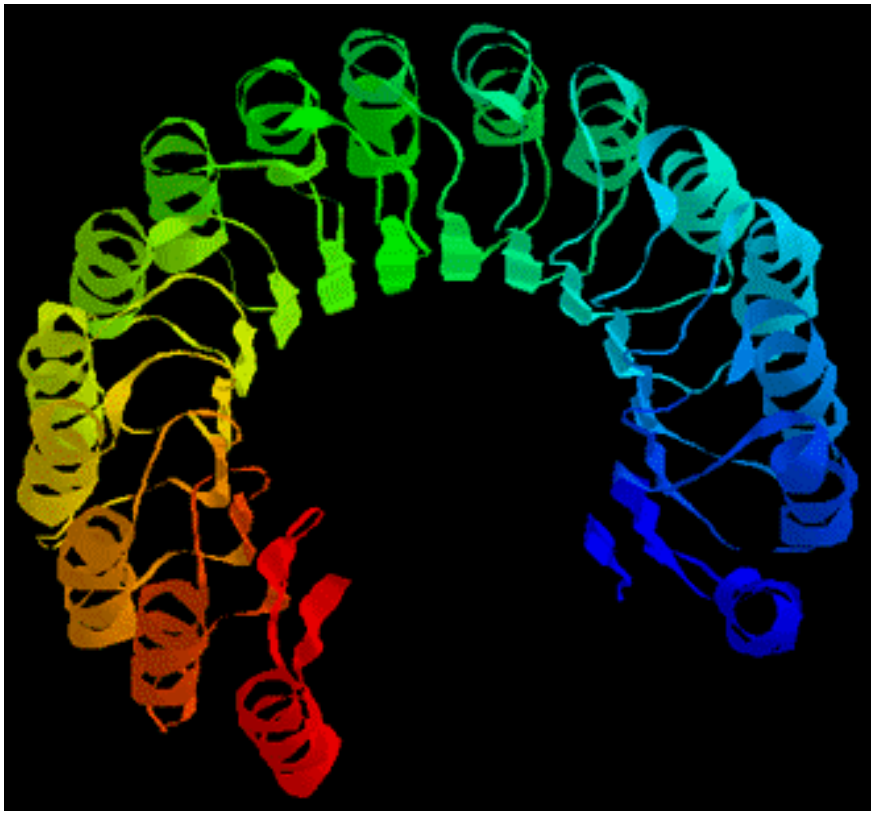




3.4. PROTEÍNAS ALFA/BETA

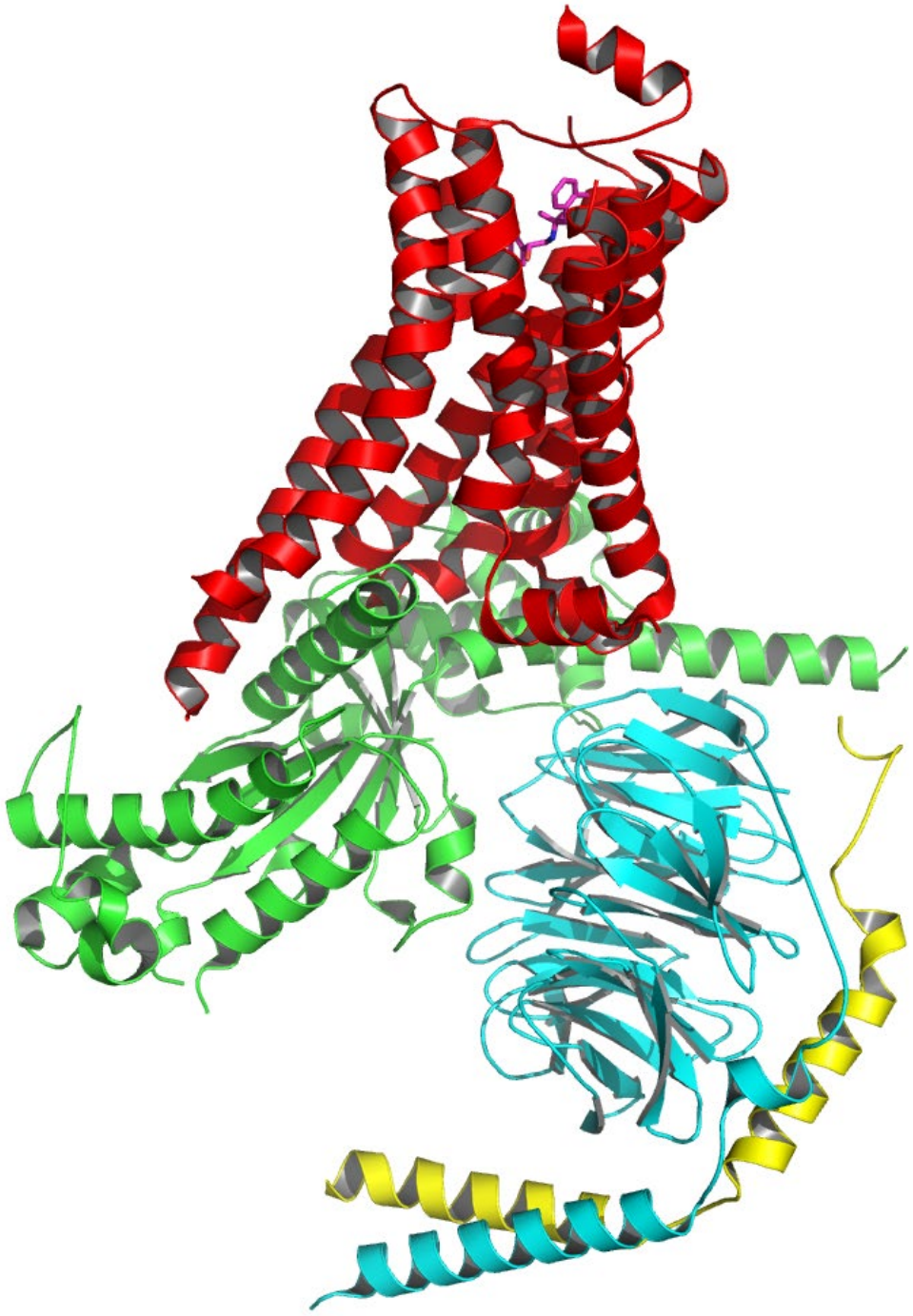
Contienen hélices alfa y láminas beta (habitualmente paralelas). Las hélices y las hebras de la lámina se alternan a lo largo de la secuencia. Con lámina cerrada (barriles alfa/beta). Láminas de 7 u 8 hebras que forman un barril sobre cuya cara externa se empaquetan las hélices. Si hay más hebras, aparece un hueco central que se rellena con un bucle o una hélice. Las hebras son esencialmente apolares.





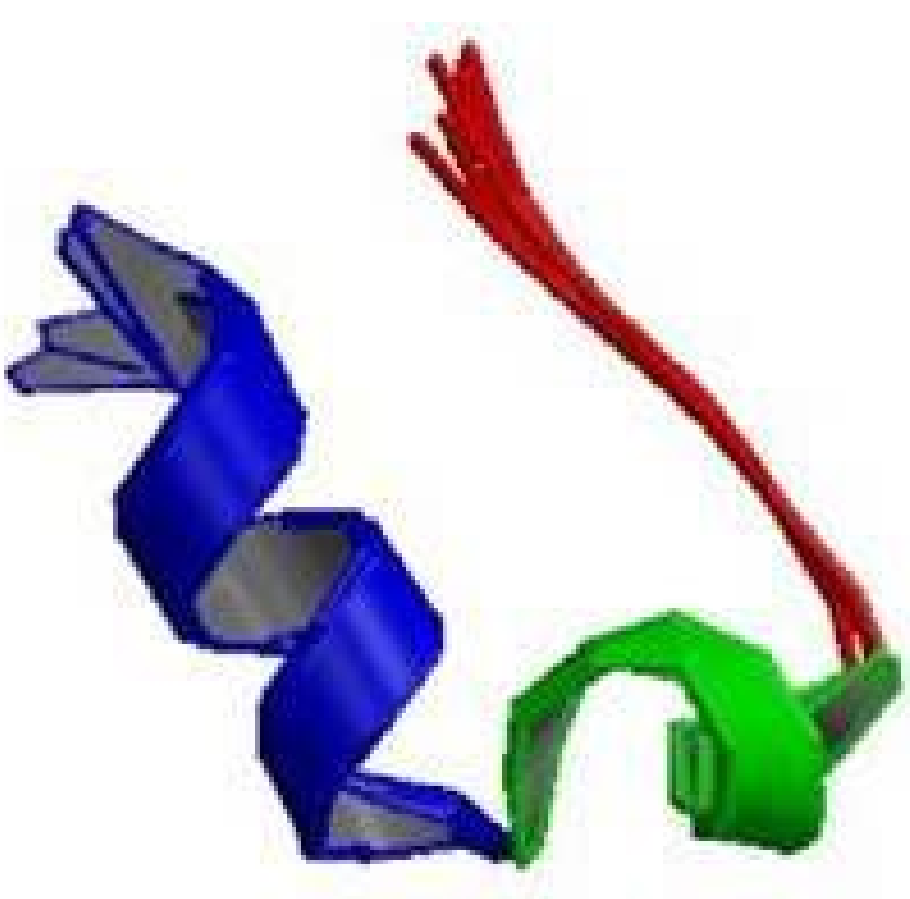
3.5. PROTEÍNAS ALFA + BETA

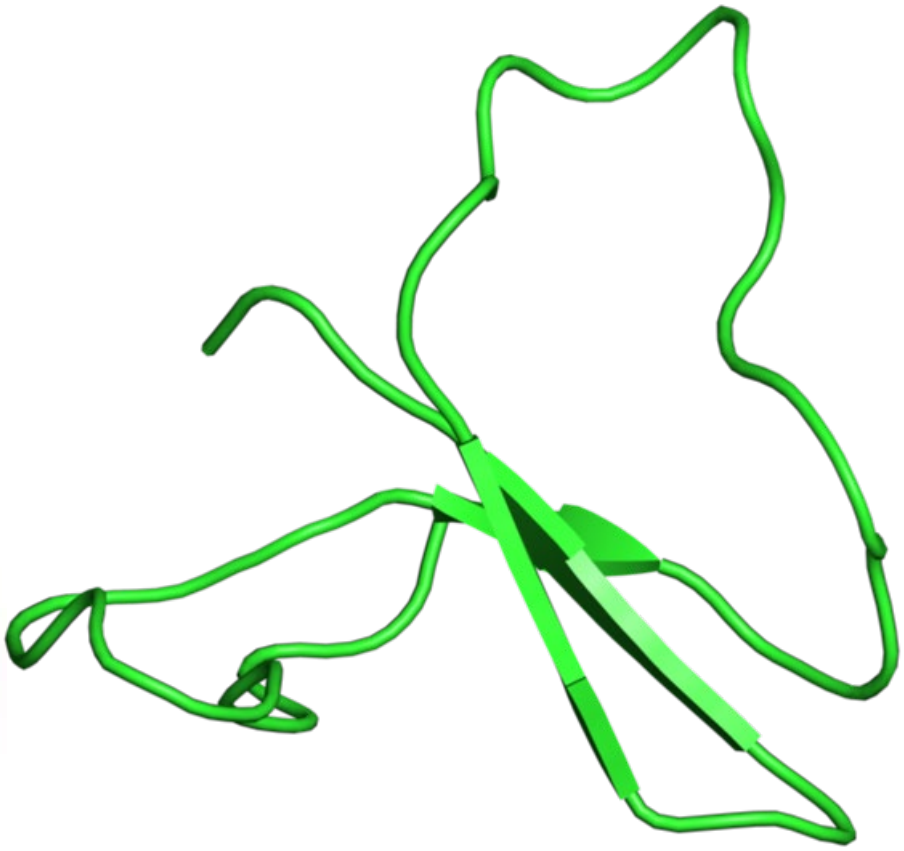
Las hélices y las hebras de las láminas tienden a estar segregadas (no alternadas) en la secuencia. Las láminas tienden a ser paralelas.



3.6.PEQUEÑAS PROTEÍNAS

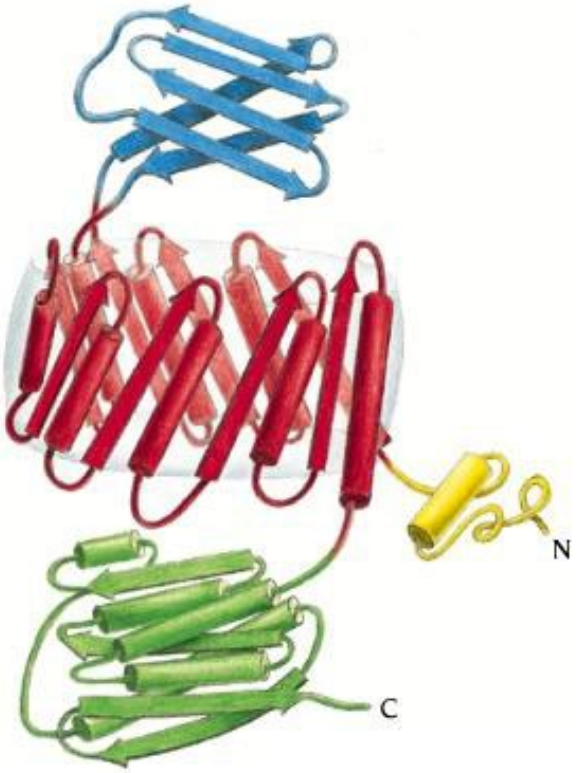
Suelen tener menos de 60 aminoácidos. Son pobres en elementos de estructura secundaria. Su estructura tridimensional está frecuentemente estabilizada por puentes disulfuro o por cationes unidos.



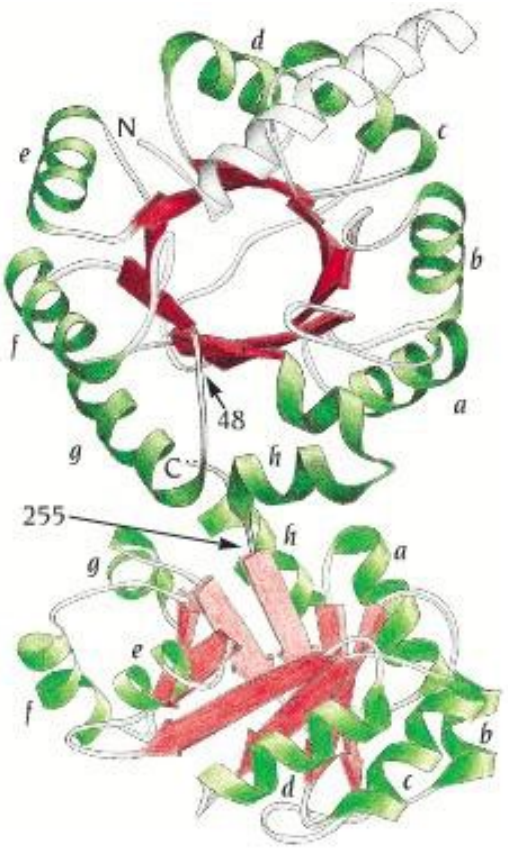


3.7.CONCEPTO DE DOMINIO. PROTEÍNAS CON VARIOS DOMINIOS

Un dominio de plegamiento es una parte de una proteína cuyos residuos forman pocos contactos con el resto. Las proteínas descritas hasta ahora constan de un solo dominio. Muchas proteínas, sin embargo, tienen varios.



piruvato quinasa



fosforibosil antranilato isomerasa/
indolglicerol fosfato sintasa de E. coli

4. PROTEÍNAS INSOLUBLES EN AGUA

table 6–1

Secondary Structures and Properties of Fibrous Proteins

Structure	Characteristics	Examples of occurrence
α Helix, cross-linked by disulfide bonds	Tough, insoluble protective structures of varying hardness and flexibility	α-Keratin of hair, feathers, and nails
β Conformation	Soft, flexible filaments	Silk fibroin
Collagen triple helix	High tensile strength, without stretch	Collagen of tendons, bone matrix

Amino Acid	α-Keratin (Wool)	Fibroin (Silk)	Collagen (Bovine Tendon)	Elastin (Pig Aorta)
Gly	8.1	44.6	32.7	32.3
Ala	5.0	29.4	12.0	23.0
Ser	10.2	12.2	3.4	1.3
Glu + Gln	12.1	1.0	7.7	2.1
Cys	11.2	0	0	—[e]
Pro	7.5	0.3	22.1[a]	10.7[c]
Arg	7.2	0.5	5.0	0.6
Leu	6.9	0.5	2.1	5.1
Thr	6.5	0.9	1.6	1.6
Asp + Asn	6.0	1.3	4.5	0.9
Val	5.1	2.2	1.8	12.1
Tyr	4.2	5.2	0.4	1.7
Ile	2.8	0.7	0.9	1.9
Phe	2.5	0.5	1.2	3.2
Lys	2.3	0.3	3.7[b]	3.6[d]
Trp	1.2	0.2	0	—[e]
His	0.7	0.2	0.3	—[e]
Met	0.5	0	0.7	—[e]

Note: The three most abundant amino acids in each protein are indicated in red. Values given are in mole percent.

[a]About 39% of this is hydroxyproline.

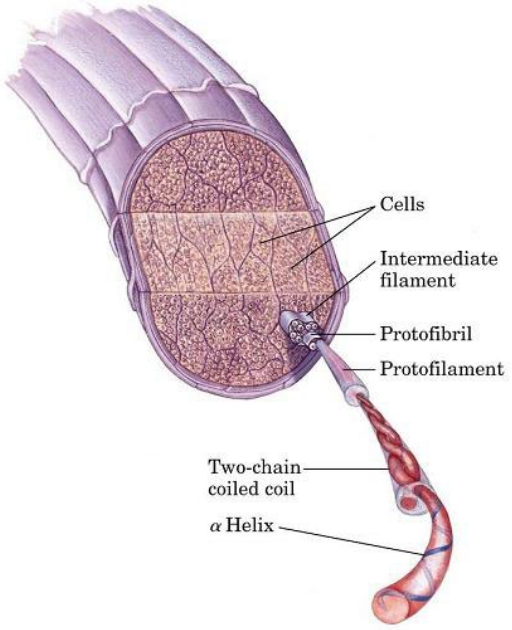
[b]About 14% of this is hydroxylysine.

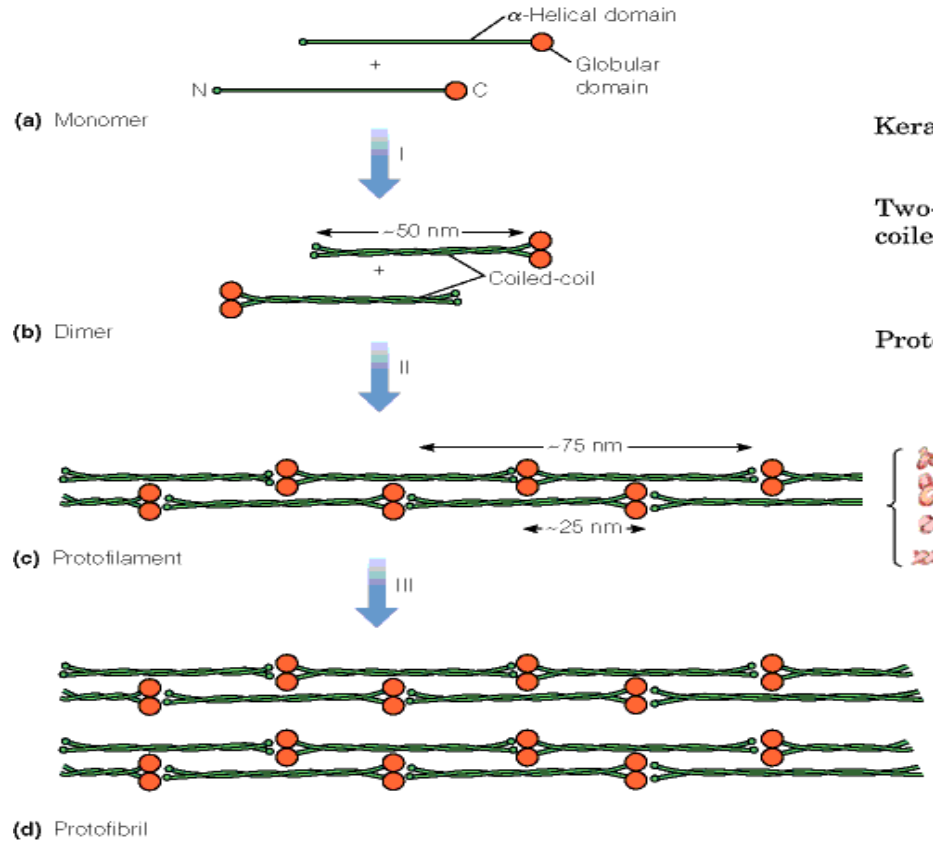
[c]About 13% of this is hydroxyproline.

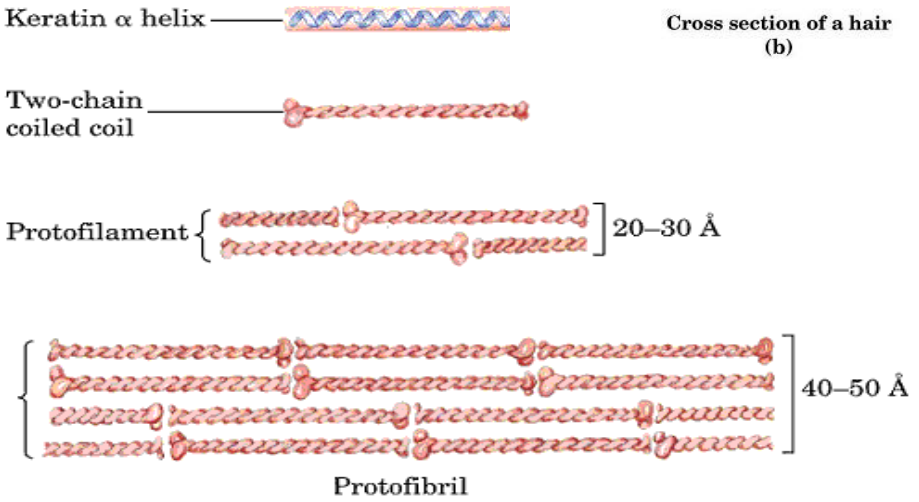
[d]Most (about 80%) is involved in cross-links.

[e]Essentially absent.

4.1.1. Alfa queratina. Presente en todos los vertebrados superiores (pelo, cuerno, uñas, plumas). Está constituida por agrupaciones de pares de hélices alfa arrolladas. Aparecen entrecruzamientos intercatenarios por puentes disulfuro que las hacen insolubles. Las queratinas duras (pelo, cuerno, uñas) contienen más disulfuros que las blandas (piel).



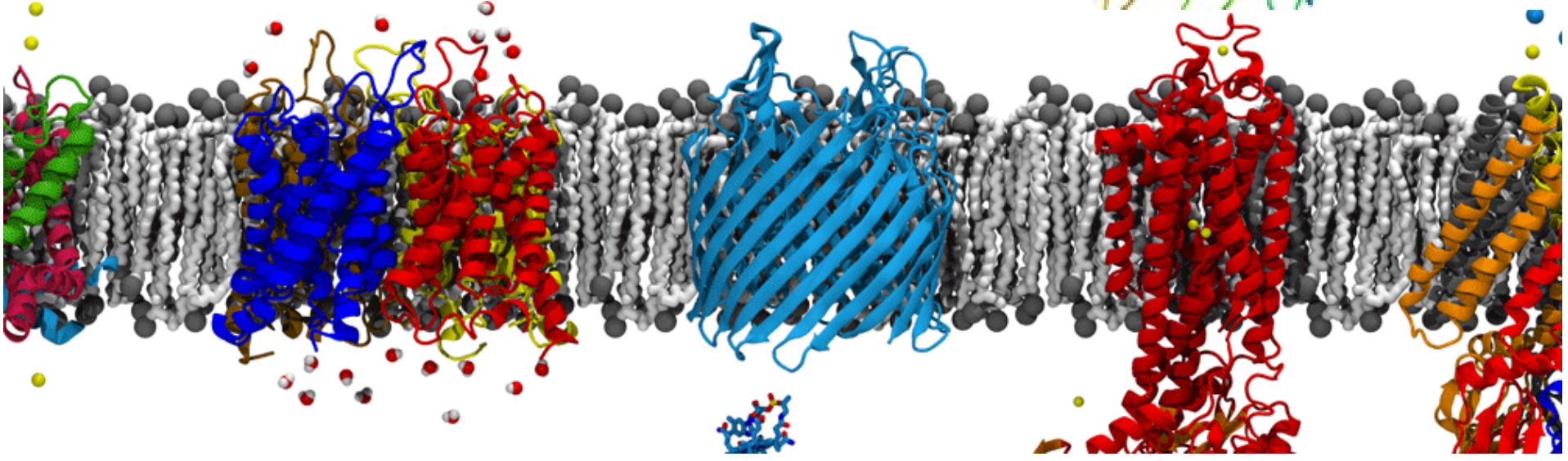




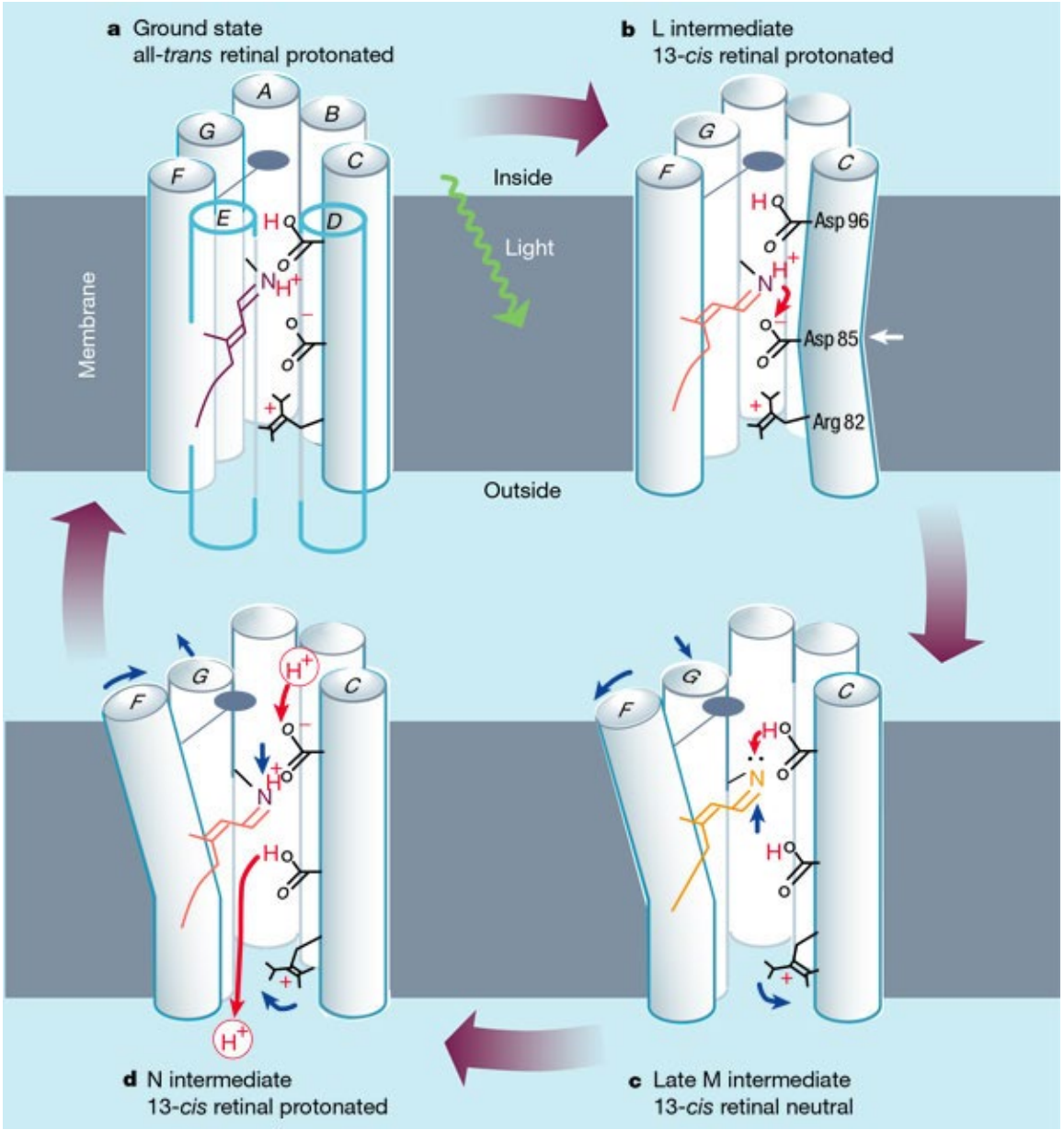
4.2. PROTEÍNAS DE MEMBRANA

En las membranas biológicas se localizan proteínas que pueden estar introducidas en ellas, ancladas, o simplemente interactuando con la parte polar de las membranas

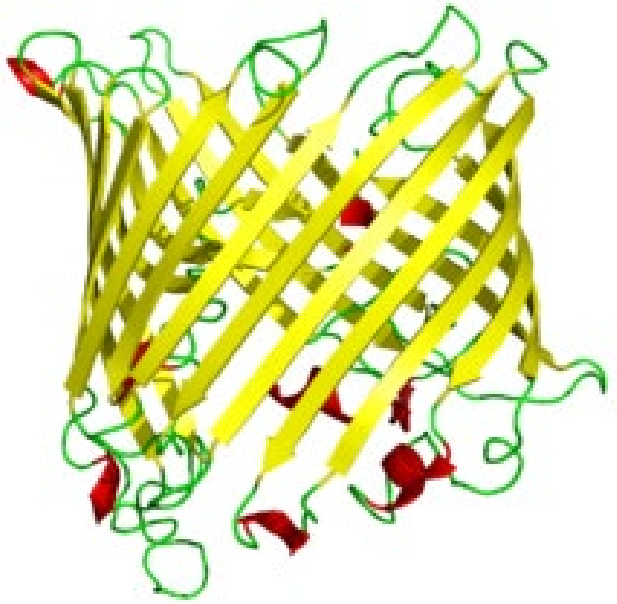


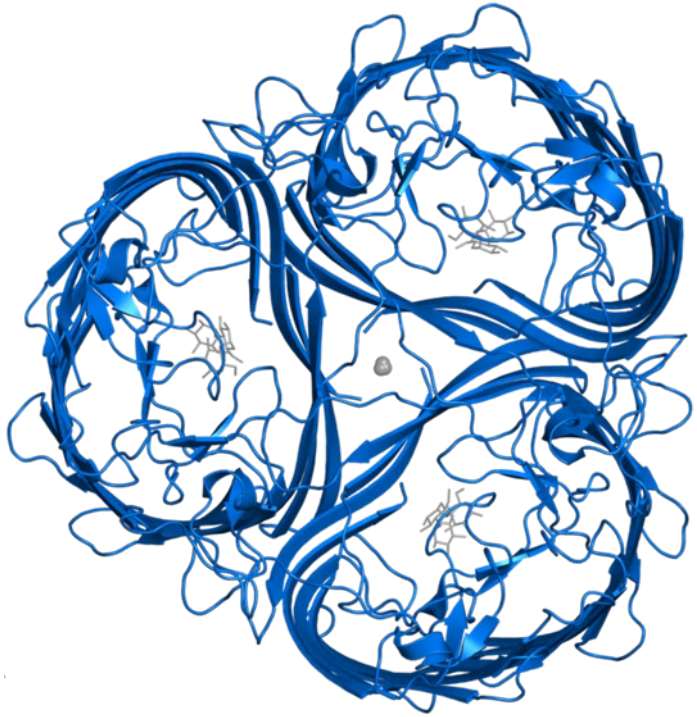


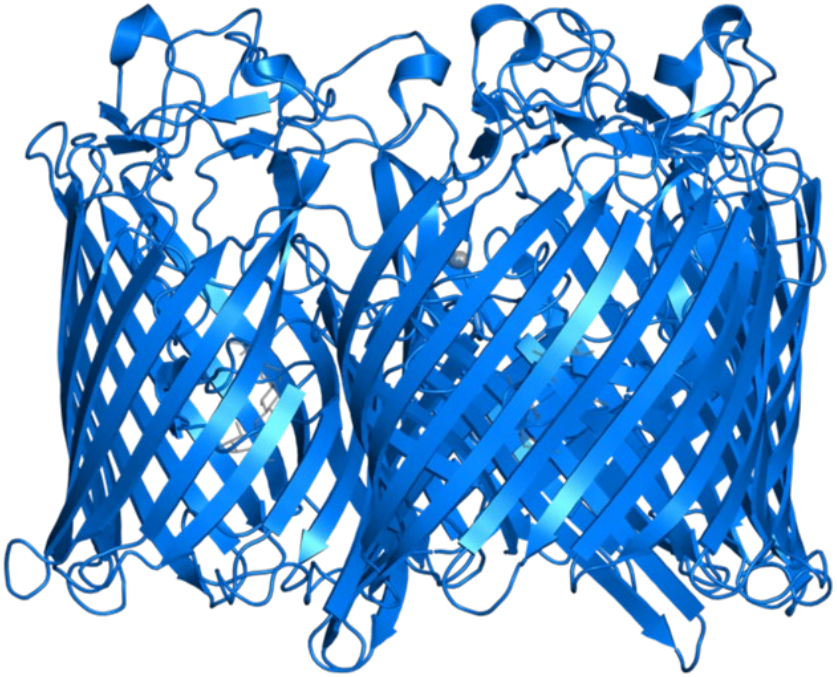
4.2.1. Generadores de diferencias de pH. La bacteriorrodopsina bombea protones a través de una membrana y la diferencia de pH



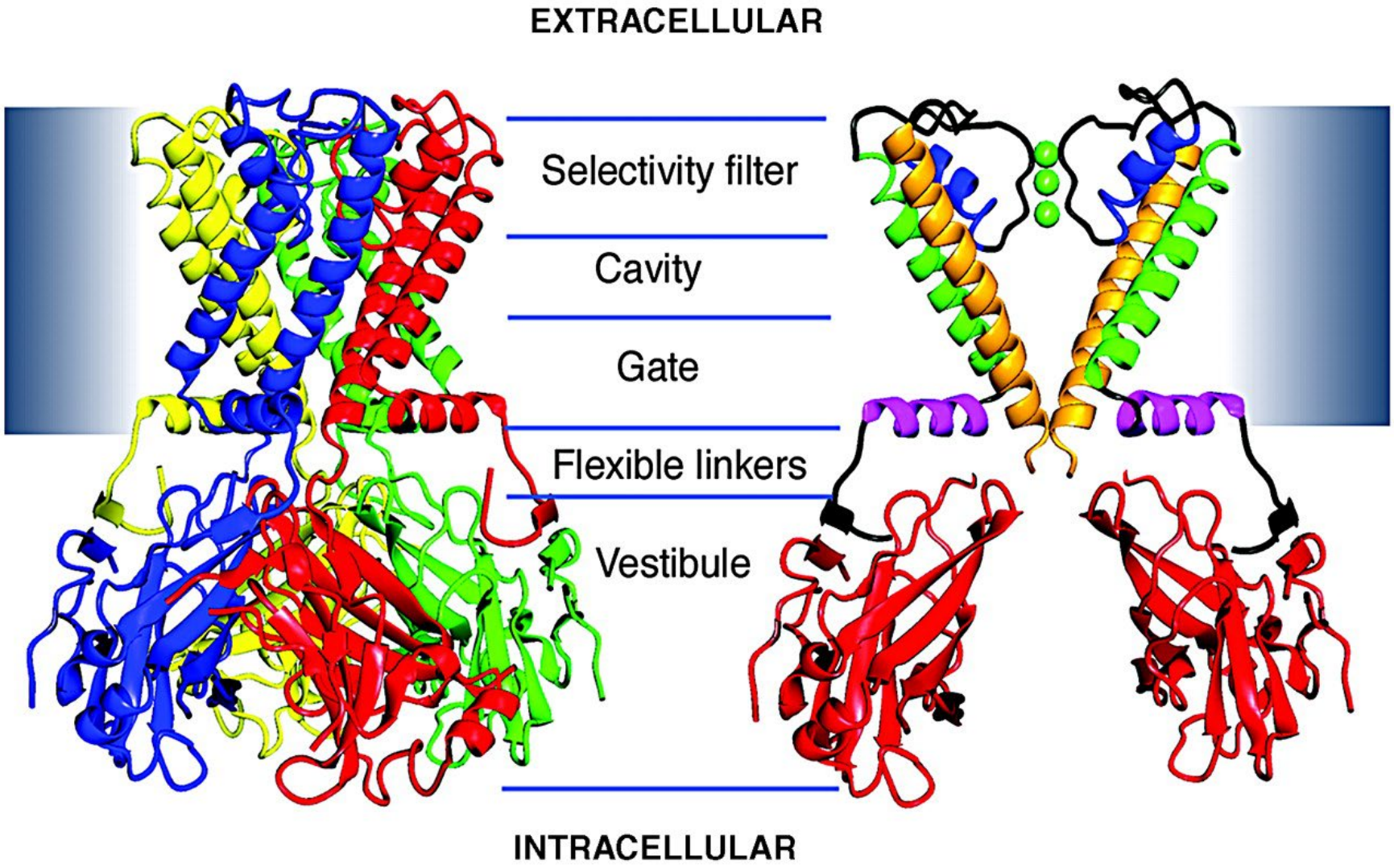
4.2.2. Transportadores de solutos. Porina: Las porinas son proteínas que forman poros en la membrana por los que son transportados solutos. Un ejemplo es la maltoporina que facilita la dilución de maltodextrinas a través de la membrana externa de bacterias Gram-negativas.







4.2.3. Canales iónicos. Canal de potasio: Transporta iones K^+ a través de la membrana celular. Tiene un filtro selectivo para el K^+, que lo deshidrata y dirige al otro extremo de la membrana



4.2.4. Toxinas. Hemolisina: La hemolisina de *Staphylococcus aureus* forma poros en la membrana de eritrocitos y otras células, permeabilizándolas y produciendo la muerte celular. El poro aparece en un barril beta heptamérico de 14 hebras formado por 7 horquillas beta de los 7 monómeros.

